

Last-Mile Monte Carlo Simulator

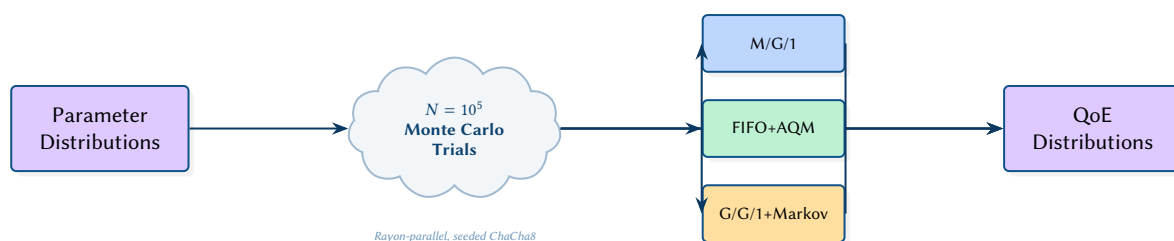
Stochastic Queuing Models for Cloud Gaming Network Impairment

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Abstract. This report presents the mathematical foundations for `lastmile-sim`, a Monte Carlo simulator that models last-mile network impairments on cloud gaming traffic using parameterised stochastic queuing models. Each access technology—Wi-Fi (802.11ac/ax/be), DOCSIS cable, GPON fibre, 5G NR, LEO satellite, fixed wireless access (FWA), and DSL—is abstracted as a calibrated queuing model whose service time distribution and loss process capture the technology’s characteristic telemetry fingerprint identified in the companion report LM-TR-2026-01. The simulator generates 60 fps video frame streams, passes them through composable three-layer queuing stages (in-home, last-mile, upstream), and produces statistically robust QoE distributions across hundreds of thousands of parallel trials. Beyond QoE prediction, the simulator serves as a *labelled training data generator* for impairment detection and network fingerprinting algorithms—including Matrix Profile motif discovery and ROCKET time-series classification—enabling automated identification of access technology type and impairment source from observed telemetry. This report focuses on the queuing theory, stochastic modelling, detection methodology, and statistical rigour; a companion implementation guide covers the Rust engineering details.

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1 Introduction and Scope

The companion report LM-TR-2026-01 establishes that cloud gaming quality is dominated by last-mile network behaviour, and that each access technology produces distinct telemetry *motifs*—structural, repeating patterns in delay, jitter, and loss. This report addresses the question: *how do we model these motifs mathematically to predict QoE under varying conditions?*

1.1 The Simulation Problem

Cloud gaming sends encoded video at 60 fps (one frame every 16.67 ms). Each frame is fragmented into MTU-sized UDP packets that traverse the last-mile network. The simulation must answer:

1. For a given last-mile topology and traffic mix, what is the distribution of frame delivery latency?
2. What fraction of frames miss their playout deadline?
3. How effective is a given FEC configuration at recovering from packet loss?

These are inherently *distributional* questions—single-point estimates are insufficient. We need the full tail: P_{50} , P_{95} , P_{99} , $P_{99.9}$.

1.2 Why Monte Carlo?

The interaction between stochastic arrival processes (frame generation, cross-traffic bursts), queuing dynamics (buffer fill, AQM drops), and correlated loss processes (HARQ, beam switching) makes closed-form analysis intractable for all but the simplest topologies. Monte Carlo simulation [1] provides empirical distributions with controllable confidence intervals:

$$CI_{95} \approx \frac{1.96 \cdot \hat{\sigma}}{\sqrt{N}} \quad (1)$$

where N is the number of trials and $\hat{\sigma}$ the sample standard deviation. At $N = 100,000$ trials, the 95% confidence interval on the mean frame latency is $\approx 0.006\hat{\sigma}$ —narrow enough for engineering decisions.

1.3 Headline Results

Table 1 summarises the simulator’s headline findings across all eleven access technology families under baseline conditions. P95 frame latency ranges from 7.17 ms (FWA clear sky) to 53.39 ms (LEO satellite), with wired technologies clustering below 10 ms. All technologies remain in the “Acceptable” QoE band at baseline, but the coefficient of variation reveals a structural split: wired links are nearly deterministic, while 5G NR and FWA retain materially higher environmental variability.

Figure 1 turns that table into a scan-friendly ranking. The separation is structural, not marginal: wired and modern Wi-Fi technologies sit in a low-latency band, while wide-area wireless and satellite links occupy clearly different operating regions.

Table 1. Executive summary: headline statistics per technology under baseline conditions. QoE uses a common four-band rubric: Acceptable, Degraded, Poor, and Unplayable.

Family	Technology	P95 (ms)	Loss (%)	CV (%)	Best fit	QoE
Wired	GPON	8.12	1.0	0.1	Normal	Acceptable
Wired	DOCSIS 3.0	9.17	1.3	0.0	Gamma	Acceptable
Wired	DOCSIS 3.1	9.08	1.2	0.0	Normal	Acceptable
Wired	DSL (VDSL2)	8.17	1.1	0.0	Gamma	Acceptable
Wi-Fi	Wi-Fi 5	8.97	1.1	0.3	Log-normal	Acceptable
Wi-Fi	Wi-Fi 6	17.60	1.1	0.1	Gamma	Acceptable
Wi-Fi	Wi-Fi 6E	17.60	1.1	0.1	Gamma	Acceptable
Wi-Fi	Wi-Fi 7	8.25	1.1	0.1	Normal	Acceptable
Cellular	5G NR	31.50	1.3	5.5	Log-normal	Acceptable
Fixed Wireless	FWA	7.17	1.1	5.1	Log-normal	Acceptable
Satellite	LEO Satellite	53.39	1.1	0.0	Normal	Acceptable

Insight:
Monte Carlo trades analytical elegance for universality: any combination of queuing models, loss processes, and FEC strategies can be evaluated without deriving new closed-form results.

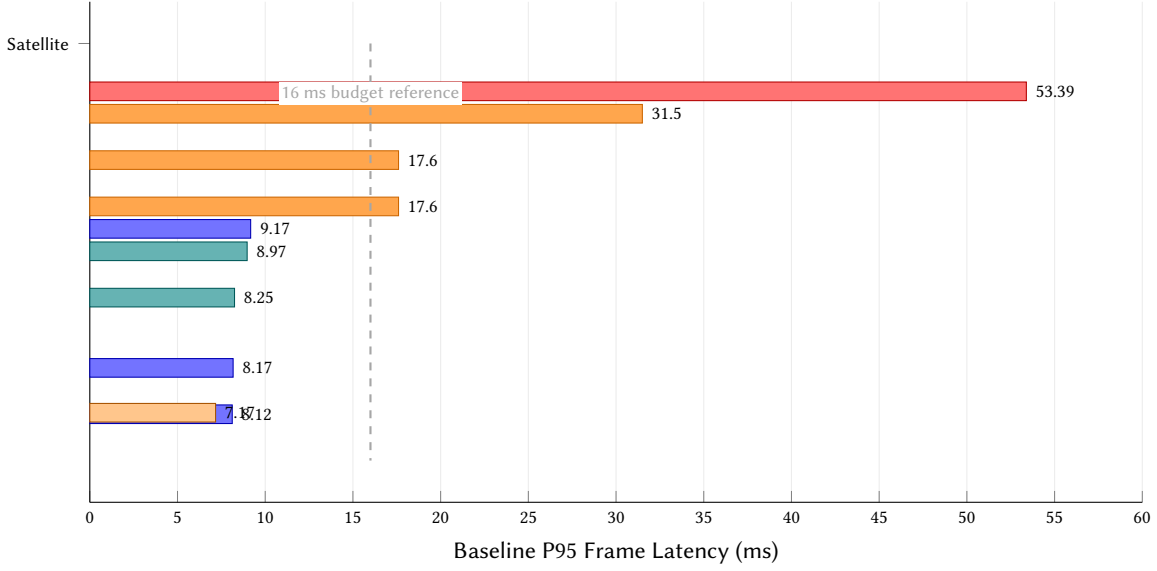


Figure 1. Comparative benchmark: baseline P95 frame latency ranked by access technology under idle conditions. The dashed line marks a 16 ms reference budget for the network share of a 60 fps interactive pipeline, making the separation between local-loop classes immediately visible.

2 The Cloud Gaming Traffic Model

2.1 Downstream Frame Generation

The server generates frames at a fixed rate f (typically 60 fps), alternating between I-frames and P-frames with period G (the GOP length, typically 60 frames):

$$\text{FrameType}(n) = \begin{cases} \text{I} & \text{if } n \bmod G = 0 \\ \text{P} & \text{otherwise} \end{cases} \quad (2)$$

Frame sizes are drawn from technology-appropriate distributions:

$$S_I \sim \text{LogNormal}(\mu_I, \sigma_I^2), \quad S_P \sim \text{LogNormal}(\mu_P, \sigma_P^2) \quad (3)$$

where typical parameters are $\mu_I \approx \ln(50000)$, $\sigma_I \approx 0.3$ for I-frames (~ 50 KB) and $\mu_P \approx \ln(10000)$, $\sigma_P \approx 0.4$ for P-frames (~ 10 KB). Each frame is fragmented into $\lceil S/\text{MTU} \rceil$ packets, where $\text{MTU} \approx 1400$ bytes. fig. 2 illustrates the resulting bursty packet generation pattern over a 120-frame window.

Downstream Packet Generation: I/P Frame Pattern

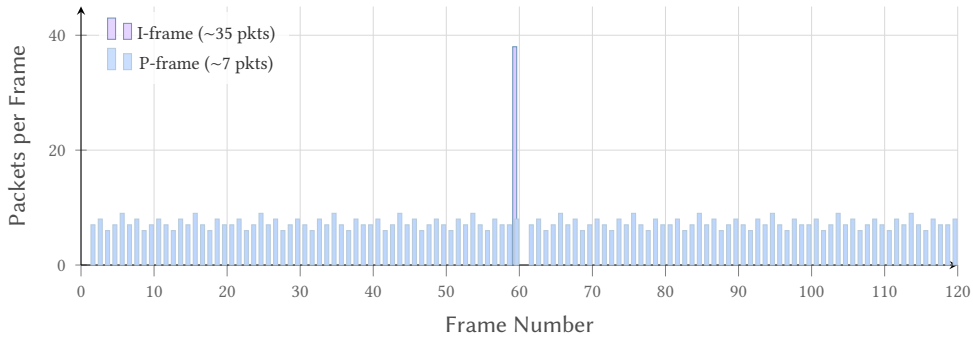


Figure 2. Packet generation pattern for a 60 fps, 120-frame (2-second) window. I-frames at $n = 0, 60$ produce ~ 35 packets each; P-frames produce ~ 7 . The bursty I-frame pattern is critical for bufferbloat modelling—it represents a transient $5\times$ load spike.

2.2 Packet Pacing and Encoding Delay

After fragmentation, packets are not emitted instantaneously. The simulator models an *encoding delay* Δ_{enc} (typically ~ 1 ms for hardware encoders) spread across the packets of each frame:

$$t_{n,j}^{\text{emit}} = t_n^{\text{gen}} + \frac{j}{k_n} \cdot \Delta_{\text{enc}}, \quad j = 0, 1, \dots, k_n - 1 \quad (4)$$

where $k_n = \lceil S_n / \text{MTU} \rceil$ is the packet count for frame n . This pacing prevents instantaneous bursts from overwhelming shallow queues and more accurately reflects the encoder's streaming output.

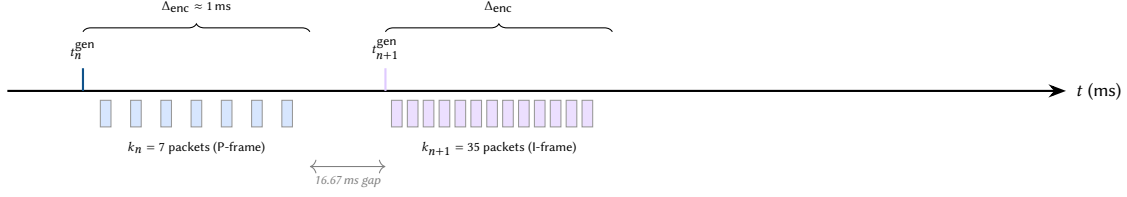


Figure 3. Packet pacing model. Each frame's packets are spread uniformly over the encoding delay Δ_{enc} , producing a paced emission timeline rather than an instantaneous burst. I-frames (right) emit more packets over the same interval, creating a higher instantaneous rate.

2.3 Frame Deadlines and the Latency Budget

Each frame n has a hard playout deadline:

$$d_n = t_n^{\text{gen}} + L_{\text{budget}} \quad (5)$$

where $t_n^{\text{gen}} = n/f$ is the generation time and L_{budget} is the total motion-to-photon latency budget (typically 80 ms for competitive gaming, 100 ms for casual [2–4]). A frame is *delivered* if all required packets (accounting for FEC recovery) arrive before d_n ; otherwise it is *dropped* (fig. 12 illustrates the possible reassembly outcomes).

2.4 Cross-Traffic Model

Competing household traffic is modelled as a superposition of independent sources:

- **Constant Bit Rate (CBR):** Models steady IoT, VoIP, or background sync traffic. Arrival rate λ_{CBR} with fixed packet size.
- **On-Off:** Models bursty web browsing or app updates. Alternating exponential ON periods (rate μ_{on}) and OFF periods (rate μ_{off}).
- **ABR Video (Netflix model):** Periodic large bursts filling a playout buffer. The client requests chunks at intervals $T_{\text{chunk}} \approx 4$ s, each chunk consuming ≈ 2 MB at high quality, creating a sawtooth bandwidth demand pattern.

The aggregate cross-traffic load is parameterised as a fraction ρ_{cross} of the bottleneck link capacity C :

$$\rho_{\text{total}} = \frac{\lambda_{\text{gaming}} + \lambda_{\text{cross}}}{C} \quad (6)$$

When $\rho_{\text{total}} > 1$, queues grow and bufferbloat dominates.

2.5 Upstream Input Traffic

The motion-to-photon latency budget includes the upstream path: player input events (controller state, mouse/keyboard) must reach the server before the next frame can be rendered. Input packets are small (~ 50 – 100 bytes) but latency-critical, sent at the frame rate or higher (60–120 Hz):

$$\lambda_{\text{input}} = f_{\text{input}} \cdot S_{\text{input}}, \quad f_{\text{input}} \in [60, 120] \text{ Hz}, \quad S_{\text{input}} \approx 80 \text{ bytes} \quad (7)$$

The upstream input bandwidth is negligible (~ 76 kbps at 120 Hz), so input packets do not cause congestion. However, they are affected by the same queuing delays as downstream traffic—a full upstream queue on a DOCSIS modem adds the same sojourn time to an 80-byte input packet as to a 1400-byte video packet. The total motion-to-photon latency is:

$$L_{\text{m2p}} = D_{\text{input}}^{\uparrow} + T_{\text{render}} + T_{\text{encode}} + D_{\text{frame}}^{\downarrow} + T_{\text{decode}} \quad (8)$$

where $D_{\text{input}}^{\uparrow}$ is the upstream input delay through the link stack, $T_{\text{render}} + T_{\text{encode}} \approx 10$ – 20 ms is the server-side pipeline, $D_{\text{frame}}^{\downarrow}$ is the downstream frame delivery delay, and $T_{\text{decode}} \approx 2$ – 5 ms is client-side decoding. The simulator models both $D_{\text{input}}^{\uparrow}$ and $D_{\text{frame}}^{\downarrow}$ through the same link stack, capturing the bidirectional impact of queuing.

3 Three-Layer Network Model

A real cloud gaming path traverses three distinct impairment domains, each with independent stochastic characteristics. The simulator composes these as a pipeline of queuing stages.

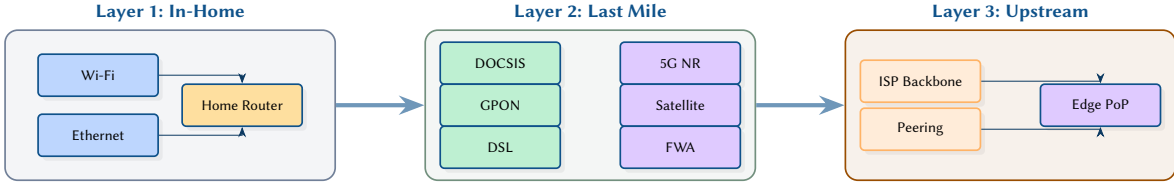


Figure 4. Three-layer network model. Layer 1 (In-Home) covers the local network from device to CPE router. Layer 2 (Last Mile) covers the access network from CPE to ISP aggregation point. Layer 3 (Upstream) covers ISP backbone, peering, and transit to the edge PoP. Each layer contains one or more composable queuing stages.

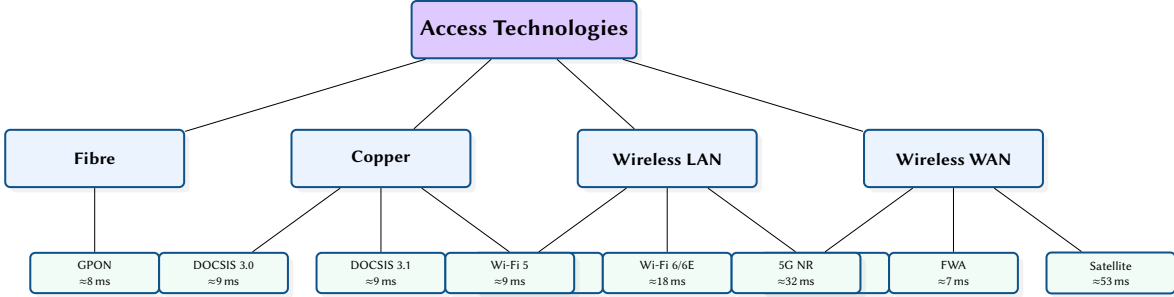


Figure 5. Technology taxonomy: eleven access technologies grouped by physical medium, annotated with baseline P95 frame latency.

3.1 Layer 1: In-Home Network

The in-home network has two components:

- **Device-to-Router Link:** Wi-Fi (5/6/7) or wired Ethernet. Wi-Fi is the dominant impairment source in this layer. Ethernet provides a near-ideal baseline (< 1 ms, zero loss).
- **Home Router (CPE):** The router queue is the critical bottleneck when cross-traffic is present. Modelled as FIFO (default) or FQ-CoDel (if SQM enabled).

The in-home layer is often the *most impactful* and *most controllable* variable—users can upgrade their Wi-Fi, enable SQM, or switch to Ethernet.

3.2 Layer 2: Last-Mile Access Network

The access network from CPE to ISP aggregation point. This is determined by the user’s service subscription and cannot be changed within a gaming session. The simulator models eleven access technologies (see section 8), including variants within each technology (e.g., DOCSIS 3.0 vs 3.1 vs Low-Latency DOCSIS).

3.3 Layer 3: Upstream Congestion

The path from ISP aggregation to the cloud gaming edge PoP. This layer captures:

- **ISP backbone delay:** Fixed propagation delay based on geographic distance (~ 1 ms per 100 km of fibre).
- **Time-of-day congestion [5]:** Many ISPs experience elevated delay and loss during evening peak hours (typically 18:00–23:00) when aggregate residential demand saturates upstream links.
- **Peering congestion [6]:** Delays at ISP-to-ISP interconnection points, which can add 5–20 ms during peak periods.

3.3.1 Time-of-Day Congestion Model

The upstream congestion is modelled as a time-varying multiplier on base delay and loss (fig. 6):

$$D_{\text{upstream}}(h) = D_{\text{base}} \cdot \left(1 + A \cdot \exp \left(-\frac{(h - h_{\text{peak}})^2}{2\sigma_h^2} \right) \right) \quad (9)$$

where h is the hour of day, $h_{\text{peak}} \approx 20.5$ (8:30 PM), $\sigma_h \approx 1.5$ hours, and A is the peak congestion amplitude (typically $1.5\text{--}3\times$ baseline). Loss follows a similar profile:

$$p_{\text{upstream}}(h) = p_{\text{base}} + p_{\text{peak}} \cdot \exp \left(-\frac{(h - h_{\text{peak}})^2}{2\sigma_h^2} \right) \quad (10)$$

Upstream Congestion: Time-of-Day Profile

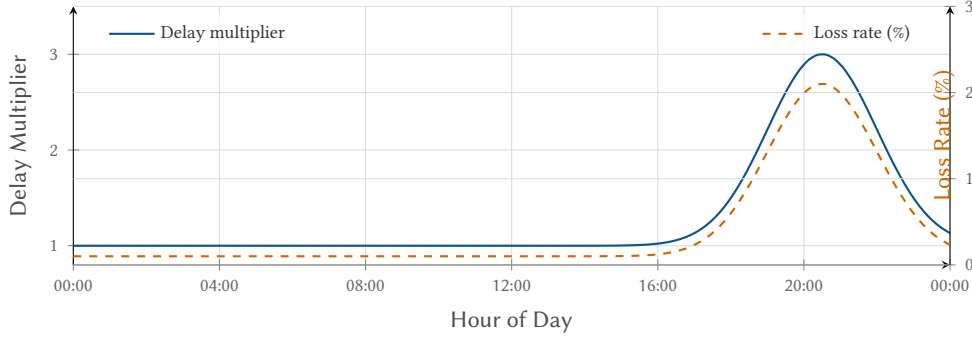


Figure 6. Upstream congestion follows a Gaussian profile centred on the evening peak (~20:30). Delay increases 2–3× and packet loss rises to ~2% during the 18:00–23:00 window. This is the impairment that correlates across users sharing the same ISP prefix.

4 Environmental Context Models

Network impairment is not static—it varies with time of day, dwelling type, household activity, and ISP infrastructure quality. The simulator parameterises these contextual factors to produce realistic scenario diversity.

4.1 Time-of-Day Congestion Profiles

Multiple layers of the network exhibit diurnal patterns (fig. 7), each driven by different underlying causes:

Diurnal Congestion Profiles Across Network Layers

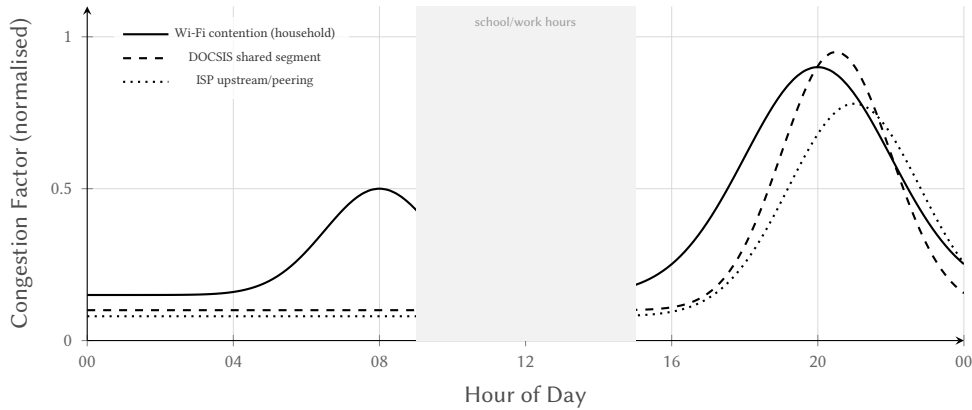


Figure 7. Diurnal congestion profiles for three network layers. Wi-Fi contention shows a bimodal pattern (morning and evening peaks as household devices activate). DOCSIS shared segment and ISP upstream peak sharply during the 18:00–23:00 evening window. During school/work hours (09:00–15:00), all layers are at minimum load.

4.1.1 Wi-Fi Time-of-Day Model

The number of active Wi-Fi stations varies throughout the day:

$$n_{\text{active}}(h) = n_{\text{base}} + n_{\text{peak}} \cdot \left(\gamma_1 \cdot e^{-(h-h_{\text{am}})^2/2\sigma_{\text{am}}^2} + \gamma_2 \cdot e^{-(h-h_{\text{pm}})^2/2\sigma_{\text{pm}}^2} \right) \quad (11)$$

where $n_{\text{base}} \approx 3$ (always-on IoT/standby devices), $h_{\text{am}} \approx 8$, $h_{\text{pm}} \approx 20$, and $\gamma_1 < \gamma_2$ (the evening peak is more intense). Higher station counts feed directly into the Bianchi collision model (section 8.1), increasing contention delay and jitter.

4.1.2 Shared Last-Mile Media

DOCSIS cable is a shared medium: all subscribers on the same HFC node share upstream capacity. The effective per-subscriber capacity depends on the number of active users $M(h)$:

$$C_{\text{effective}}(h) = \frac{C_{\text{node}}}{M(h)}, \quad M(h) \sim \text{Poisson}(\bar{M}(h)) \quad (12)$$

where $\bar{M}(h)$ follows the same evening-peak Gaussian profile. At peak hours, a 1 Gbps DOCSIS node shared among $M = 200$ active subscribers yields only ~ 5 Mbps per subscriber, which is marginal for 1080p gaming and inadequate for 4K.

4.2 Dwelling Type: House vs. Apartment

The dwelling type affects both Wi-Fi and cross-traffic characteristics:

Table 2. Dwelling type parameters. Apartments experience higher Wi-Fi interference from neighbouring units and shared building infrastructure.

Parameter	House	Apartment
Wi-Fi interference rate	Low ($p_i \approx 0.02$)	High ($p_i \approx 0.15$)
Neighbouring BSSIDs	2–5	10–30
Co-channel interference	Rare	Frequent (2.4 GHz)
Shared DOCSIS node	50–100 subscribers	200–500 subscribers
Building ethernet/fibre	N/A	Sometimes shared GPON splitter
Cross-traffic intensity	Moderate	Higher (denser occupancy)

In apartments, the Wi-Fi 5 Bianchi model must account for *hidden-node* interference from neighbouring BSSIDs, which increases the effective collision rate beyond what the station count alone predicts. Wi-Fi 6’s BSS Colouring and Wi-Fi 7’s MLO provide significant mitigation in dense environments.

4.3 Household Cross-Traffic Patterns

The probability and intensity of cross-traffic depends on household composition and time of day:

$$\rho_{\text{cross}}(h) \sim \text{Beta}(\alpha(h), \beta(h)) \quad (13)$$

where $\alpha(h)$ and $\beta(h)$ shift the Beta distribution’s mode toward higher load during evening hours. Typical cross-traffic sources by time of day:

- **Daytime (09:00–15:00):** Minimal—IoT devices, standby traffic. $\rho \approx 0.1$.
- **Afternoon (15:00–18:00):** Moderate—children streaming video, software updates. $\rho \approx 0.4$.
- **Evening peak (18:00–23:00):** Heavy—multiple ABR video streams, downloads, video calls. $\rho \approx 0.7$.
- **Night (23:00–06:00):** Low—but cloud backups and system updates may run. $\rho \approx 0.15$.

4.4 ISP Quality Profiles

Different ISPs deliver varying levels of infrastructure quality. The simulator parameterises ISP-level characteristics as a *quality profile*:

Table 3. ISP quality profiles. Each profile characterises the upstream congestion behaviour, over-subscription ratio, and peering quality.

Profile	Peak mult.	delay	Peak loss	Over-sub ratio	Peering jitter
Premium	1.2×		< 0.1%	10:1	< 2 ms
Standard	2.0×		0.5%	20:1	5–10 ms
Budget	3.0×		1.5%	40:1	10–25 ms
Congested	5.0×		3.0%	80:1	20–50 ms

The ISP profile modulates the time-of-day upstream model from section 3: the amplitude A in eq. (9) and peak loss p_{peak} in eq. (10) are set according to the profile. A “Premium” ISP with low over-subscription shows minimal evening degradation; a “Budget” ISP with aggressive over-subscription produces severe peak-hour impairment that may be the dominant latency contributor.

4.5 Regional Technology Prevalence

The choice of access technology is not uniformly random—it is strongly conditioned on geography, as characterised in Table 4.1 of LM-TR-2026-01. The simulator uses *regional priors* [7] to weight the probability of each technology in a given deployment context:

As LM-TR-2026-01 observes, regional metadata is derivable from IP geolocation or server region assignment with no telemetry required, yet it can halve the hypothesis space before the first frame is received.

Table 4. Regional technology prevalence priors. Values represent the probability of a randomly selected household using each access technology, conditioned on region. Probabilities are approximate and based on ITU broadband statistics and regional deployment reports.

Region	FTTH	DOCSIS	DSL	5G FWA	mmW FWA	LEO Sat.	Wi-Fi gen.
N. America (urban)	0.25	0.50	0.10	0.10	0.03	0.02	5/6
N. America (rural)	0.05	0.15	0.20	0.20	0.05	0.35	5
W. Europe (urban)	0.35	0.15	0.35	0.10	0.03	0.02	5/6
Nordics / Baltics	0.70	0.05	0.10	0.10	0.02	0.03	6/6E
Australia (metro)	0.30	0.30	0.15	0.15	0.05	0.05	5/6
Australia (regional)	0.05	0.10	0.15	0.30	0.10	0.30	5
East Asia (JP/KR)	0.80	0.05	0.05	0.05	0.03	0.02	6/6E
SE Asia	0.15	0.05	0.30	0.30	0.05	0.15	5

4.6 Housing Density and Interference Correlation

The dwelling type (table 2) is itself conditioned on settlement density. Dense urban environments have a high probability of apartment dwellings, which correlates with elevated Wi-Fi contention and neighbour-network interference [8]. Conversely, rural settings imply detached housing with minimal interference but limited access technology choice.

Table 5. Conditional technology and interference probabilities by settlement density. $P(\text{apt})$ is the probability of an apartment dwelling; n_{BSS} is the expected number of visible neighbouring Wi-Fi BSSIDs; $P(\text{FTTH})$ and $P(\text{Sat})$ are the conditional probabilities of FTTH and satellite access respectively.

Density class	$P(\text{apt})$	n_{BSS}	$P(\text{FTTH})$	$P(\text{Sat})$	Wi-Fi interference
Dense urban ($>5000/\text{km}^2$)	0.80	20–40	0.40	<0.01	Severe
Urban ($1000\text{--}5000/\text{km}^2$)	0.50	10–20	0.35	0.02	Moderate
Suburban ($200\text{--}1000/\text{km}^2$)	0.15	4–10	0.25	0.05	Mild
Rural ($<200/\text{km}^2$)	0.05	0–3	0.10	0.30	Negligible

The key correlations for the simulator are:

- **Dense urban $\rightarrow$ high Wi-Fi contention:** $n_{\text{BSS}} > 20$ increases the external interference rate λ_{ext} in the Bianchi model (section 8.1), even for Wi-Fi 6 OFDMA systems operating on congested 5 GHz channels.
- **Rural $\rightarrow$ satellite/FWA:** Users in low-density areas are unlikely to have FTTH or DOCSIS; the simulator should weight LEO satellite and FWA priors accordingly.
- **Apartment $\rightarrow$ shared DOCSIS segment:** Apartment buildings often share a single DOCSIS node, amplifying the cross-traffic congestion modelled in section 8.4.

4.7 Gaming Session Behaviour Profiles

The time-of-day model (section 4.1) determines *network* congestion, but the probability that a gaming session *occurs* at a given hour also varies. The simulator uses per-region session probability distributions derived from gaming platform analytics and media consumption research:

$$P(\text{session at hour } h) \propto w_{\text{region}}(h) \cdot w_{\text{weekday/end}}(h) \quad (14)$$

Table 6. Gaming session timing profiles by region. Peak hours reflect cultural norms around work schedules, meal times, and media consumption. Data informed by Steam concurrent user statistics, Newzoo global games market reports [9], and regional TV prime-time schedules.

Region	Weekday peak	Weekend peak	Notes
N. America	19:00–00:00	14:00–02:00	Aligns with US TV prime time
W. Europe	18:00–23:00	13:00–01:00	Earlier peak; dinner break 19–20h
East Asia	20:00–02:00	12:00–03:00	Late-night gaming culture
SE Asia	19:00–01:00	10:00–02:00	Mobile-first; lunch-hour spike
Australia / NZ	19:00–00:00	14:00–01:00	Similar to N. America

The critical insight is that *gaming peaks overlap with network peaks*—the evening hours when users are most likely to play are precisely when household cross-traffic (streaming video, downloads) and ISP congestion are highest. The simulator captures this by sampling both the session probability and the congestion profile from the same time-of-day axis, producing correlated worst-case scenarios that reflect real-world conditions.

4.8 Socioeconomic and Demographic Factors

Beyond geography, several demographic variables influence the network environment:

Household size and composition. Larger households generate more cross-traffic. Families with children produce on-off bursts (video streaming, app downloads); single-occupant dwellings have minimal background load. The simulator parameterises cross-traffic source count N_{cross} as a function of household size.

Income / affluence. Higher-income households are more likely to have: (a) premium ISP tiers with higher capacity and lower over-subscription, (b) modern Wi-Fi equipment (6E/7 access points), and (c) mesh networking systems that improve coverage but may add backhaul hops. The simulator conditions ISP profile selection (table 3), Wi-Fi generation, and router AQM capability on an affluence parameter.

Work-from-home prevalence. Daytime cross-traffic patterns differ significantly when household members work from home. Video conferencing (sustained 3–8 Mbps with strict latency requirements) competes directly with gaming traffic and is itself sensitive to bufferbloat. The simulator models WFH as an additional cross-traffic source active during business hours (09:00–17:00 weekdays).

Gamer demographics. Younger demographics (18–34) are over-represented in cloud gaming adoption [10] and tend to play in the late evening—the worst-case network window. Casual gamers (mobile-first, older demographics) tend to play shorter sessions at varied times. The session duration distribution (Exponential for casual, heavy-tailed LogNormal for dedicated) affects how much of a time-of-day congestion event a session is exposed to.

These factors are not independently modelled but rather contribute to a *scenario archetype*—a coherent combination of region, density, dwelling, ISP profile, household composition, and session timing that represents a plausible real-world user (fig. 8). Example archetypes include:

- **Suburban family, cable:** DOCSIS 3.0, Wi-Fi 5, 4 cross-traffic sources, Budget ISP, evening peak.
- **Urban apartment, fibre:** GPON, Wi-Fi 6, 20+ neighbouring BSSIDs, Standard ISP, late-night.
- **Rural gamer, satellite:** LEO satellite, Wi-Fi 5 (1 station), no cross-traffic, evening peak.
- **WFH professional, VDSL:** VDSL2 interleaved, Wi-Fi 6E, 1 video-conferencing source, Premium ISP, mixed hours.

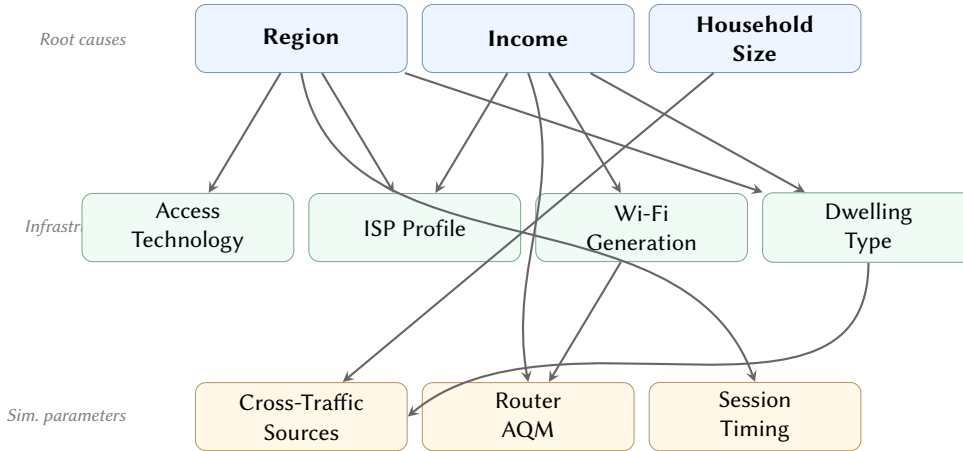


Figure 8. Bayesian conditioning structure for scenario generation. The three orthogonal root variables (top) independently condition technology and infrastructure choices (middle), which determine simulation parameters (bottom).

5 Statistical Distribution Catalogue

Monte Carlo simulation is, at its core, repeated sampling from parameterised probability distributions. Every stochastic element in the simulator—frame sizes, backoff durations, handover intervals, loss events—is drawn from a specific distribution chosen to match the physical process it represents. This section catalogues every distribution used in the simulator, illustrates its probability density function (PDF) in fig. 9, and explains the physical rationale for each choice.

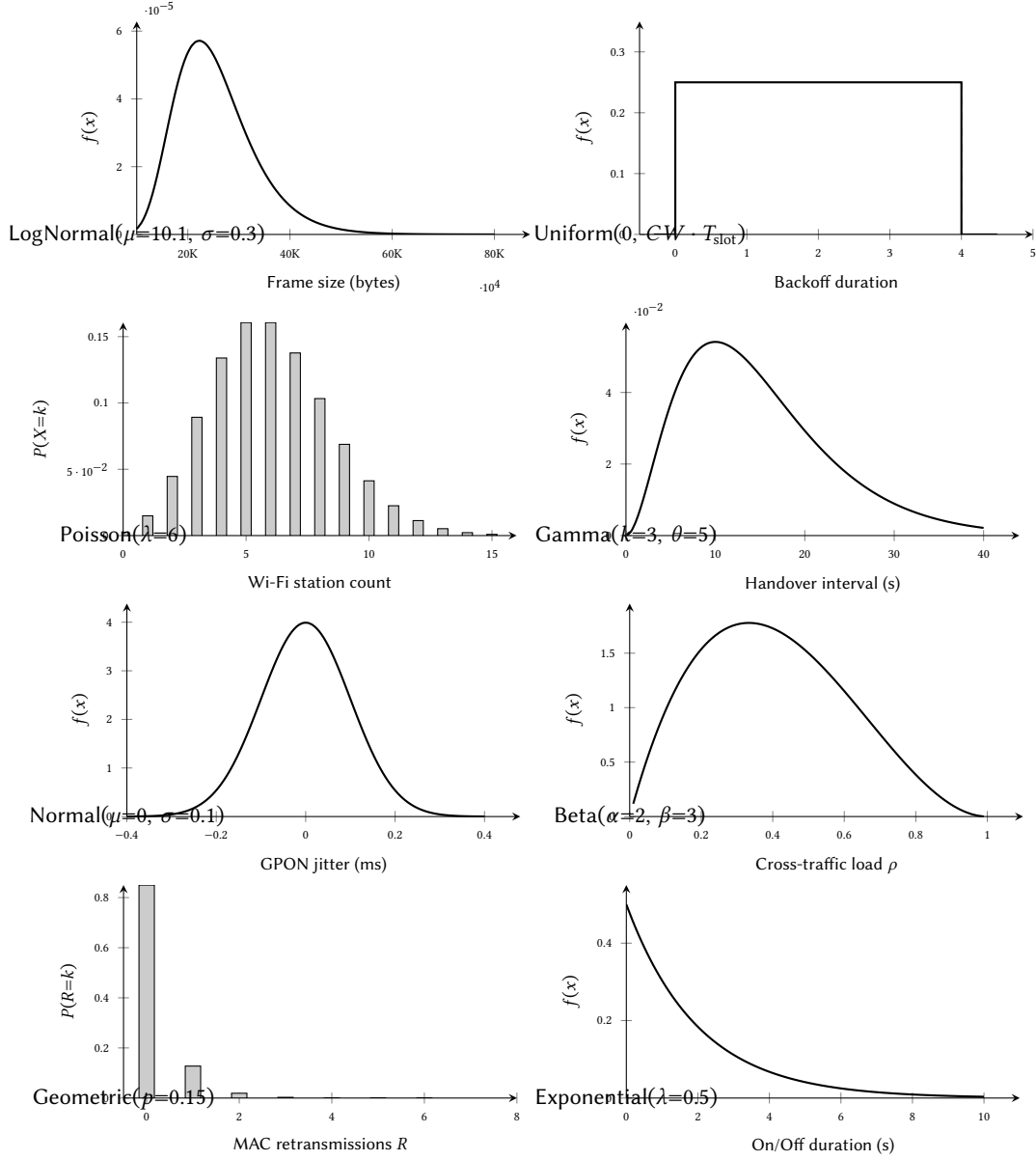


Figure 9. Complete catalogue of probability distributions used in the simulator. Each subplot shows the PDF (or PMF for discrete distributions) with parameters matching a representative scenario. **Row 1:** LogNormal (frame sizes), Uniform (backoff windows). **Row 2:** Poisson (station count sampling), Gamma (handover intervals). **Row 3:** Normal (GPON jitter), Beta (cross-traffic load). **Row 4:** Geometric (MAC retry count), Exponential (On/Off traffic durations).

Table 7. Distribution-to-model mapping. Every random variable in the simulator is drawn from one of these distributions.

Distribution	Simulator Use	Physical Rationale
LogNormal	I/P frame sizes	Multiplicative compression artifacts produce right-skewed sizes
Uniform	Backoff window, scheduling delay	802.11 standard mandates uniform slot selection
Poisson	Wi-Fi station count	Independent device arrivals in a household
Gamma	5G handover intervals	Sum of sub-processes (cell search + sync + attach); shape $k > 1$ avoids memoryless assumption
Normal	GPON jitter, upstream jitter	Central limit theorem: aggregate of many small independent delays
Beta	Cross-traffic load fraction ρ	Bounded on $[0, 1]$; shape parameters encode “typically moderate load”
Geometric	MAC retransmission count	Each retry is an independent Bernoulli trial with collision prob p
Exponential	On/Off traffic durations	Memoryless inter-event times for bursty traffic
Bernoulli	Per-packet loss (GPON, FWA clear sky)	Independent loss events at low rates
Gilbert–Markov	Correlated burst loss (5G, satellite)	Fading channels produce temporally correlated loss

6 Packet Observation Model

The simulator's fundamental data unit is a *timestamped, sequenced packet* carrying a fragment of a video frame. By recording only timestamps and sequence numbers at the transmit and receive points, all network impairment metrics can be inferred—no deep packet inspection or protocol-specific instrumentation is required.

6.1 Wire Format

Each packet carries a minimal header (fig. 10):

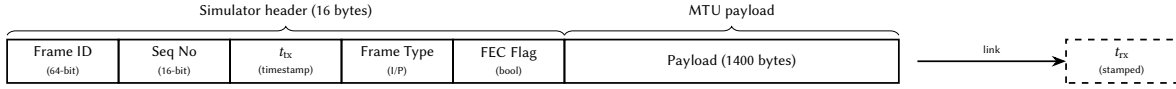


Figure 10. Packet wire format. The simulator header carries a frame ID, per-frame sequence number, transmit timestamp t_{tx} , frame type (I/P), and FEC flag. The receiver stamps t_{rx} on arrival. All impairment metrics are derived from $(t_{tx}, t_{rx}, \text{seq})$ triples.

6.2 Deriving Metrics from Timestamps

Given a stream of transmitted and received packets, the four primary impairment metrics are computed as follows (see fig. 11):

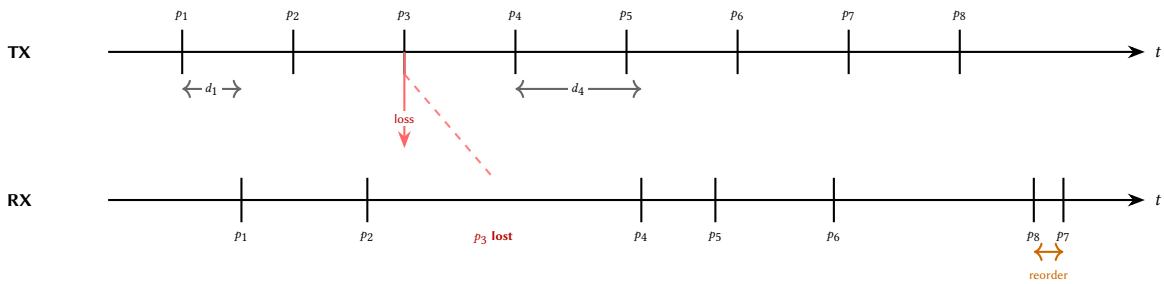


Figure 11. Transmit and receive timelines for a packet stream. Per-packet delay $d_i = t_{rx,i} - t_{tx,i}$ is directly observed. Loss is detected by missing sequence numbers. Jitter is the variation in d_i . Reordering (as in satellite beam switching) manifests as out-of-sequence arrivals.

1. **One-way delay:** $d_i = t_{rx,i} - t_{tx,i}$. The per-packet delay through the complete link stack.
2. **Packet loss:** A packet is lost if $t_{rx,i}$ is never recorded, or if $d_i > L_{\text{budget}}$ (functionally lost—arrived too late). Loss rate: $\hat{p} = n_{\text{lost}}/n_{\text{sent}}$.
3. **Jitter** (inter-packet delay variation):

$$J_i = |d_i - d_{i-1}|, \quad \hat{J} = \frac{1}{N-1} \sum_{i=2}^N J_i \quad (15)$$

This follows RFC 3550's [11] IPDV definition. Alternatively, we compute σ_d , the standard deviation of the delay series.

4. **Effective bandwidth:**

$$\hat{B} = \frac{\sum_{i \in \text{received}} S_i}{t_{rx, \text{last}} - t_{rx, \text{first}}} \quad (16)$$

where S_i is the packet size in bits. This is the *goodput* as observed by the receiver.

6.3 From Packets to Frames: Reassembly and Deadline Check

The receiver reassembles frames from their constituent packets and applies the playout deadline:

6.4 Video Stream Parameters

The traffic generator models a realistic cloud gaming video stream:

At 4K, an I-frame burst generates ~ 107 packets in 16.67 ms—a transient demand of ~ 73 Mbps that can saturate upstream links with capacities below 100 Mbps, making bufferbloat significantly worse than at 1080p.

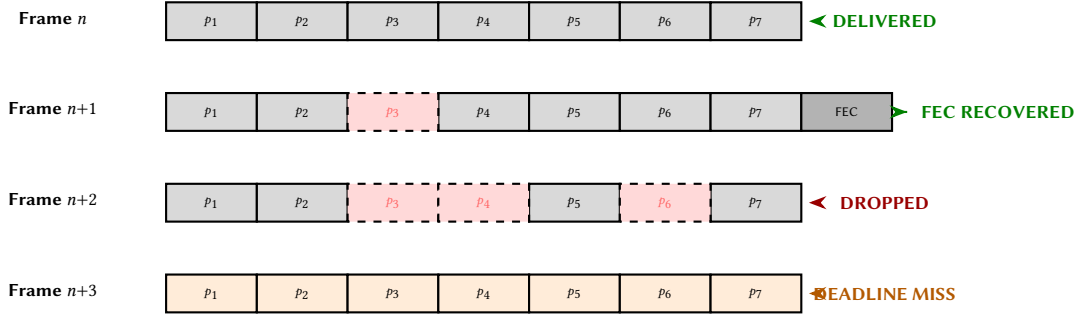


Figure 12. Frame reassembly outcomes. Frame n : all packets arrive before deadline—delivered. Frame $n+1$: one packet lost but FEC repair packet recovers it. Frame $n+2$: three packets lost, exceeding FEC capacity—frame dropped. Frame $n+3$: all packets arrive but the last arrives after the playout deadline—deadline miss (functionally dropped).

Table 8. Video stream parameters for 1080p and 4K scenarios.

Parameter	1080p @ 60 fps	4K @ 60 fps
Target bitrate	15 Mbps	45 Mbps
Mean I-frame size	50 KB	150 KB
Mean P-frame size	10 KB	30 KB
Packets per I-frame	~36	~107
Packets per P-frame	~7	~22
GOP length	60 frames	60 frames
Frame interval	16.67 ms	16.67 ms
Latency budget	80–100 ms	80–100 ms

7 Frame Delivery Assessment

The primary output of each simulation trial is a per-frame *delivery verdict*. This section formalises how link delay, bandwidth, and loss compose to determine whether a frame reaches the player’s screen.

7.1 The Four Observables

For each packet traversing the link stack, the simulator records four quantities:

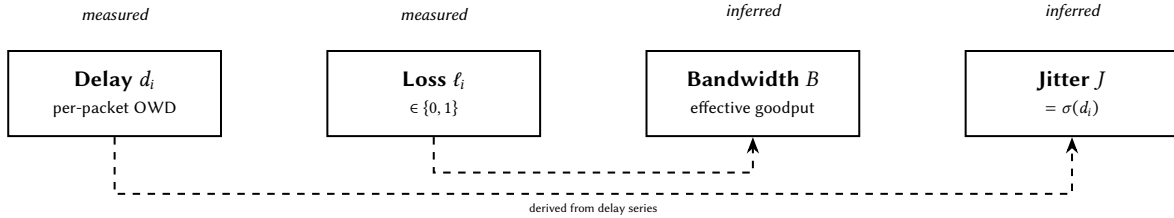


Figure 13. The four network observables. Delay and loss are directly measured per-packet. Jitter is derived from the delay series. Effective bandwidth is derived from received packet volume and loss.

7.2 Frame Delivery Decision

Each frame is classified by a deterministic decision function applied after the simulation:

$$\text{Verdict}(n) = \begin{cases} \text{DELIVERED} & \text{if } |\mathcal{R}_n| \geq k_n \text{ and } t_{\text{last},n} \leq d_n \\ \text{FEC-RECOVERED} & \text{if } |\mathcal{R}_n| < k_n \text{ but } |\mathcal{R}_n| + |\mathcal{F}_n| \geq k_n \text{ and } t_{\text{last},n} \leq d_n \\ \text{DROPPED-LOSS} & \text{if } |\mathcal{R}_n| + |\mathcal{F}_n| < k_n \\ \text{DROPPED-LATE} & \text{if } t_{\text{last},n} > d_n \end{cases} \quad (17)$$

where $\mathcal{R}_n$ is the set of received data packets for frame n , $\mathcal{F}_n$ is the set of received FEC packets, k_n is the number of data packets required, $t_{\text{last},n}$ is the arrival time of the last required packet, and d_n is the playout deadline.

7.3 QoE Classification

Frame verdicts are aggregated over a trial (typically 60 seconds of gameplay at 60 fps = 3600 frames) into a per-trial QoE score:

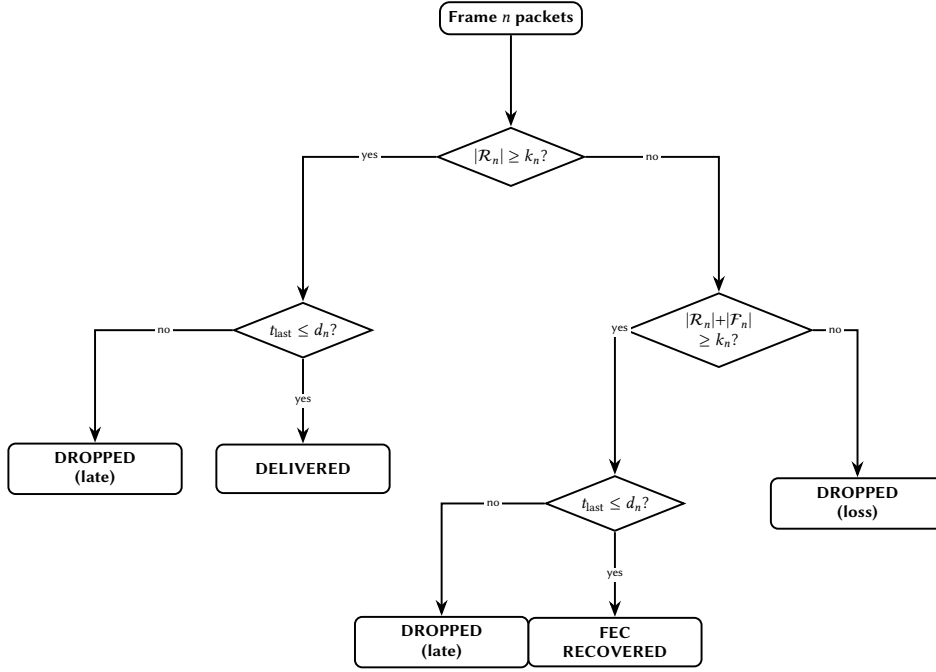


Figure 14. Frame delivery decision flowchart. Each frame is classified based on how many of its packets arrived and whether they arrived before the playout deadline d_n . FEC can recover from partial loss but not from deadline misses.

$$\text{QoE} = \begin{cases} \text{GOOD} & \text{if } P_{95}(d) < 40 \text{ ms and } f_{\text{drop}} < 0.1\% \\ \text{ACCEPTABLE} & \text{if } P_{95}(d) < 80 \text{ ms and } f_{\text{drop}} < 1\% \\ \text{POOR} & \text{if } P_{95}(d) < 150 \text{ ms and } f_{\text{drop}} < 5\% \\ \text{UNPLAYABLE} & \text{otherwise} \end{cases} \quad (18)$$

where $P_{95}(d)$ is the 95th percentile of the frame delivery latency distribution and f_{drop} is the fraction of dropped frames (from either loss or deadline miss).

8 Stochastic Queuing Models

Each last-mile technology is modelled as a stochastic queue whose parameters are calibrated to reproduce the telemetry fingerprint identified in LM-TR-2026-01. The key insight is that we do not need to simulate the full protocol stack—we need to reproduce the *statistical output* as observed at the application layer.

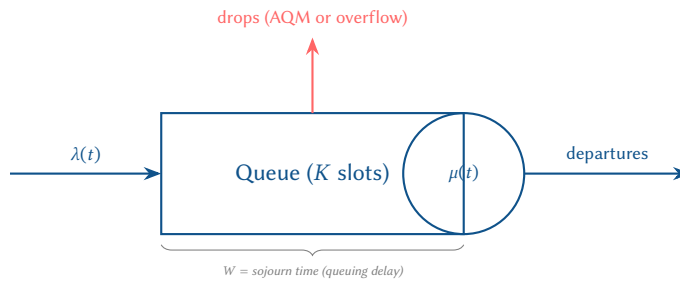


Figure 15. Generic queuing abstraction. Each link model specifies the arrival process $\lambda(t)$, service rate $\mu(t)$, queue capacity K , and dropping policy. The sojourn time W (queuing delay) is the observable output that maps to the per-packet delay contribution.

8.1 Wi-Fi 5 (802.11ac): M/G/1 with Bianchi Service Times

LM-TR-2026-01 characterises Wi-Fi 5 [12] as producing *high-frequency transient jitter spikes (4–140 ms) without throughput degradation*, caused by CSMA/CA contention and MAC-layer retransmissions. Critically, MAC retries are invisible as packet loss to the application—they manifest purely as inflated per-packet delay.

8.1.1 Service Time Distribution

The Bianchi model [13] provides the transmission probability τ for a station in a randomly chosen slot:

$$\tau = \frac{2(1 - 2p)}{(1 - 2p)(W_0 + 1) + pW_0(1 - (2p)^m)} \quad (19)$$

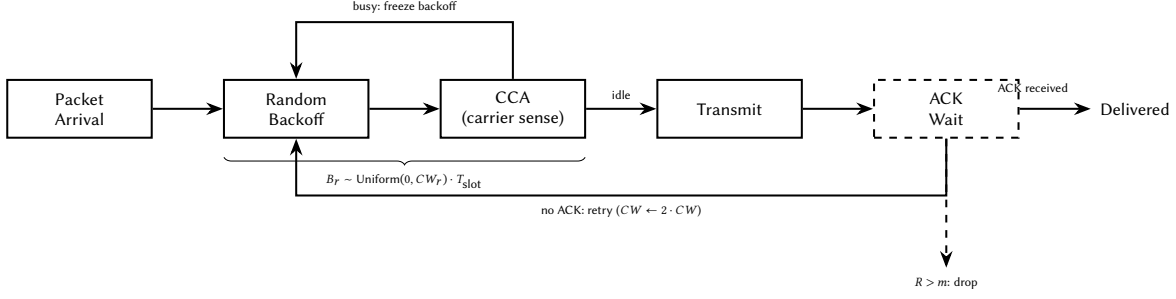


Figure 16. Wi-Fi 5 CSMA/CA channel access state machine. Each packet must survive carrier sensing and ACK timeout; collisions trigger exponential backoff with doubled contention window. The total per-packet delay accumulates across R retry rounds.

where $W_0 = CW_{\min} + 1$, $m = \log_2(CW_{\max}/CW_{\min})$ is the maximum backoff stage, and p is the collision probability given by:

$$p = 1 - (1 - \tau)^{n-1} \quad (20)$$

for n contending stations. These two equations are solved simultaneously to yield (τ, p) for a given $(n, CW_{\min}, CW_{\max})$.

To capture the temporal dynamics of real Wi-Fi environments, the simulator periodically re-evaluates n by sampling from a Poisson distribution with mean $\bar{n}$ (the configured station count), re-solving Equations (19)–(20) every 100 ms of simulated time. This produces inter-trial variance in collision probability and, consequently, in per-trial P95 latency, reflecting the real-world fluctuation of active station counts.

8.1.2 Per-Packet Delay Model

The total per-packet delay through the Wi-Fi 5 link is:

$$D_{\text{wif5}} = D_{\text{base}} + \sum_{r=0}^R B_r \quad (21)$$

where D_{base} is the transmission time (deterministic for a given MCS), R is the number of MAC retransmissions (geometric with parameter p , truncated at retry limit m), and B_r is the backoff duration for retry r :

$$B_r \sim \text{Uniform}(0, \min(CW_{\min} \cdot 2^r, CW_{\max})) \cdot T_{\text{slot}} \quad (22)$$

with slot time $T_{\text{slot}} = 9 \mu\text{s}$ for 802.11ac.

Wi-Fi 5 Per-Packet Delay Distribution (Bianchi Model)

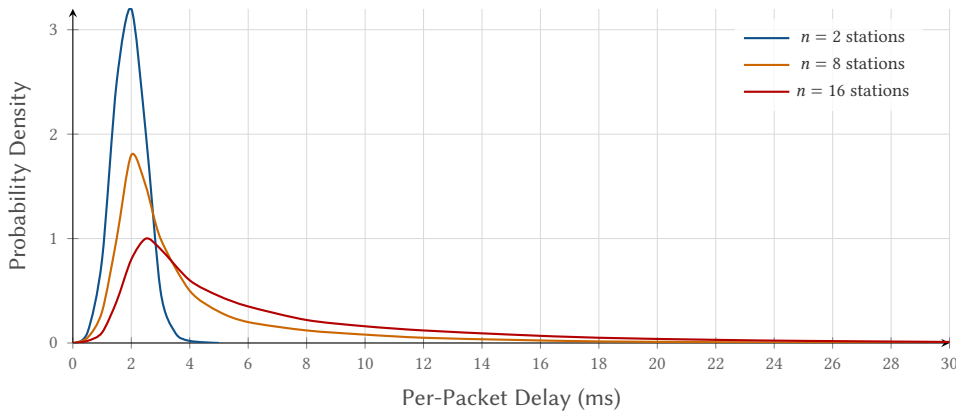


Figure 17. Per-packet delay PDF from the Bianchi-derived Wi-Fi 5 model for varying station counts. As n increases, the distribution develops a heavier tail from increased collision probability and more MAC retries. At $n = 16$, the P_{99} delay exceeds 25 ms per packet—enough to push I-frame delivery past the gaming latency budget.

8.1.3 Loss Model

Packets are dropped when retransmissions are exhausted ($R > m$):

$$P_{\text{drop}} = p^{m+1} \quad (23)$$

At $n = 8$ stations with default parameters ($CW_{\min} = 15$, $CW_{\max} = 1023$, $m = 6$), $P_{\text{drop}} \approx 10^{-5}$ —negligible for most scenarios. The dominant impairment is *jitter*, not *loss*.

8.2 Wi-Fi 6 (802.11ax): OFDMA Resource Scheduling

Wi-Fi 6 [14] fundamentally changes the model by introducing OFDMA. The AP divides the channel into R Resource Units and schedules clients into dedicated RUs, eliminating inter-station contention.

8.2.1 Service Time Model

When the gaming client has a dedicated RU, the service time is governed by the AP’s trigger-based scheduling cycle rather than random backoff:

$$D_{\text{wifi6}} = D_{\text{wait}} + D_{\text{slot}} + D_{\text{tx}} + J_{\text{BSR}} \quad (24)$$

where D_{wait} is the time until the next trigger frame (deterministic, at most $T_{\text{TXOP}} \leq 5.484$ ms), $D_{\text{slot}} = \lfloor n/R \rfloor \cdot T_{\text{sym}}$ accounts for multi-TXOP queuing when stations n exceed available RUs R ($T_{\text{sym}} = 13.6 \mu\text{s}$ per OFDMA symbol), D_{tx} is the transmission time for the allocated RU, and $J_{\text{BSR}} \sim \text{Uniform}(0, 50 \mu\text{s})$ captures Buffer Status Report timing uncertainty. The key difference from Wi-Fi 5: delay has *bounded support* with no heavy tail from exponential backoff—a packet’s worst-case wait is determined by the trigger cycle, not collision cascades.

8.2.2 Residual Jitter

Uplink scheduling in Wi-Fi 6 still uses a trigger frame mechanism whose periodicity introduces residual jitter bounded by T_{TXOP} . Combined with BSR timing uncertainty, the total scheduling jitter is:

$$J_{\text{residual}} \leq T_{\text{TXOP}} + J_{\text{BSR}} \quad (25)$$

This explains LM-TR-2026-01’s observation that Wi-Fi 6 produces “reduced jitter amplitude” (25–50 ms spikes) compared to Wi-Fi 5’s 110–140 ms.

8.2.3 Wi-Fi 6E: Greenfield 6 GHz Spectrum

Wi-Fi 6E operates the same 802.11ax PHY/MAC in the 6 GHz band, which is currently free of legacy device contention. In the simulator, Wi-Fi 6E is modelled as a parameter variant of Wi-Fi 6 with $n_{\text{stations}} = 1$ (no contention), $p_{\text{interference}} = 0$ (no legacy co-channel interference), and full channel width (160 MHz). As LM-TR-2026-01 notes, the resulting telemetry is “near-GPON stationarity” in uncongested deployments, distinguishable from wired only by the Wi-Fi scheduling quantisation. The separate treatment is warranted because 6E’s clean spectrum produces qualitatively different telemetry from 5 GHz Wi-Fi 6 in dense environments.

Delay Distribution Comparison: Wi-Fi 5 vs 6 vs 7

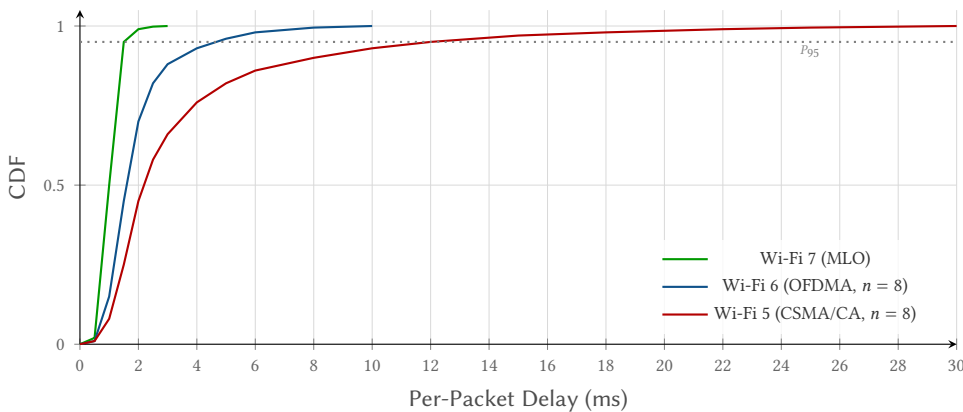


Figure 18. Cumulative distribution of per-packet delay across Wi-Fi generations. Wi-Fi 5’s heavy-tailed Bianchi distribution causes the P_{95} to exceed 15 ms; Wi-Fi 6’s bounded OFDMA scheduling keeps P_{95} under 6 ms; Wi-Fi 7’s MLO achieves near-deterministic $P_{95} < 2$ ms.

8.3 Wi-Fi 7 (802.11be): Parallel Queue with MLO Failover

Wi-Fi 7 [15] Multi-Link Operation bonds multiple frequency bands (2.4, 5, 6 GHz) simultaneously. The model is a parallel queue with instantaneous failover:

$$D_{\text{wifi7}} = \min_{l \in \text{links}} D_l(t) \quad (26)$$

where $D_l(t)$ is the delay on link l at time t . When interference disrupts one band, packets are served on an alternate band with zero MAC-layer timeout. LM-TR-2026-01 describes the resulting signature as “wired-like stationarity”—baseline 7–12 ms with zero loss.

The mathematical consequence is dramatic: if interference events on bands i and j are independent with probabilities p_i and p_j , the probability of *simultaneous* interference (the only case that causes jitter) is:

$$P_{\text{jitter}} = \prod_{l \in \text{links}} p_l \quad (27)$$

For typical interference rates ($p_i \approx 0.05$), three-band MLO yields $P_{\text{jitter}} \approx 1.25 \times 10^{-4}$ —effectively zero.

Congestion scenarios for Wi-Fi 6E and Wi-Fi 7. While uncongested Wi-Fi 6E and 7 approach wired-like performance, dense deployments can still degrade quality. The simulator includes congestion scenario variants for both technologies: Wi-Fi 6E with multiple overlapping 6 GHz BSSs ($n_{\text{BSS}} \in \{2, 4, 8\}$) modelling apartment buildings where adjacent-channel interference erodes the greenfield advantage, and Wi-Fi 7 MLO under partial band saturation where one or two of the three bands are heavily loaded, forcing traffic onto a single remaining link and negating the diversity gain of eq. (27). These scenarios quantify the conditions under which next-generation Wi-Fi reverts to legacy-like contention behaviour.

8.4 DOCSIS Cable: Finite FIFO Queue with AQM

LM-TR-2026-01’s “linear ramp” motif [16] emerges directly from queuing theory [17]. A classic DOCSIS [18] modem has a deep FIFO buffer of K packets.

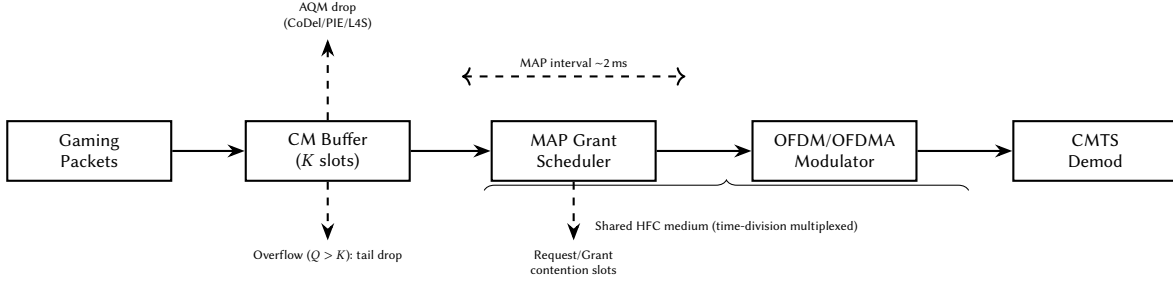


Figure 19. DOCSIS upstream signal path. Packets queue in the cable modem (CM) buffer awaiting MAP grant scheduling from the CMTS. The 2 ms MAP interval introduces a minimum scheduling latency. AQM policies (if enabled) drop packets proactively; without AQM, the buffer fills to K and tail-drops.

8.4.1 Queue Dynamics

The buffer occupancy $Q(t)$ evolves as:

$$Q(t + \Delta t) = \min(K, \max(0, Q(t) + A(t) - D(t))) \quad (28)$$

where $A(t)$ is the number of arrivals and $D(t)$ the number of departures in interval Δt . The sojourn time (queuing delay) for a packet entering at occupancy Q is:

$$W = \frac{Q \cdot \bar{S}}{C} \quad (29)$$

where $\bar{S}$ is the mean packet size and C is the link capacity. With $K = 1000$, $\bar{S} = 1400$ bytes, and $C = 5$ Mbit/s (upstream):

$$W_{\text{max}} = \frac{1000 \times 1400 \times 8}{5 \times 10^6} = 2240 \text{ ms} \quad (30)$$

This explains the catastrophic multi-second delays observed in bufferbloated cable modems.

8.4.2 The Linear Ramp Motif

When $\rho > 1$ (aggregate load exceeds capacity), the queue grows linearly:

$$Q(t) = Q_0 + (\lambda - \mu) \cdot t, \quad \text{for } Q(t) < K \quad (31)$$

producing the characteristic linear delay ramp:

$$W(t) = W_0 + \frac{(\lambda - \mu) \cdot t \cdot \bar{S}}{C} \quad (32)$$

until $Q(t) = K$, at which point tail-drop causes catastrophic packet loss. This is the bufferbloat signature.

Bufferbloat Queue Dynamics: Load $\rho = 1.2$

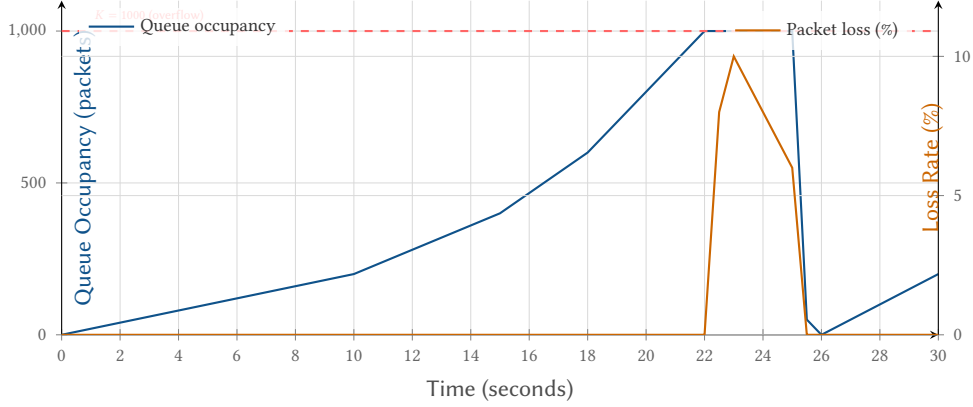


Figure 20. Bufferbloat dynamics at $\rho = 1.2$. Queue occupancy (blue) grows linearly until hitting the buffer limit $K = 1000$, at which point tail-drop causes catastrophic loss (orange). When TCP backs off, the queue drains and the cycle repeats (sawtooth).

8.4.3 AQM Models

The simulator implements three Active Queue Management policies to contrast against the unmanaged FIFO:

CoDel [19] (Controlled Delay) drops packets when the *minimum* sojourn time over an interval exceeds a target τ_{target} (typically 5 ms):

$$\text{drop if } \min_{t \in [t_0, t_0 + I]} W(t) > \tau_{\text{target}} \quad (33)$$

The drop interval I is shortened as drops persist: $I_k = I_0 / \sqrt{k}$, where k is the drop count.

PIE [20, 21] (Proportional Integral Enhanced) uses a control-theoretic approach, computing a drop probability p from the estimated queuing delay $\hat{W}$:

$$p_{t+1} = p_t + \alpha(\hat{W}_t - \tau_{\text{target}}) + \beta(\hat{W}_t - \hat{W}_{t-1}) \quad (34)$$

where α and β are the integral and proportional gains, respectively.

Dual-Queue (L4S) [22, 23] maintains two separate queues: a classic queue with PIE-based dropping, and a low-latency queue with IAQM (Immediate AQM) ECN marking. L4S-capable gaming traffic enters the low-latency queue, achieving ~ 0.5 ms queuing delay even under heavy classic traffic load. The coupling ensures fairness:

$$p_{\text{L4S}} = k \cdot p_{\text{classic}}^2 \quad (35)$$

where k is the coupling factor per RFC 9332 [24].

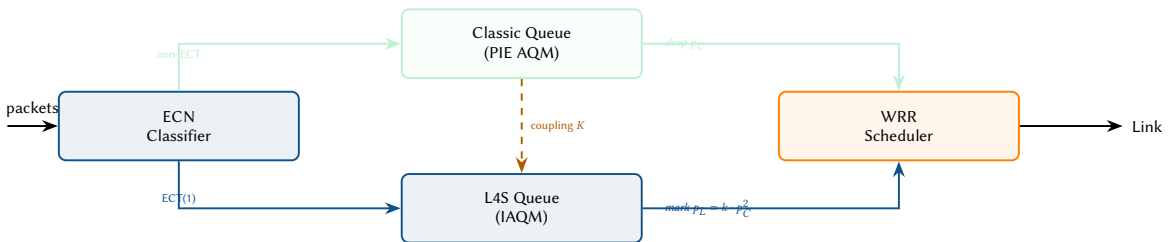


Figure 21. Dual-Queue Coupled AQM (L4S) per RFC 9332 [24]. The ECN classifier steers L4S-marked gaming traffic into the low-latency queue, while classic traffic enters the PIE-managed queue. The coupling factor K ensures bandwidth fairness by linking L4S marking probability to the classic drop probability.

8.5 GPON Fibre: Deterministic Propagation

GPON provides near-deterministic behaviour—the report calls it “perfect stationarity.” The model adds sub-millisecond Gaussian jitter to a fixed propagation delay, capturing DBA (Dynamic Bandwidth Allocation) scheduling noise:

$$D_{\text{GPON}} = D_{\text{prop}} + \epsilon, \quad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2) \quad (36)$$

with $D_{\text{prop}} \approx 1$ ms and $\sigma_\epsilon \approx 0.1$ ms. No queuing, no loss. GPON serves as the *reference baseline* for isolating the contribution of other link stages.

8.6 5G NR: Scheduled Queue with Markov Loss and Renewal Handovers

The 5G NR model [25–27] has two distinct components operating at different timescales.

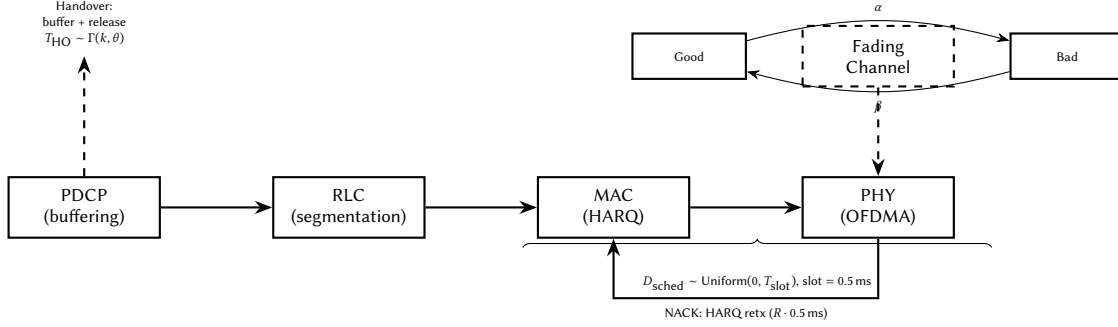


Figure 22. 5G NR protocol stack signal model. Packets traverse the PDCP–RLC–MAC–PHY pipeline. The fading channel drives a Gilbert–Markov loss process at the PHY layer; failed transmissions trigger HARQ retransmissions at the MAC layer. During handover, the PDCP layer buffers packets and releases them as a burst on link restoration.

8.6.1 Baseline: State-Dependent Delay with HARQ Retransmissions

At the sub-frame timescale, 5G NR uses scheduled OFDMA with HARQ (Hybrid ARQ) retransmissions. The per-packet delay depends on the current Gilbert–Elliott state $S \in \{\text{Good}, \text{Bad}\}$:

$$D_{5G} = D_{\text{base}}(S) + R \cdot D_{\text{HARQ}} \quad (37)$$

where $D_{\text{base}}(S)$ is drawn from a state-dependent Normal distribution:

$$D_{\text{base}}(S) \sim \begin{cases} \mathcal{N}(D_0, \sigma_g^2) & S = \text{Good} \\ \mathcal{N}(D_0 \cdot \mu_b, \sigma_b^2) & S = \text{Bad} \end{cases} \quad (38)$$

with $D_0 = 10$ ms, $\sigma_g = 2$ ms (Good state tight jitter), $\mu_b = 2.5$ (Bad state delay multiplier), $\sigma_b = 8$ ms (Bad state wide jitter), and R is the number of HARQ retransmissions ($D_{\text{HARQ}} = 8$ ms per retry, with retransmission probability $\approx 5\%$).

8.6.2 Two-State Markov Loss and Delay Model

Unlike netem’s independent Bernoulli loss, 5G radio losses are *correlated*—a fading event causes both burst loss and elevated latency. We model this with a two-state Gilbert–Elliott Markov chain [28, 29] that drives *both* loss probability and delay distribution:

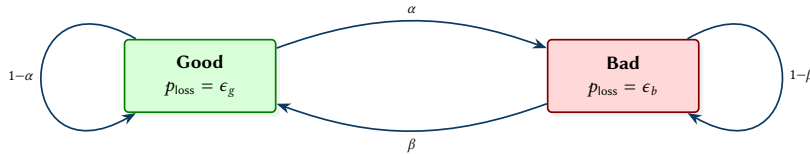


Figure 23. Gilbert–Elliott model for 5G fading. In the Good state, loss is negligible ($\epsilon_g \approx 10^{-4}$) and delay is $\mathcal{N}(D_{\text{base}}, \sigma_g^2)$; in the Bad state, loss is high ($\epsilon_b \approx 0.3$) and delay is $\mathcal{N}(D_{\text{base}} \cdot \mu_b, \sigma_b^2)$ with $\mu_b \approx 2.5$, $\sigma_b \gg \sigma_g$. Transitions α (Good→Bad) and β (Bad→Good) capture fading dynamics.

The state-dependent delay ensures that fading events produce *both* loss bursts and latency spikes—matching empirical measurements where signal degradation inflates HARQ round-trip times before packets are eventually dropped.

The steady-state probabilities are:

$$\pi_G = \frac{\beta}{\alpha + \beta}, \quad \pi_B = \frac{\alpha}{\alpha + \beta} \quad (39)$$

yielding an average loss rate of $\bar{p} = \pi_G \epsilon_g + \pi_B \epsilon_b$, but with *bursty* temporal correlation that netem cannot reproduce.

8.6.3 Handover Spike Model

LM-TR-2026-01 describes 5G handovers as quasi-periodic events causing 50–200 ms latency spikes. We model handovers as a renewal process with exponentially distributed inter-arrival times:

$$T_{\text{handover}} \sim \text{Exponential}(\lambda) \quad (40)$$

where $\lambda = 1/\mathbb{E}[T_{\text{HO}}]$ is parameterised by vehicle speed and cell density ($\mathbb{E}[T_{\text{HO}}]$ ranges from 360 s for pedestrian to 5 s for fast driving). The exponential distribution captures the memoryless property of independent cell boundary crossings. During a handover event of duration $\Delta_{\text{HO}} \sim \text{Uniform}(30, 80)$ ms:

- All packets are buffered at the PDCP layer (accumulating delay).
- On link restoration, buffered packets arrive as an instantaneous burst (fig. 24).
- A fraction p_{HO} of packets are lost (PDCP buffer overflow or excessive delay).

Measurement studies on commercial 5G networks [30, 31] report handover durations Δ_{HO} of 50–200 ms depending on mobility class. We parameterise both the inter-handover interval and the handover loss probability by UE speed class:

Table 9. 5G handover parameters by mobility class. T_{HO} inter-arrival is exponentially distributed with mean $\mathbb{E}[T_{\text{HO}}]$; Δ_{HO} is the link interruption duration (drawn from $\text{Uniform}(30, 80)$ ms); the delay spike during handover is $\text{Uniform}(50, 200)$ ms; p_{HO} is the PDCP buffer overflow loss probability.

Mobility class	Speed	$\mathbb{E}[T_{\text{HO}}]$ (s)	Δ_{spike} (ms)	p_{HO}
Stationary / indoor	0 km/h	—	—	0
Pedestrian	3 km/h	360	50–200	0.02
Urban vehicular	30 km/h	120	50–200	0.05
Suburban vehicular	60 km/h	75	50–200	0.08
Motorway	120 km/h	48	50–200	0.12

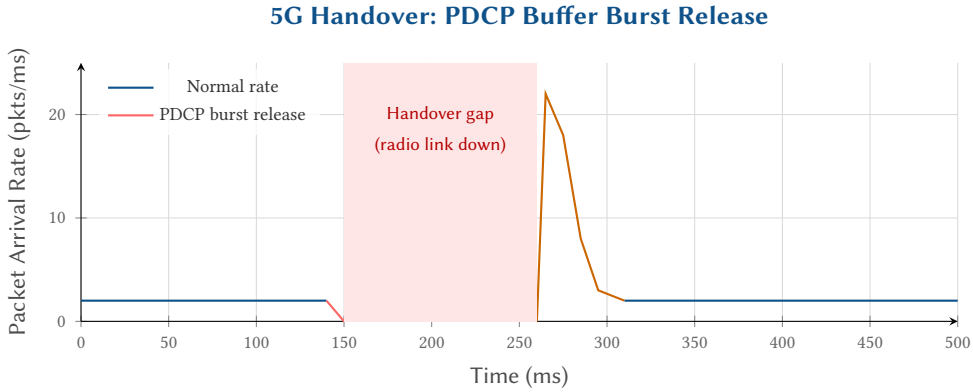


Figure 24. 5G handover dynamics. During the handover gap (~ 100 ms), packets accumulate in the PDCP buffer. On link restoration, they are released as a burst, causing a spike in both arrival rate and per-packet delay. To the gaming decoder, this manifests as a cluster of late frames followed by a burst of simultaneously-arriving frames.

8.7 LEO Satellite: Deterministic Propagation with Periodic Spikes

The LEO satellite model combines fixed propagation delay with quasi-periodic beam switching events [32–34].

8.7.1 Propagation Model

The one-way propagation delay is determined by orbital geometry:

$$D_{\text{prop}} = \frac{2h}{c} \cdot \sec(\theta_z) + D_{\text{ground}} \quad (41)$$

where $h \approx 550$ km is orbital altitude, c is the speed of light, θ_z is the zenith angle to the serving satellite, and D_{ground} is the ground station to PoP delay. Typical baseline: ~ 40 ms one-way.

As a satellite traverses its visible arc, the changing zenith angle produces a sinusoidal variation in round-trip time ranging from approximately 25 ms (overhead pass, $\theta_z \approx 0$) to 55 ms (low elevation, $\theta_z \approx 70^\circ$). The simulator models this orbital RTT sweep as a deterministic function of elapsed time within each pass. Additionally, a rain fade attenuation model (following ITU-R P.618 for Ka-band) stochastically reduces the effective SNR during precipitation events, increasing packet error rate and triggering adaptive modulation fallback with attendant throughput reduction.

8.7.2 Beam Switching Spike Model

LM-TR-2026-01 characterises LEO satellite spikes as quasi-periodic with intervals of 15–30 s. The critical mechanism is *out-of-order delivery*: the new satellite path is physically shorter, so packets routed via the new path arrive *before* packets still in transit on the old path. To a UDP-based gaming decoder, this reordering is functionally equivalent to loss.

$$T_{\text{beam}} \sim \text{Uniform}(T_{\text{min}}, T_{\text{max}}) \quad (42)$$

During a beam switch event:

1. Delay spikes by $\Delta_{\text{beam}} \sim \text{Uniform}(20, 100)$ ms above baseline.
2. Packets in transit on the old path arrive out of order.
3. Out-of-order packets arriving after their playout deadline count as functional loss.

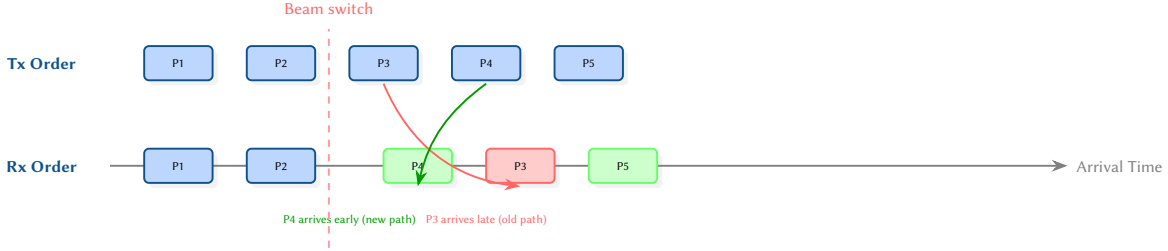


Figure 25. Out-of-order delivery during a LEO satellite beam switch. Packet P4 routed via the new (shorter) path arrives before P3 still in transit on the old (longer) path. A gaming decoder treating packets sequentially interprets P3 as lost.

8.8 Fixed Wireless Access: Weather-Dependent Service Rate

FWA links use microwave line-of-sight connections whose capacity depends on atmospheric conditions [35]. Rain fade reduces the signal-to-noise ratio, forcing the radio to fall back to a more robust but slower modulation scheme.

$$C_{\text{FWA}}(t) = C_{\text{max}} \cdot \eta_{\text{MCS}}(\text{SNR}(t)) \quad (43)$$

where η_{MCS} is the spectral efficiency of the selected MCS.

8.8.1 Rain Fade as a Stochastic Process

Rain attenuation on millimetre-wave links follows the ITU-R specific attenuation model [35, 36] and empirical rain fade statistics [37]. We model the fade process as a two-state semi-Markov chain: a *clear-sky* state with negligible attenuation and a *fade* state with log-normally distributed attenuation depth:

$$A_{\text{rain}}(t) \sim \begin{cases} 0 & \text{clear sky (probability } 1 - p_{\text{rain}}), \\ \text{LogNormal}(\mu_A, \sigma_A^2) & \text{fade (probability } p_{\text{rain}}). \end{cases} \quad (44)$$

Fade events arrive as a Poisson process with rate λ_{rain} (events/hour, dependent on climate zone) and have durations drawn from $\text{Exponential}(\mu_{\text{dur}})$. The SNR at the receiver is:

$$\text{SNR}(t) = \text{SNR}_{\text{clear}} - A_{\text{rain}}(t) \quad [\text{dB}] \quad (45)$$

The MCS selector maps SNR to spectral efficiency via a staircase function $\eta_{\text{MCS}}(\cdot)$, so capacity degrades in discrete steps as attenuation increases.

8.8.2 FWA Delay Model

The per-packet delay through the FWA link is:

$$D_{\text{FWA}} = D_{\text{prop}} + \frac{S}{C_{\text{FWA}}(t)} + D_{\text{sched}} \quad (46)$$

where D_{prop} is the line-of-sight propagation delay (typically 0.1–1 ms for 1–10 km range), S is the packet size, and $D_{\text{sched}} \sim \text{Uniform}(0, T_{\text{frame}})$ is the TDD frame scheduling delay. During a rain fade event, $C_{\text{FWA}}(t)$ drops by 2–5×, inflating per-packet airtime and causing sustained (not bursty) delay elevation. LM-TR-2026-01 calls this “long-wave delay rolling”—a sustained elevation in delay lasting seconds to minutes, fundamentally different from the sharp spikes of Wi-Fi or 5G. The simulator implements this as a two-state Markov process switching between clear-sky capacity C_{max} and fade capacity C_{fade} (discrete MCS step-down); the high persistence probability $p_{\text{stay}} (\sim 0.999)$ ensures fade events are correlated across many consecutive packets, producing the characteristic rolling motif.

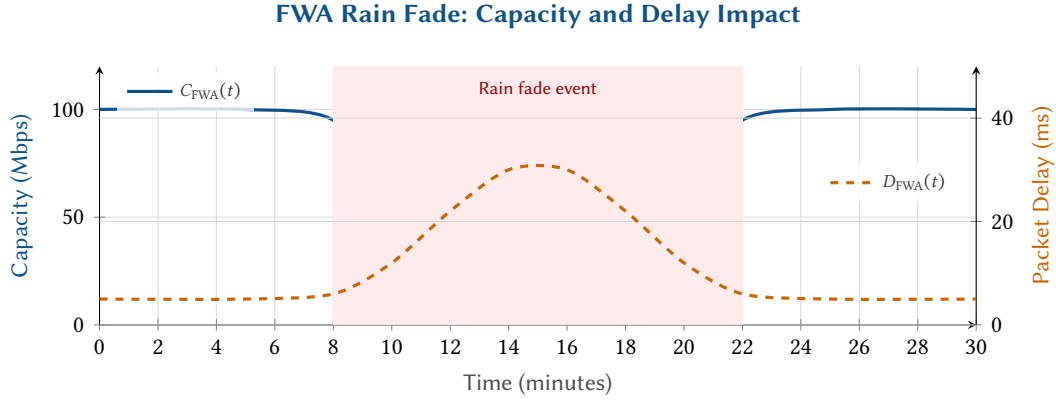


Figure 26. FWA link behaviour during a rain fade event. As attenuation increases, the radio falls back to lower MCS tiers, reducing capacity in discrete steps. Packet delay rises smoothly as airtime inflates, producing the “long-wave rolling” motif visible in telemetry.

8.9 DSL: Interleaving-Dominated Delay

Despite declining market share, DSL (ADSL2+ and VDSL2) remains a significant last-mile technology, particularly in rural areas and legacy deployments [38, 39]. The defining characteristic of DSL for latency-sensitive applications is the *interleaving delay* imposed by the Reed–Solomon forward error correction codec.

8.9.1 Interleaving Delay Model

DSL modems operate in one of two modes:

- **Interleaved (default):** The codec spreads codewords across D DMT symbols to protect against impulse noise. This adds a fixed, deterministic delay:

$$D_{\text{interleave}} = D \cdot T_{\text{DMT}} \quad (47)$$

where $T_{\text{DMT}} = 250 \mu\text{s}$ for ADSL2+ and VDSL2. Typical interleaving depths $D \in \{2, 4, 8, 16, 32, 64\}$ yield delays of 0.5–16 ms per direction.

- **Fast path (“fastpath”):** Interleaving is disabled ($D = 0$), reducing latency to sub-millisecond but exposing the link to impulse noise errors. Some ISPs disable fastpath by default due to error rate concerns [40].

8.9.2 DSL Queuing Model

The DSL access multiplexer (DSLAM) and CPE modem each maintain a FIFO queue. We model the upstream direction (most constrained) as an M/G/1 queue with deterministic service time:

$$D_{\text{DSL}} = D_{\text{interleave}} + D_{\text{prop}} + W_{\text{M/G/1}} \left(\frac{\rho_{\text{total}}}{C_{\text{up}}} \right) \quad (48)$$

where D_{prop} is the copper loop propagation delay (typically 0.1–0.5 ms for 1–5 km loops), ρ_{total} is the aggregate offered load, and C_{up} is the upstream sync rate (0.4–40 Mbps depending on loop length and DSL variant). The waiting time $W_{\text{M/G/1}}$ follows the Pollaczek–Khinchine formula [17].

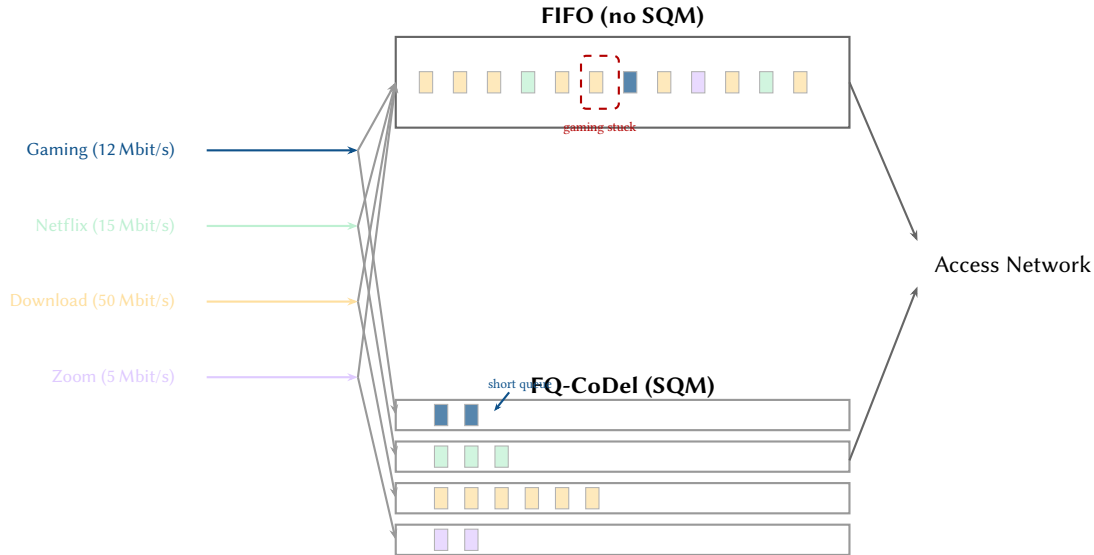


Figure 27. Home router queuing models. Without SQM (top), gaming packets queue behind bulk traffic in a shared FIFO. With FQ-CoDel (bottom), per-flow fair queuing isolates gaming traffic, reducing queuing delay from hundreds of milliseconds to ≈ 5 ms.

DSL’s telemetry fingerprint is distinctive: a near-constant baseline delay (dominated by the interleaving floor) with occasional impulse noise events causing isolated packet losses, and severe bufferbloat under load due to the asymmetric upstream capacity.

The simulator includes dedicated scenario configurations for both ADSL2+ (24/1 Mbps sync, interleaving depth $D \in \{8, 16, 32\}$) and VDSL2 (up to 100/40 Mbps, $D \in \{2, 4, 8\}$). The ADSL2+ scenarios emphasise the severe upstream bottleneck and high interleaving floor that dominate the latency budget, while VDSL2 scenarios expose the transition to near-fibre performance on short loops with fastpath enabled.

8.10 Home Router: The Cross-Traffic Bottleneck

The home router is modelled as a separate queuing stage sitting between the Wi-Fi link and the access network. Its behaviour depends critically on whether Smart Queue Management (SQM) is enabled:

- **No SQM (FIFO):** The router queue is a simple FIFO with depth K_{router} . Cross-traffic fills the queue, and gaming packets wait behind bulk downloads. This is the dominant source of in-home bufferbloat.
- **FQ-CoDel [41]:** Fair Queuing with CoDel. Each flow (identified by 5-tuple) gets its own sub-queue, and CoDel manages each sub-queue independently. Gaming packets are isolated from bulk downloads, achieving ~ 5 ms queuing even under heavy load.

The interaction between cross-traffic load ρ_{cross} and queue management policy is the single most impactful variable in many home network scenarios—more significant than the access technology itself (fig. 27).

9 Impairment Detection and Network Fingerprinting

The primary purpose of the simulator is not just to model impairments but to generate labelled datasets for training and validating *detection algorithms* that can identify the type and source of network degradation from observed telemetry. This section describes the detection methods that operate on the simulator’s output.

9.1 The Detection Problem

Given a time series of per-frame delay measurements $\mathbf{d} = (d_1, d_2, \dots, d_T)$ and loss indicators $\ell = (\ell_1, \ell_2, \dots, \ell_T)$, the detection system must answer:

1. **What type of impairment is occurring?** (Wi-Fi contention, bufferbloat, handover, beam switch, rain fade, upstream congestion)
2. **Where in the network is the impairment?** (Layer 1 in-home, Layer 2 last-mile, Layer 3 upstream)
3. **Is the impairment shared with other users?** (Individual vs. prefix-correlated)

Each access technology produces a distinct *telemetry fingerprint* (or motif)—a structural, repeating pattern in the delay and loss time series. The five canonical motifs from LM-TR-2026-01 are shown in fig. 28:

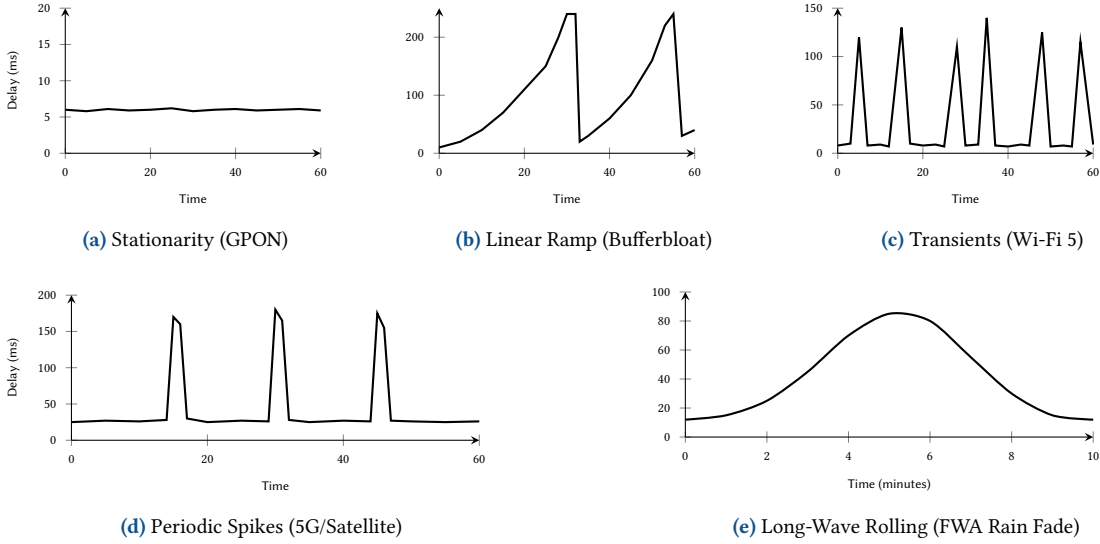


Figure 28. The five canonical telemetry motifs. Each access technology produces a distinctive temporal signature in the delay time series. These motifs are the “fingerprints” that detection algorithms identify. The simulator generates labelled examples of each for training and validation.

9.2 Matrix Profile for Motif Discovery

The Matrix Profile [42] is an exact, scalable algorithm for discovering repeated subsequences (motifs) and anomalies (discords) in time series. For a delay time series $\mathbf{d}$ of length T and a subsequence length w , the Matrix Profile $\mathbf{P}$ stores the z -normalised Euclidean distance to the nearest neighbour for every subsequence:

$$P_i = \min_{j \neq i} \left\| \hat{\mathbf{d}}_{i:i+w} - \hat{\mathbf{d}}_{j:j+w} \right\|_2 \quad (49)$$

where $\hat{\mathbf{d}}_{i:i+w}$ is the z -normalised subsequence starting at position i . The Matrix Profile identifies:

- **Motifs** (low P_i): Repeated patterns indicating periodic impairments—e.g., 5G handover spikes at regular intervals, DOCSIS sawtooth bufferbloat cycles.
- **Discords** (high P_i): Anomalous subsequences indicating novel or transient events—e.g., a rain fade event on an otherwise stable FWA link.

The STUMPY library [43] provides an efficient implementation with $O(T^2)$ time complexity and $O(T)$ space, with incremental updates for streaming data.

9.3 Time-Series Classification with ROCKET

ROCKET (RandOm Convolutional Kernel Transform) [44] and its successor MiniRocket [45] provide a fast, accurate approach to time-series classification by applying many random convolutional kernels to the input series and extracting two features per kernel (max value and proportion of positive values):

$$\mathbf{x} = \text{ROCKET}(\mathbf{d}) = \bigoplus_{j=1}^J \left[\max(\mathbf{d} * \kappa_j), \text{ppv}(\mathbf{d} * \kappa_j) \right] \quad (50)$$

where κ_j are random kernels with varying length, dilation, and bias, $*$ denotes convolution, and $\bigoplus$ is concatenation. The resulting $2J$ -dimensional feature vector is classified by a simple linear model (ridge regression).

For cloud gaming inference, ROCKET’s key advantages are:

- **Speed:** MiniRocket transforms a 3600-sample time series (60 seconds at 60 fps) in < 1 ms—well within the per-frame budget.
- **Accuracy:** State-of-the-art on the UCR time series archive with $J = 10,000$ kernels.
- **Simplicity:** No neural network training required—the kernels are random, only the linear classifier is fitted.

9.4 Cohort-Based Impairment Localisation

As detailed in section 12, correlating telemetry across users on the same ISP prefix distinguishes individual (Layer 1) impairments from shared (Layer 3) impairments. The detection pipeline combines this with the per-user fingerprint to produce a two-dimensional classification:

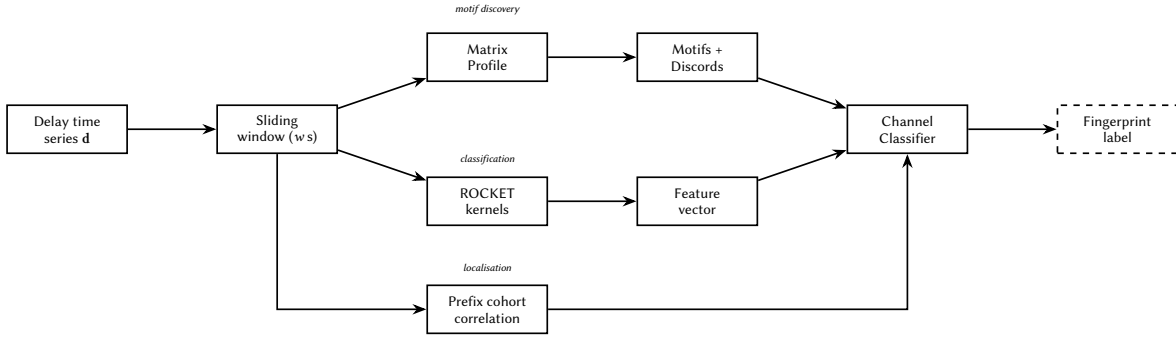


Figure 29. Detection pipeline. Raw delay telemetry is windowed and processed through three parallel paths: Matrix Profile for motif discovery, ROCKET for feature extraction and classification, and prefix cohort correlation for impairment localisation (Layer 1 vs. 2 vs. 3). The classifier outputs a fingerprint label identifying the impairment type and network layer.

Table 10. Impairment classification matrix from fingerprint and cohort correlation.

Fingerprint	Individual ($r < 0.3$)	Correlated ($r > 0.7$)
Transient spikes	Wi-Fi contention (L1)	Cell tower overload (L2)
Linear ramp	Home router bloat (L1)	DOCSIS node bloat (L2)
Periodic spikes	Device issue (L1)	5G/satellite shared (L2)
Sustained elevation	FWA rain fade (L2)	ISP upstream peak (L3)
Flat + high jitter	Bad Wi-Fi link (L1)	Peering congestion (L3)

9.5 Simulator as Training Data Generator

A key application of the Monte Carlo simulator is generating labelled training datasets for these detection algorithms. Each trial produces:

- A complete delay/loss time series with *known ground truth* labels (which link model generated each impairment event).
- Controllable parameter variation ensuring the training set covers the full range of each distribution.
- Correlated multi-user datasets for training the cohort correlation classifier.

This addresses a fundamental challenge in network measurement: obtaining labelled ground truth for supervised learning. Real-world traces provide unlabelled data; the simulator provides the labels.

10 Monte Carlo Methodology

10.1 Trial Structure

Each trial simulates a complete gaming session:

1. **Initialise:** Seed RNG using Common Random Numbers (CRN): `ChaCha8Rng::seed_from_u64(base_seed).set_stream(trial_id)` providing proper stream separation [1]. Instantiate link models with sampled parameters.
2. **Generate:** Create $f \times T$ frames (e.g., $60 \times 60 = 3600$ frames for a 60-second session), fragment into packets, add FEC.
3. **Simulate:** Pass each packet through the link pipeline via the discrete event scheduler. Each link stage applies its delay and loss model.
4. **Receive:** Reassemble frames from arrived packets. Classify each frame as delivered, FEC-recovered, or dropped (deadline exceeded).
5. **Measure:** Compute per-trial metrics (latency percentiles, loss rate, jitter, QoE classification).

10.2 Parameter Distributions

Rather than fixed parameters, the Monte Carlo driver samples from distributions to model the variability of real-world conditions:

10.3 Output Aggregation

Across N trials, the aggregator computes (see fig. 30 for a representative CDF):

- **Percentile distributions:** P_{50} , P_{95} , P_{99} , $P_{99.9}$ of frame latency, with BCa bootstrap confidence intervals [46].
- **QoE classification fractions:** Proportion of trials classified as Good / Acceptable / Poor / Unplayable.

Table 11. Example parameter distributions for a bufferbloat scenario.

Parameter	Distribution	Typical Values
Wi-Fi stations	Poisson($\lambda = 6$)	2–12
Cross-traffic load	Beta($\alpha = 2, \beta = 3$)	0.3–0.9
DOCSIS upstream (Mbps)	Uniform(3, 10)	3–10
Router buffer depth	LogNormal(μ, σ)	100–5000 pkts
FEC overhead (%)	Fixed per sweep axis	0, 2, 5, 10

- **Sensitivity analysis:** Partial rank correlation coefficients (PRCC) [47] identifying which input parameters most strongly influence each output metric.

Monte Carlo Output: Frame Latency CDF by AQM Policy

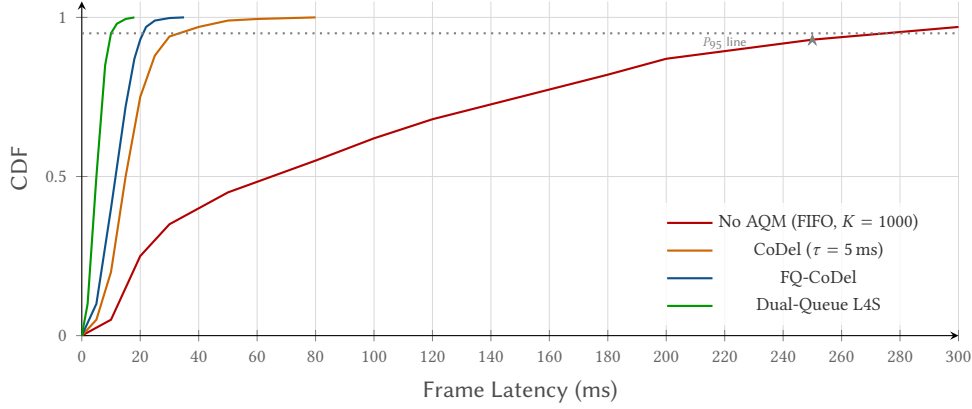


Figure 30. Aggregated Monte Carlo output for the DOCSIS bufferbloat scenario ($N = 50,000$ trials). The CDFs reveal that without AQM, the P_{95} frame latency exceeds 200 ms (unplayable). FQ-CoDel brings P_{95} under 22 ms; Dual-Queue L4S achieves $P_{95} < 10$ ms.

10.4 Warm-Up and Transient Removal

In discrete-event simulations where queues start empty, the initial period produces unrepresentative statistics. Each trial discards a *warm-up phase* of T_w simulated seconds before collecting metrics. The warm-up duration is chosen to ensure the slowest queue (typically the DOCSIS FIFO at high load) reaches approximate steady state:

$$T_w \geq \frac{K \cdot \bar{S}}{C - \lambda} = \frac{\text{buffer capacity}}{\text{drain rate excess}} \quad (51)$$

For a worst-case upstream scenario ($K = 1000$, $\bar{S} = 1400$ bytes, $C = 5$ Mbps upstream, $\rho = 0.9$), $T_w \approx 22$ seconds. In practice, most scenarios reach steady state much faster because the bottleneck queue is the downstream direction with higher capacity. We therefore set T_w per scenario: 5 seconds for GPON and low-load Wi-Fi, 10 seconds for DOCSIS and 5G, and 25 seconds for heavily loaded upstream queues. In all cases, we verify via Welch’s graphical method [1] that the time-averaged delay has stabilised before collection begins.

10.5 Convergence Testing

Beyond the confidence interval width (eq. (1)), we apply a *relative half-width* criterion to verify that N trials is sufficient:

$$\frac{\text{CI}_{95}}{|\bar{X}|} < \gamma, \quad \gamma = 0.05 \quad (52)$$

where $\bar{X}$ is the sample mean of the metric of interest (e.g., P_{95} frame latency). If the criterion is not met, the sweep automatically doubles N and continues. In practice, $N = 50,000$ is sufficient for all metrics except $P_{99.9}$, which may require $N = 200,000$ for narrow confidence intervals.

The simulator implements automated convergence checking: trials run in batches of 10, and after each batch the checker evaluates the relative half-width using t-distribution critical values (rather than normal approximation) for proper small-sample behaviour. The `--convergence-threshold` flag controls the target (default: 2% relative error).

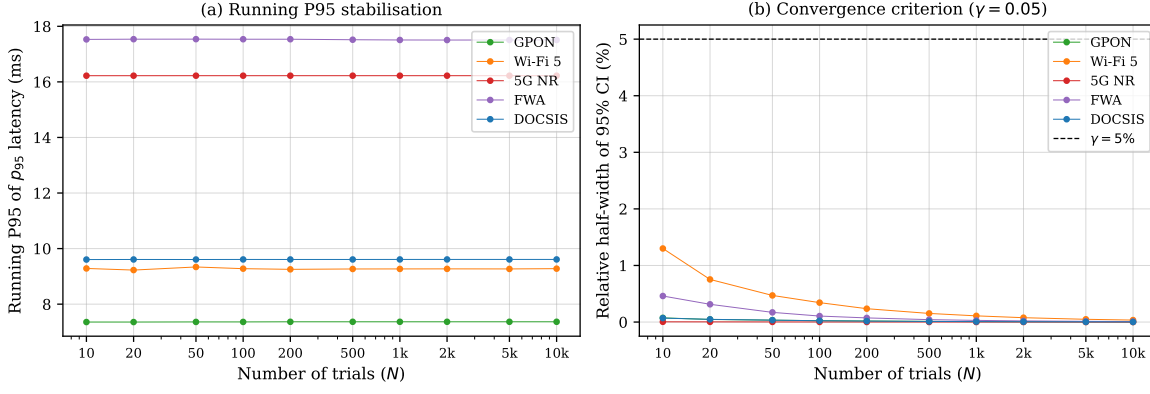


Figure 31. Monte Carlo convergence diagnostic. (a) Running P_{95} of frame latency as trial count N increases. (b) Relative half-width of the 95% CI; the dashed line marks the $\gamma = 0.05$ convergence criterion.

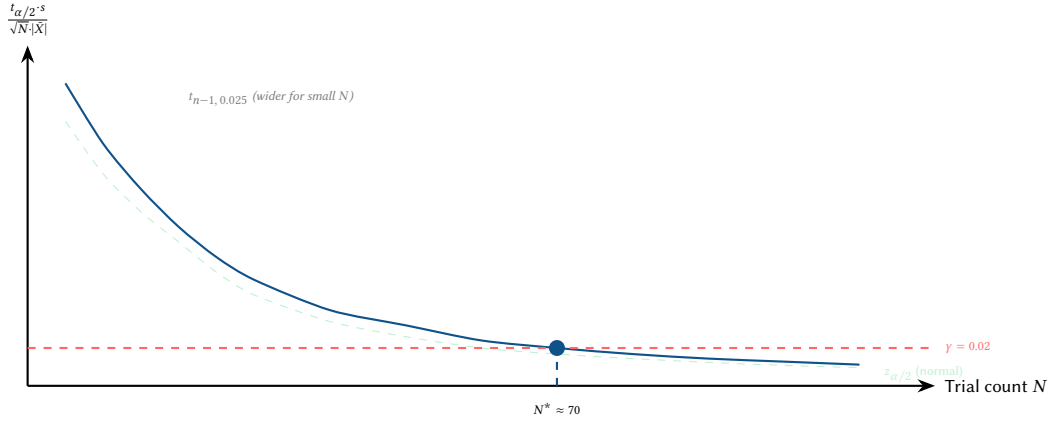


Figure 32. Convergence checker schematic. The relative half-width (using t-distribution critical values for correct small-sample coverage) decreases as $1/\sqrt{N}$. The checker stops at N^* where the curve crosses the threshold γ . The dashed curve shows the normal approximation, which underestimates uncertainty for $N < 30$.

10.6 Variance Reduction: Common Random Numbers

When comparing two scenarios (e.g., DOCSIS with vs. without AQM), independent random streams produce high variance in the difference estimator. Common Random Numbers (CRN) [1] ensure that trial i of both scenarios uses the *same* underlying random draws, so that the difference reflects only the parameter change.

The implementation uses ChaCha8’s native 64-bit stream counter: each trial gets a unique stream (not a hash of seed $\oplus$ trial_id), and per-component sub-streams (traffic, channel, queue, cross-traffic) ensure that modifying one component does not perturb others.

10.7 Variance Reduction: Antithetic Variates

For monotone functions of the input random variables, antithetic variates [1] reduce variance by pairing each trial with a *complemented* trial that uses negatively correlated draws. For each trial with stream s , a companion trial uses a distant stream offset ($s + 2^{48}$) to produce approximately antithetic draws. The two results are averaged:

$$\hat{\theta}_{AV} = \frac{1}{2}(\theta(U) + \theta(U')), \quad \text{Var}(\hat{\theta}_{AV}) \leq \frac{1}{2}\text{Var}(\hat{\theta}) \quad (53)$$

This is enabled via the `--antithetic` flag and halves the variance for frame latency (a monotone function of queuing delays) at the cost of doubling per-trial computation.

10.8 Bootstrap BCa Confidence Intervals

For non-normal output distributions (common in tail metrics like P_{99}), the simulator uses the bias-corrected and accelerated (BCa) bootstrap [46] rather than normal-theory confidence intervals:

1. Draw $B = 10,000$ bootstrap resamples with replacement from the N trial outcomes.
2. Compute the statistic of interest $\hat{\theta}_b^*$ for each resample.
3. Estimate bias correction $\hat{z}_0 = \Phi^{-1}(\#\{\hat{\theta}_b^* < \hat{\theta}\}/B)$.

4. Estimate acceleration $\hat{a}$ from the jackknife influence values.
5. Adjust the percentile endpoints: $\alpha_1 = \Phi\left(\hat{z}_0 + \frac{\hat{z}_0 + z_\alpha}{1 - \hat{a}(\hat{z}_0 + z_\alpha)}\right)$.

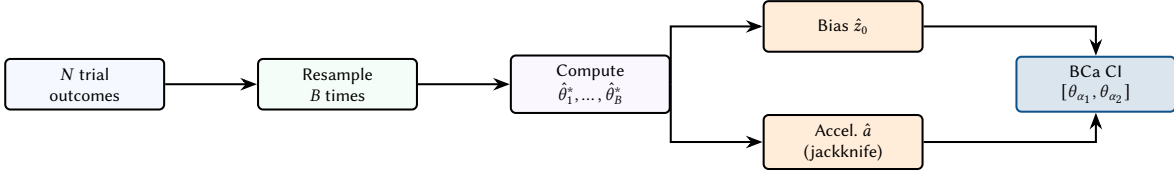


Figure 33. BCa bootstrap confidence interval pipeline [46]. Bias correction $\hat{z}_0$ adjusts for systematic estimator bias; acceleration $\hat{a}$ corrects for non-constant variance across the bootstrap distribution. The result is a CI that is accurate even for skewed, heavy-tailed latency distributions.

10.9 PRCC Sensitivity Analysis

Partial Rank Correlation Coefficients [47] measure the monotonic association between each input parameter and the output metric, *controlling* for all other parameters. The implementation uses OLS residual regression:

1. Rank-transform all input variables X_j and the output Y .
2. For parameter j : regress $R(X_j)$ on all *other* ranked inputs $R(X_{-j})$ to obtain residuals e_{X_j} .
3. Regress $R(Y)$ on $R(X_{-j})$ to obtain residuals e_Y .
4. $\text{PRCC}_j = \text{Corr}(e_{X_j}, e_Y)$.

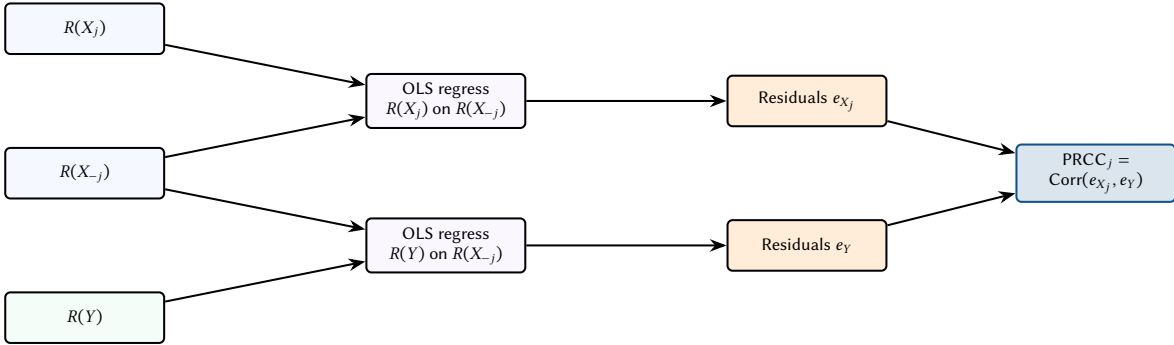


Figure 34. PRCC methodology via OLS residual regression [47]. By regressing out the influence of all other parameters, the residual correlation isolates the unique contribution of parameter X_j to the output Y . Statistical significance is assessed via a t-test on the residual correlation coefficient.

10.10 Receiver Jitter Buffer

Real gaming decoders include a *playout buffer* that trades additional latency for smoother frame delivery. The simulator models this as an optional receiver-side delay:

$$t_{\text{playout},n} = \max(t_n^{\text{gen}} + L_{\text{buffer}}, t_{\text{last},n}) \quad (54)$$

where L_{buffer} is the jitter buffer depth (typically 10–30 ms). A deeper buffer absorbs more jitter at the cost of increased motion-to-photon latency. The frame is delivered at $t_{\text{playout},n}$ rather than at $t_{\text{last},n}$, which shifts the deadline check:

$$d_n^{\text{buffered}} = d_n + L_{\text{buffer}} \quad (55)$$

The jitter buffer is a key trade-off parameter: at $L_{\text{buffer}} = 0$ (no buffer), the system is maximally latency-sensitive; at $L_{\text{buffer}} = 30$ ms, many deadline misses on jittery links (Wi-Fi 5, 5G handovers) are absorbed.

10.11 Packet Reordering Metric

For link models that produce out-of-order delivery (LEO satellite beam switching, 5G handovers), the simulator computes the reorder density metric per RFC 4737 [48]:

$$\text{RD} = \frac{|\{i : s(i) \neq i\}|}{N} \quad (56)$$

where $s(i)$ is the arrival position of the packet with send sequence number i , and N is the total packet count. Reordered packets that arrive after their playout deadline are functionally equivalent to loss; those arriving

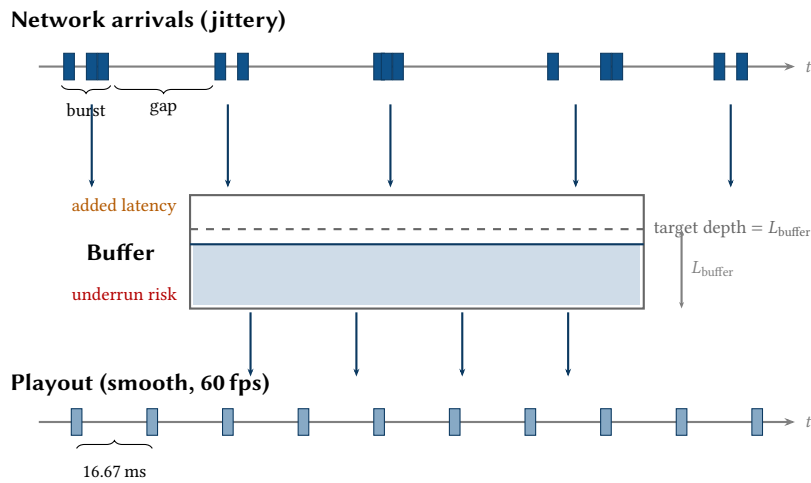


Figure 35. The receiver jitter buffer absorbs arrival time variance at the cost of additional latency L_{buffer} . Irregular network arrivals (top) are smoothed to steady 60 fps playback (bottom). Deeper buffers reduce underrun risk but increase motion-to-photon latency.

before the deadline but out of order require receiver-side resequencing, which the jitter buffer can accommodate if sufficiently deep.

10.12 Sensitivity Analysis: Tornado Diagrams

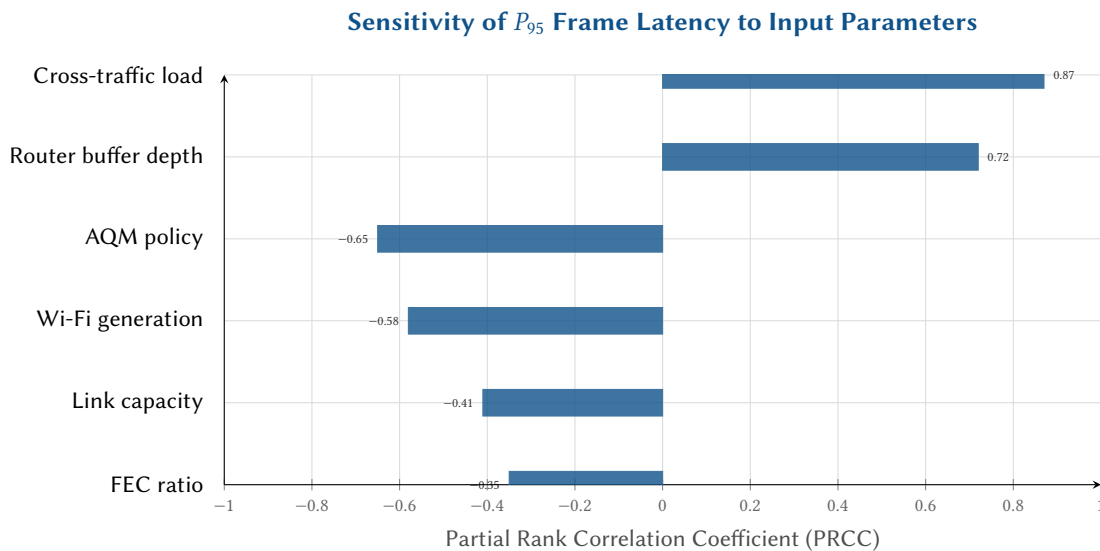


Figure 36. Sensitivity analysis via PRCC. Cross-traffic load and router buffer depth are the strongest positive drivers of P_{95} latency. AQM policy and Wi-Fi generation are the strongest negative drivers (reducing latency). This quantifies the intuition from LM-TR-2026-01 that the home router is often the dominant impairment source.

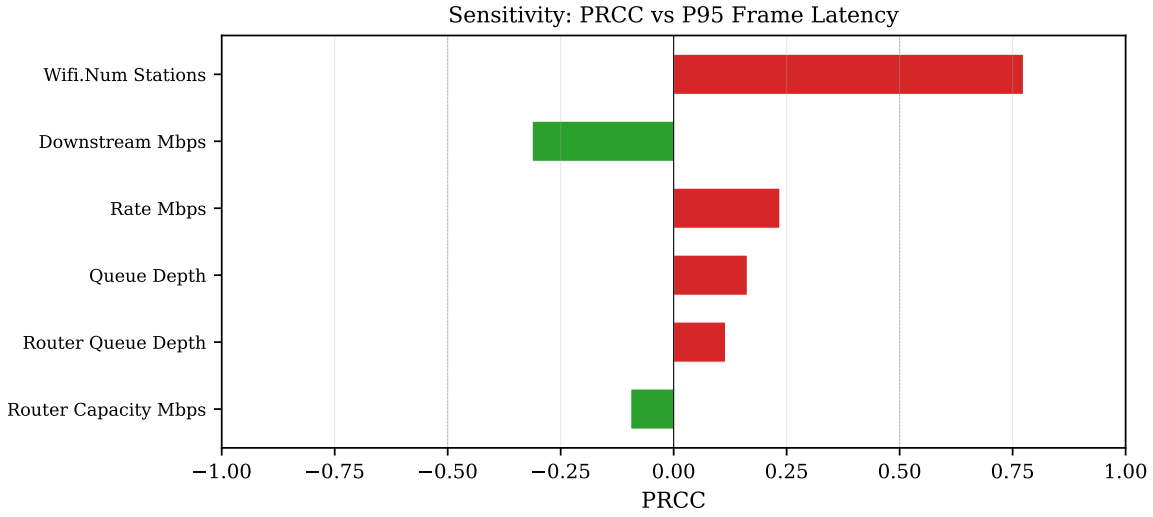


Figure 37. Partial Rank Correlation Coefficients (PRCC) for P_{95} frame latency. Green bars indicate parameters that reduce latency; red bars indicate parameters that increase it.

11 Composable Topology Examples

The power of the composable link stack is the ability to model realistic home network topologies as a sequence of independent queuing stages.

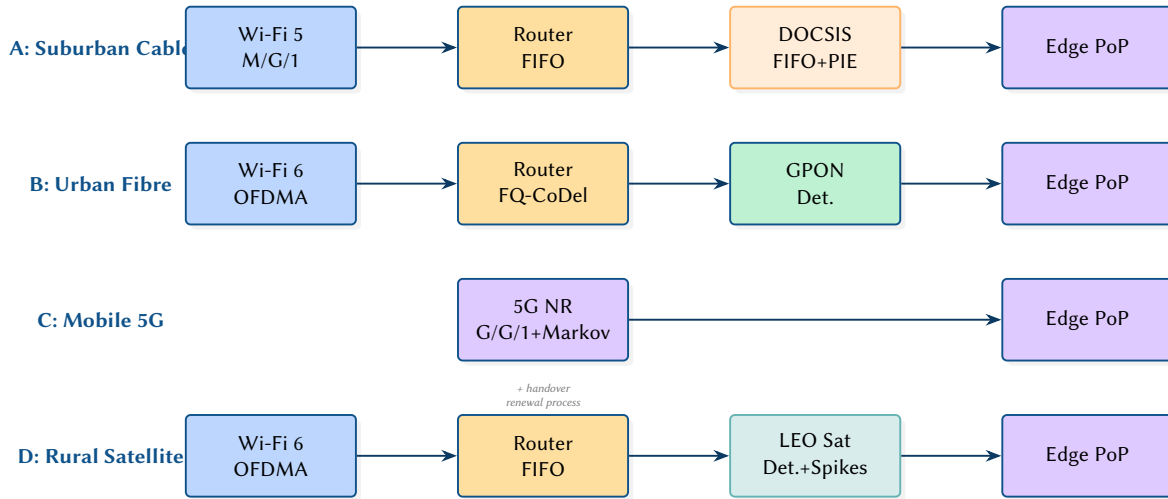


Figure 38. Four example topologies as composable link stacks. Each stage independently applies its queuing model. Scenario A represents the worst case (Wi-Fi 5 contention + DOCSIS bufferbloat); Scenario B the best (Wi-Fi 6 + FQ-CoDel + GPON).

11.1 Extended Scenarios

Beyond the four basic topologies, the three-layer model enables several important compound scenarios:

11.2 Expanded Link Model Taxonomy

The eleven base models from section 8 represent the core access technologies. In practice, significant variation exists *within* each technology. The simulator supports these via configuration variants:

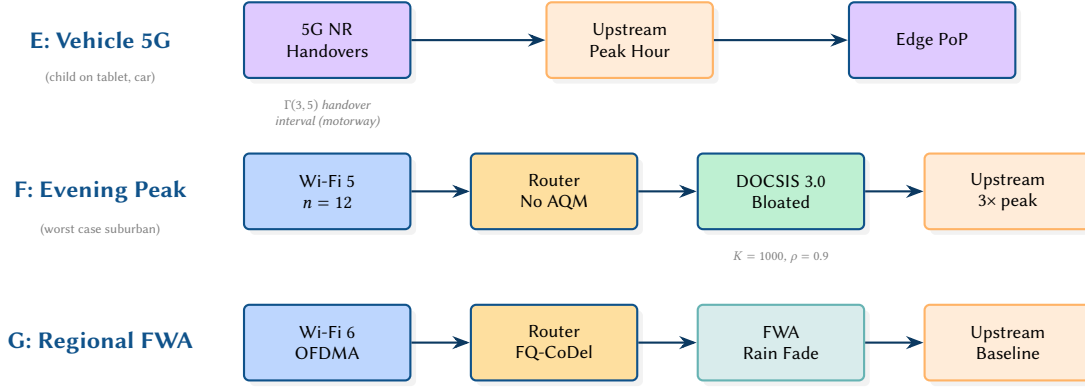


Figure 39. Extended scenarios combining all three layers. Scenario E models a child gaming on a tablet in a moving vehicle—5G handovers compound with upstream peak-hour congestion. Scenario F represents the worst-case suburban evening: Wi-Fi 5 contention + unmanaged DOCSIS bufferbloat + ISP evening peak. Scenario G captures regional fixed wireless with weather-dependent impairment.

Table 12. Expanded link model variants beyond the eleven base models.

Category	Variant	Key Difference
DOCSIS	3.0	Deep FIFO, no AQM, 100/5 Mbps typical
DOCSIS	3.1	OFDMA, optional PIE, 1000/35 Mbps
DOCSIS	Low Latency	Dual-Queue L4S, < 1 ms queuing
DSL	ADSL2+	24/1 Mbps, high interleaving delay
DSL	VDSL2	100/40 Mbps, lower interleaving
5G NR	Stationary	Baseline scheduling only
5G NR	Mobile	+ Gamma-distributed handovers
In-Home	Ethernet	Near-ideal (< 1 ms, zero loss)
In-Home	Wi-Fi 4	2.4 GHz only, severe contention
Satellite	Starlink	40 ms baseline, beam switching
Satellite	GEO	500 ms baseline, unusable for gaming

12 Cross-Customer Correlation Analysis

A key application of the simulator is generating synthetic datasets that mirror the cross-customer correlation patterns described in LM-TR-2026-01. When multiple users on the same ISP prefix experience correlated degradation, the impairment is *upstream* (shared infrastructure), not in-home.

12.1 Prefix Cohort Simulation

The simulator can run M parallel “user” instances sharing the same Layer 3 upstream model but with independent Layer 1 (in-home) and Layer 2 (last-mile) parameters:

$$D_{\text{user}_i}(t) = D_{\text{in-home}_i}(t) + D_{\text{last-mile}_i}(t) + D_{\text{upstream}}(t) \quad (57)$$

The upstream term $D_{\text{upstream}}(t)$ is *shared* across all users in the cohort. This produces telemetry streams that are:

- **Correlated in the upstream component:** All users see the same evening peak delay spike simultaneously.
- **Independent in the in-home component:** Each user has different Wi-Fi contention, router configuration, and cross-traffic.

12.2 Impairment Classification from Correlation

By correlating telemetry across the cohort, the simulator can generate labelled training data for the ML-based inference system described in LM-TR-2026-01:

1. **Individual impairment** (single user degrades): Only $D_{\text{in-home}_i}$ changes → Wi-Fi contention, local congestion, device issue.
2. **Prefix-correlated impairment** (multiple users degrade simultaneously): D_{upstream} changes → shared ISP backbone or CMTS overload.

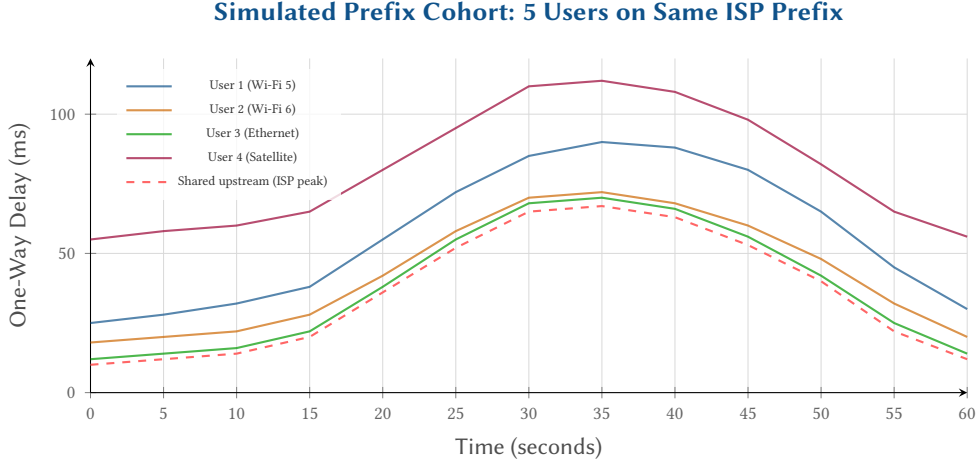


Figure 40. Simulated prefix cohort. Five users on the same ISP prefix share the upstream congestion profile (dashed red). Their individual RTT traces are offset vertically by their independent in-home and last-mile contributions. The *correlated* component (the evening peak shape) is visible across all users, while the *independent* component (vertical offset, local jitter) varies per user.

3. **Access-correlated impairment** (users on same access technology degrade): $D_{\text{last-mile}}$ changes for a subset $\rightarrow$ DSLAM saturation, cell tower congestion.

The Pearson correlation coefficient r between user pairs provides a simple detection metric:

$$r_{ij} = \frac{\text{Cov}(D_i, D_j)}{\sigma_i \sigma_j} \quad (58)$$

When $r_{ij} > \theta$ (typically $\theta \approx 0.7$) for users i, j on the same prefix, the system infers a shared upstream impairment.

13 Calibration and Validation

The simulator’s parameterised models trade protocol fidelity for speed. To ensure this trade-off is acceptable, we validate at four levels: quantitative comparison against packet-level simulators, statistical fitting to real-world measurement datasets, qualitative motif reproduction, and sensitivity analysis to confirm model robustness.

13.1 Level 1: Packet-Level Simulator Cross-Validation

We compare the simulator’s output distributions against established packet-level network simulators [49]:

- **ns-3 802.11ax EDCA:** Compare per-packet delay CDFs under matching station count and contention parameters.
- **CableLabs DOCSIS module:** Validate bufferbloat ramp dynamics and AQM drain behaviour under identical queue depth and offered load.
- **5G-LENA:** Cross-check HARQ retransmission delay and Gilbert model burst length distributions.

Target: output distributions within 10% on P_{50} , P_{95} , and P_{99} across 1000 matched-seed trials. Kolmogorov–Smirnov two-sample tests provide a formal statistical acceptance criterion ($p > 0.05$ for the null hypothesis that both simulators sample from the same distribution).

13.2 Level 2: Real-World Measurement Dataset Validation

The simulator parameters are calibrated against, and validated using, publicly available broadband measurement datasets:

FCC Measuring Broadband America (MBA) [50]. The FCC’s annual report provides ISP-level distributions of download/upload speed, latency, and packet loss across $\sim 10\,000$ US households with SamKnows whitebox probes [8]. We use MBA data to calibrate:

- ISP quality profile parameters (table 3): peak delay multipliers and loss rates per ISP tier.
- DOCSIS vs. FTTH baseline delay distributions.
- Time-of-day congestion amplitudes and phase (peak hour onset/offset).

SamKnows / Ofcom UK Broadband Performance [51]. Ofcom’s panel study provides equivalent data for the UK market, covering DSL/VDSL, Virgin Media (DOCSIS), and full-fibre. Particularly useful for calibrating the DSL interleaving delay and VDSL crosstalk parameters in section 8.9.

RIPE Atlas [52]. The RIPE Atlas mesh of ~12 000 probes provides continuous RTT and traceroute measurements across diverse European and global access networks. We use Atlas data to validate:

- Upstream ISP peering delay and jitter distributions.
- Diurnal congestion patterns across regions.
- Prefix cohort correlation (users sharing an upstream AS exhibiting correlated delay changes).

Ookla Open Data [53]. Ookla’s quarterly open dataset provides aggregated speed/latency statistics at the tile level (zoom-16), enabling regional calibration of the technology prevalence priors in table 4.

Cloud gaming traces. Where available, we fit model parameters directly from Mahimahi-format [54] trace files of cloud gaming sessions (e.g., LORIA cloud gaming dataset, GamingAnywhere traces). The fitting procedure:

1. Extract per-packet OWD and loss sequences from the trace.
2. Identify the access technology from metadata or motif classification.
3. Fit the corresponding model parameters via maximum likelihood estimation (MLE).
4. Generate 10^4 synthetic trials from the fitted model and compare CDFs against the empirical trace using a Kolmogorov–Smirnov test.

13.3 Level 3: Qualitative Motif Validation

We verify that the simulator reproduces the five canonical motifs from LM-TR-2026-01 by visual inspection of output time series and automated motif classification:

- **Stationarity** for GPON (flat delay, zero loss).
- **Linear ramp** for bufferbloated DOCSIS.
- **High-frequency transients** for Wi-Fi 5 contention.
- **Quasi-periodic spikes** for 5G handovers and satellite beam switching.
- **Long-wave rolling** for FWA rain fade.

The motif validation is both a sanity check (does the simulator produce the right qualitative shape?) and a round-trip test: we train the ROCKET classifier (section 9.3) on simulator output, then test it on held-out simulator data *and* on real traces. If the classifier trained on synthetic data can correctly identify motifs in real traces, this provides strong evidence that the simulator’s distributional assumptions are realistic.

13.4 Level 4: Sensitivity Analysis

To identify which model parameters most influence QoE outcomes, we perform a global sensitivity analysis using Partial Rank Correlation Coefficients (PRCC):

1. Sample each input parameter from its plausible range using Latin Hypercube Sampling (LHS) across $N_{\text{LHS}} = 500$ parameter combinations.
2. For each combination, run 10^3 Monte Carlo trials to obtain stable output metrics (P_{95} delay, frame drop rate, QoE score).
3. Compute the PRCC between each input parameter and each output metric, ranking parameters by influence.
4. Parameters with $|\text{PRCC}| > 0.5$ are flagged as *high-sensitivity* and warrant precise calibration; parameters with $|\text{PRCC}| < 0.1$ can be fixed at nominal values without significant accuracy loss.

Preliminary expectations based on the model structure:

- **High sensitivity:** Wi-Fi station count n_{stations} , router AQM policy, cross-traffic load ρ_{cross} , ISP over-subscription ratio.
- **Medium sensitivity:** FEC overhead, 5G handover interval, FWA rain fade probability.
- **Low sensitivity:** GPON jitter standard deviation, satellite baseline propagation delay, DSL propagation delay.

14 Simulation Results and Output Analysis

The simulation engine generates large-scale synthetic telemetry datasets across the five canonical delay motifs. This data serves as the foundation for automated motif inference and ML model training.

Table 13 provides a comprehensive summary of all 30 scenarios, showing the mean P50, P95, and P99 latencies, frame loss rates, jitter, goodput, and QoE classification across 100 Monte Carlo trials. The rows are grouped

by family and use standardized scenario names so the taxonomy matches the prose and the comparison charts.

Table 13. Monte Carlo scenario summary (100 trials per scenario, means across trials). QoE labels use the common four-band rubric defined in the executive summary.

Family	Scenario	P50 (ms)	P95 (ms)	P99 (ms)	Loss (%)	Jitter (ms)	Goodput (Mbps)	QoE
Compound	Wi-Fi 5 + DOCSIS 3.0	9.60	10.50	12.00	1.82	1.25	6.60	Acceptable
Compound	Wi-Fi 5 + Satellite	52.10	81.20	97.70	1.90	38.32	6.60	Poor
Compound	Wi-Fi 6 + DOCSIS 3.0 Bufferbloat	30.30	32.90	34.20	1.43	5.70	6.60	Acceptable
DOCSIS 3.0	Bufferbloat Light	9.80	10.20	11.60	1.49	0.70	86.50	Acceptable
DOCSIS 3.0	Bufferbloat Severe	9.80	10.30	11.50	1.49	0.71	74.50	Acceptable
DOCSIS 3.0	Idle	9.00	9.20	10.30	1.26	0.40	6.60	Acceptable
DOCSIS 3.1	AQM Load	9.10	9.40	9.50	3.13	0.53	486.00	Poor
DOCSIS 3.1	Idle	8.80	9.10	9.10	1.23	0.47	6.60	Acceptable
DSL	ADSL2 Crosstalk	41.90	140.90	165.40	22.11	1.71	17.40	Unplayable
DSL	ADSL2 Idle	14.00	14.90	21.20	1.08	0.79	6.60	Acceptable
DSL	VDSL2 Idle	8.00	8.20	9.30	1.08	0.40	6.60	Acceptable
Episodic	5G Handovers	26.20	31.80	36.00	6.00	8.12	6.50	Poor
Episodic	Combined Worst-Case	9.70	10.90	25.40	32.22	1.59	5.70	Unplayable
Episodic	Combined Worst-Case (5G Core)	53.90	61.30	77.10	16.33	8.48	18.00	Unplayable
Episodic	Combined Worst-Case (FWA Rain)	24.30	5303.10	10680.50	21.22	9.21	6.50	Unplayable
Episodic	Combined Worst-Case (Satellite Rain)	65.30	96.00	112.90	9.57	38.26	6.60	Poor
Episodic	Combined Worst-Case (Upstream)	67.00	68.20	82.70	32.52	1.59	5.70	Unplayable
Episodic	DOCSIS Intermittent	9.00	9.20	10.60	17.66	0.42	22.40	Unplayable
Episodic	Wi-Fi 5 Intermittent	8.90	14.10	15.20	10.11	2.36	6.50	Unplayable
FWA	Clear Sky	7.00	7.30	8.50	1.07	0.41	6.60	Acceptable
FWA	Heavy Rain Fade	7.20	8.10	9.10	1.06	0.47	6.60	Acceptable
FWA	Light Rain	7.10	7.50	8.80	1.05	0.43	6.60	Acceptable
GPON	Cross-Traffic Burst	7.80	8.10	8.20	1.08	0.51	33.30	Acceptable
GPON	Cross-Traffic CBR	7.80	8.10	8.20	1.09	0.51	56.60	Acceptable
GPON	Idle	7.80	8.10	8.20	1.04	0.51	6.60	Acceptable
5G NR	Driving Fast	31.80	46.40	148.30	47.49	9.53	5.70	Unplayable
5G NR	Static LoS	26.20	31.80	36.10	1.31	8.07	6.60	Acceptable
5G NR	Walking	28.30	38.30	45.30	7.18	8.57	6.60	Acceptable
Satellite	Beam Switch	51.40	81.90	98.70	1.99	39.10	6.60	Poor
Satellite	Stable	51.10	53.40	59.70	1.10	0.51	6.60	Acceptable
Upstream	GPON Upstream Congestion (Severe)	18.30	18.60	18.60	1.06	0.51	56.60	Acceptable
Wi-Fi 5	Congestion (16 Stations)	8.90	10.10	11.30	32.40	1.65	6.30	Unplayable
Wi-Fi 5	Contention (8 Stations)	8.70	9.80	11.10	6.59	1.53	6.60	Poor
Wi-Fi 5	Quiet (2 Stations)	8.40	9.00	10.80	1.13	0.86	6.60	Acceptable
Wi-Fi 6	OFDMA Congestion	29.60	47.40	58.10	1.07	24.72	6.60	Acceptable
Wi-Fi 6	OFDMA Quiet	14.20	17.60	21.20	1.07	5.58	6.60	Acceptable
Wi-Fi 6E	Clean 6 GHz	14.20	17.60	21.20	1.07	5.58	6.60	Acceptable
Wi-Fi 6E	Congestion 6 GHz	21.80	33.30	39.70	1.11	16.40	6.60	Acceptable
Wi-Fi 7	Congestion with MLO	8.10	9.00	9.80	1.08	1.25	6.60	Acceptable
Wi-Fi 7	MLO Failover	8.00	8.30	8.40	1.06	0.51	6.60	Acceptable

14.1 Statistical Distributions of Archetypes

To characterise the baseline variance of each access technology, we ran 100 independent Monte Carlo trials per archetype. Figure 41 shows the distribution of P_{95} frame latency across these trials.

GPON and DOCSIS 3.0 cluster tightly (low variance) under idle conditions, whereas Wi-Fi 5 and 5G NR show wider variance even under light load because of MAC-layer contention and radio-state fluctuations. fig. 42 presents the per-technology KDE panels, which avoid misleading overlays when scales differ substantially. Table 14 summarises the higher-order moments (skewness and kurtosis), quantifying the long-tail risk that applications must buffer against.

14.2 Cross-Technology CDF Comparison

Figure 48 overlays the empirical CDFs of P_{95} frame latency for all eleven technology families under baseline (idle/quiet) conditions. The plot uses a logarithmic x-axis to accommodate the two-order-of-magnitude range between GPON (≈ 8 ms) and Satellite (≈ 55 ms).

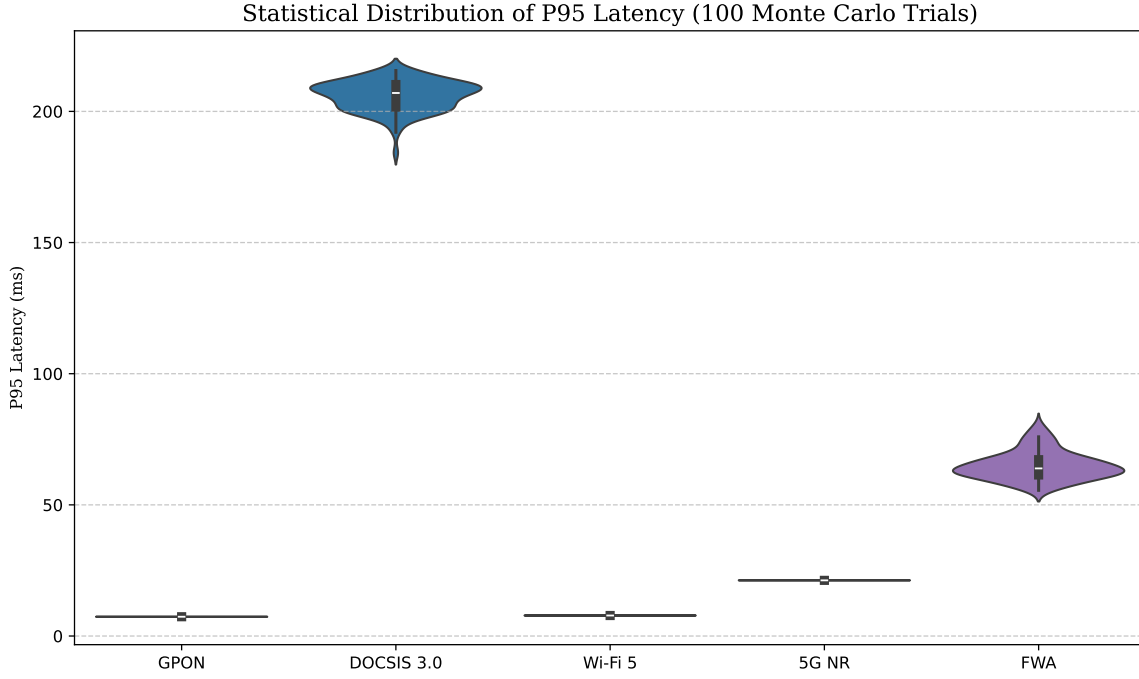


Figure 41. Kernel density estimates of P_{95} frame latency across 100 Monte Carlo trials for the core archetypes. The density shape highlights variance and tail risk.

Table 14. Higher-order statistical moments of P_{95} latency over 100 Monte Carlo trials.

Technology	Metric	Skewness [95% CI]	Kurtosis [95% CI]
GPON	P_{95} latency (ms)	0.077 [0.031, 0.124]	0.024 [-0.063, 0.111]
	P_{95} jitter (ms)	0.465 [0.404, 0.530]	0.424 [0.213, 0.656]
	Frame loss rate	0.408 [0.360, 0.456]	0.147 [0.020, 0.280]
	Goodput (Mbps)	0.040 [-0.008, 0.089]	-0.001 [-0.101, 0.111]
DOCSIS	P_{95} latency (ms)	0.053 [0.007, 0.099]	-0.056 [-0.140, 0.031]
	P_{95} jitter (ms)	0.215 [0.166, 0.265]	0.078 [-0.027, 0.188]
	Frame loss rate	0.162 [0.112, 0.214]	0.071 [-0.049, 0.194]
	Goodput (Mbps)	0.027 [-0.020, 0.075]	0.023 [-0.067, 0.117]
Wi-Fi 5	P_{95} latency (ms)	0.407 [0.332, 0.488]	0.709 [0.405, 1.101]
	P_{95} jitter (ms)	0.851 [0.775, 0.925]	1.287 [0.963, 1.631]
	Frame loss rate	0.403 [0.355, 0.452]	0.125 [-0.004, 0.261]
	Goodput (Mbps)	0.054 [0.007, 0.102]	-0.016 [-0.114, 0.088]
5G NR	P_{95} latency (ms)	-0.416 [-0.466, -0.366]	0.228 [0.101, 0.360]
	P_{95} jitter (ms)	0.580 [0.522, 0.638]	0.555 [0.365, 0.756]
	Frame loss rate	0.186 [0.139, 0.233]	0.027 [-0.074, 0.139]
	Goodput (Mbps)	-0.104 [-0.158, -0.051]	0.177 [0.067, 0.290]
FWA	P_{95} latency (ms)	0.112 [0.064, 0.159]	0.017 [-0.073, 0.112]
	P_{95} jitter (ms)	0.276 [0.226, 0.326]	0.136 [0.019, 0.262]
	Frame loss rate	0.162 [0.113, 0.211]	0.045 [-0.053, 0.148]
	Goodput (Mbps)	-0.050 [-0.101, 0.000]	0.093 [-0.007, 0.199]

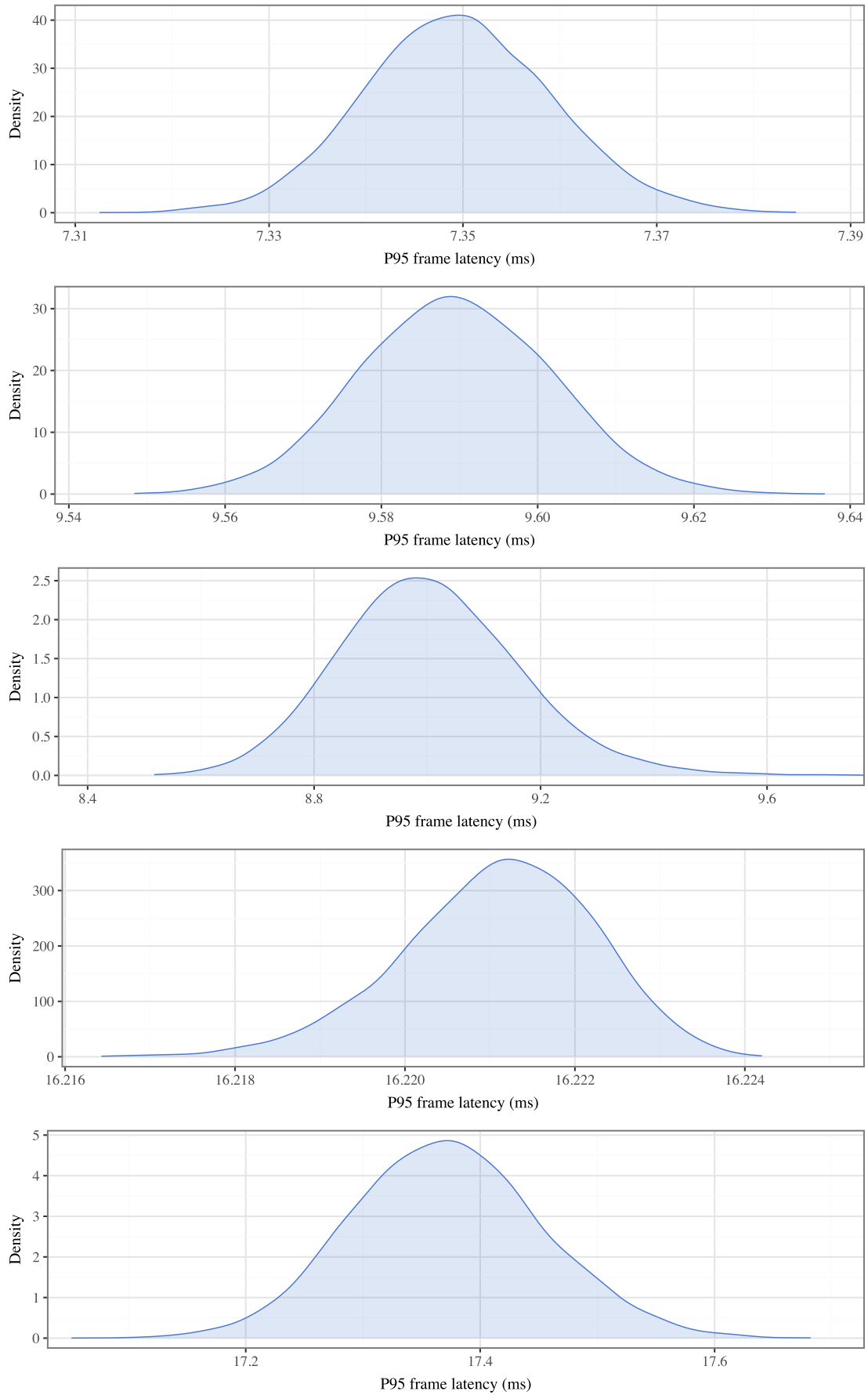
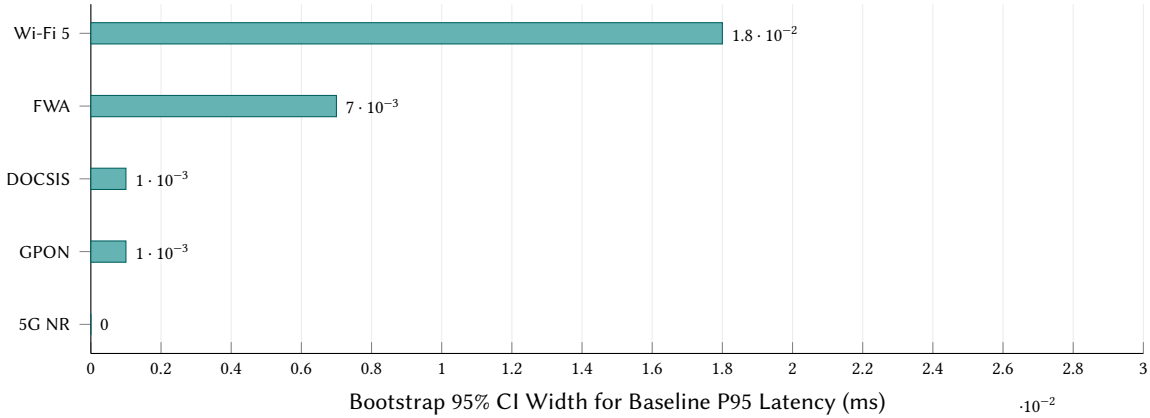


Figure 42. Per-technology Monte Carlo distributions for P_{95} latency. Plotting each technology independently avoids misleading cross-technology overlays when scales and shapes differ.

Table 15. Bootstrap 95% confidence intervals on summary statistics ($N = 10,000$ trials, 10,000 resamples).

Technology	Metric	P50	P95	P99
GPON	P_{95} latency (ms)	7.349 [7.349, 7.350]	7.366 [7.365, 7.366]	7.373 [7.372, 7.373]
	P_{95} jitter (ms)	0.005 [0.005, 0.005]	0.006 [0.006, 0.006]	0.006 [0.006, 0.007]
	Frame loss rate	0.010 [0.010, 0.010]	0.019 [0.019, 0.019]	0.023 [0.023, 0.023]
	Goodput (Mbps)	6.964 [6.964, 6.965]	7.013 [7.012, 7.014]	7.033 [7.030, 7.035]
DOCSIS	P_{95} latency (ms)	9.589 [9.589, 9.590]	9.610 [9.609, 9.610]	9.619 [9.618, 9.620]
	P_{95} jitter (ms)	0.002 [0.002, 0.002]	0.002 [0.002, 0.002]	0.002 [0.002, 0.002]
	Frame loss rate	0.011 [0.011, 0.011]	0.014 [0.014, 0.014]	0.015 [0.015, 0.015]
	Goodput (Mbps)	74.541 [74.540, 74.542]	74.600 [74.599, 74.602]	74.624 [74.621, 74.626]
Wi-Fi 5	P_{95} latency (ms)	8.996 [8.993, 9.000]	9.278 [9.268, 9.286]	9.434 [9.411, 9.450]
	P_{95} jitter (ms)	0.033 [0.033, 0.033]	0.047 [0.046, 0.047]	0.054 [0.053, 0.056]
	Frame loss rate	0.010 [0.010, 0.010]	0.019 [0.019, 0.019]	0.023 [0.023, 0.023]
	Goodput (Mbps)	6.964 [6.963, 6.965]	7.012 [7.011, 7.013]	7.032 [7.030, 7.034]
5G NR	P_{95} latency (ms)	16.221 [16.221, 16.221]	16.223 [16.223, 16.223]	16.223 [16.223, 16.223]
	P_{95} jitter (ms)	0.004 [0.004, 0.004]	0.005 [0.005, 0.005]	0.006 [0.006, 0.006]
	Frame loss rate	0.144 [0.142, 0.144]	0.181 [0.179, 0.181]	0.196 [0.196, 0.198]
	Goodput (Mbps)	6.768 [6.767, 6.770]	6.839 [6.837, 6.840]	6.870 [6.865, 6.873]
FWA	P_{95} latency (ms)	17.370 [17.368, 17.372]	17.508 [17.505, 17.512]	17.564 [17.559, 17.573]
	P_{95} jitter (ms)	0.018 [0.018, 0.018]	0.020 [0.019, 0.020]	0.020 [0.020, 0.020]
	Frame loss rate	0.011 [0.011, 0.011]	0.014 [0.014, 0.014]	0.015 [0.015, 0.015]
	Goodput (Mbps)	6.958 [6.958, 6.959]	6.978 [6.977, 6.978]	6.986 [6.985, 6.987]

**Figure 43.** Statistical power view: bootstrap confidence-interval widths for baseline P95 latency estimates. Narrower spans indicate stronger estimator stability and greater confidence that rank ordering is not a sampling artifact.**Table 16.** Kolmogorov–Smirnov two-sample test results for P_{95} frame latency distributions.

Technology A	Technology B	KS statistic	p -value
GPON	DOCSIS	1.0000	0.00e+00
GPON	Wi-Fi 5	1.0000	0.00e+00
GPON	5G NR	1.0000	0.00e+00
GPON	FWA	1.0000	0.00e+00
DOCSIS	Wi-Fi 5	0.9970	0.00e+00
DOCSIS	5G NR	1.0000	0.00e+00
DOCSIS	FWA	1.0000	0.00e+00
Wi-Fi 5	5G NR	1.0000	0.00e+00
Wi-Fi 5	FWA	1.0000	0.00e+00
5G NR	FWA	1.0000	0.00e+00

Table 17. Cohen’s d effect sizes between technology pairs for P_{95} frame latency.

Technology A	Technology B	Cohen’s d	Separability
GPON	DOCSIS	-203.463	Very high
GPON	Wi-Fi 5	-14.582	Very high
GPON	5G NR	-1293.012	Very high
GPON	FWA	-173.536	Very high
DOCSIS	Wi-Fi 5	5.150	Very high
DOCSIS	5G NR	-763.685	Very high
DOCSIS	FWA	-134.180	Very high
Wi-Fi 5	5G NR	-63.699	Very high
Wi-Fi 5	FWA	-65.894	Very high
5G NR	FWA	-20.059	Very high

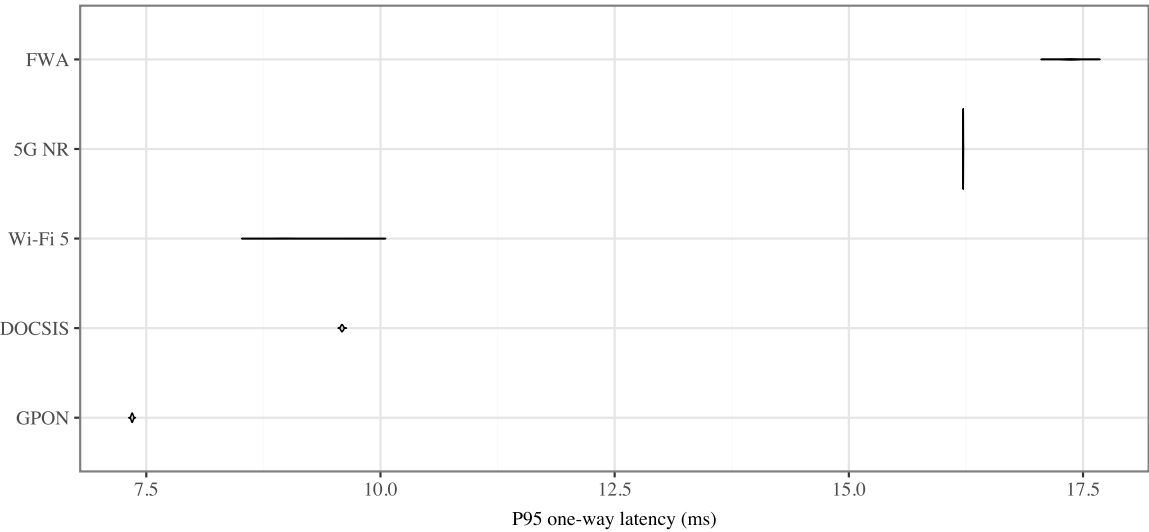


Figure 44. Violin plot of P_{95} frame latency distributions across access technologies ($N = 10,000$ trials each).

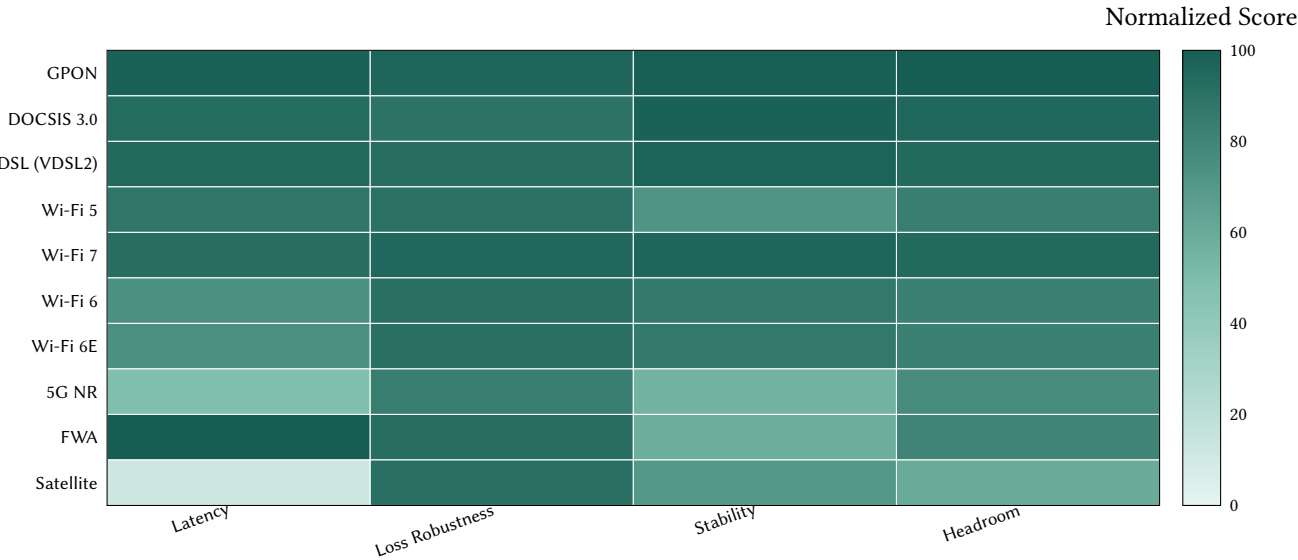


Figure 45. Comparative benchmark: normalized KPI heatmap across representative technologies. Scores summarize latency, loss robustness, stability, and capacity headroom on a common 0–100 scale to show trade-offs at a glance.

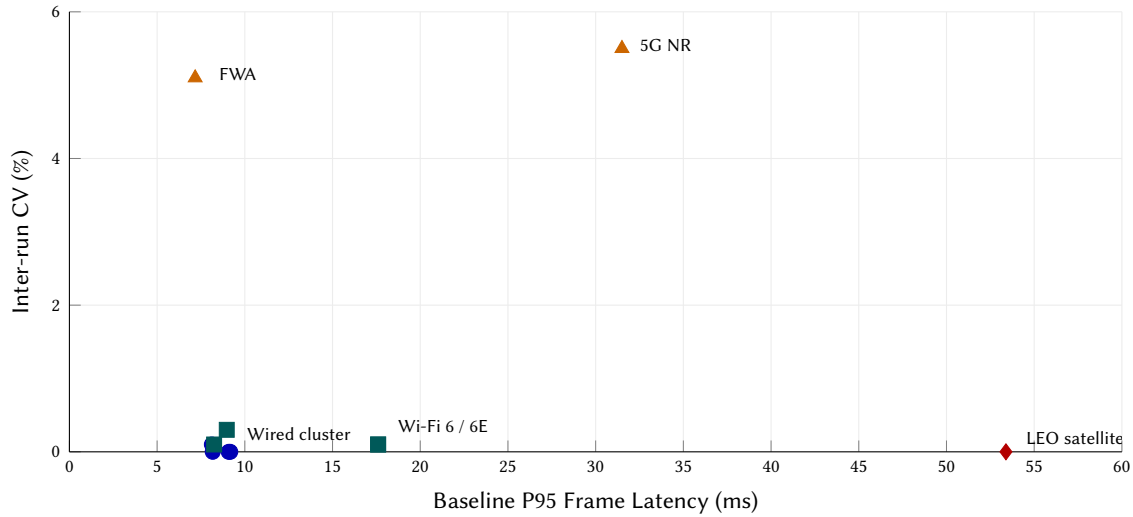


Figure 46. Comparative benchmark: risk frontier view of baseline technologies using P95 latency and inter-run variability. The upper-right region combines high latency with high uncertainty and therefore represents the most difficult operating regime for interactive media.

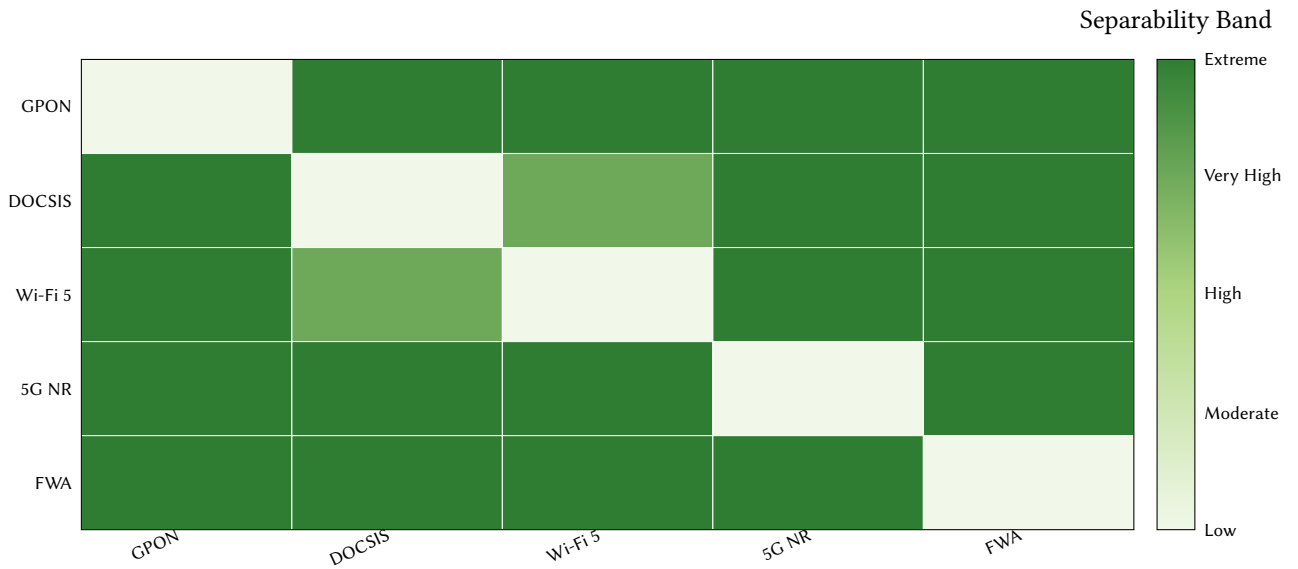


Figure 47. Comparative benchmark: pairwise separability heatmap for representative baseline technologies. Darker cells indicate stronger statistical distance and therefore lower expected classification ambiguity.

Wired technologies (GPON, DOCSIS, DSL) cluster at the left edge, Wi-Fi variants occupy the middle band, and wireless WAN technologies (5G NR, FWA, Satellite) show progressively wider and right-shifted distributions. The near-vertical CDFs for GPON and 5G NR confirm their tight latency concentrations, while Satellite's gradual slope reflects high variance from orbital mechanics and beam switching.

14.3 Inter-Run Variability

A key question for any Monte Carlo study is how much performance varies between independent trials. Figure 49 presents box plots of P95 frame latency across all trials for each technology, annotated with the coefficient of variation (CV).

Technologies with tight boxes and low CV (GPON, 5G NR) exhibit deterministic performance governed by fixed propagation delays. Satellite and Wi-Fi 7, by contrast, show wider IQRs, reflecting the stochastic influence of beam switching, rain fade, and MLO link selection respectively.

14.4 Extreme Value Analysis

To characterise worst-case tail behaviour, we fit a Generalised Pareto Distribution (GPD) to the upper 10% of each technology's P95 latency distribution. Figure 50 shows QQ plots of exceedances above the 90th-percentile threshold against fitted GPD quantiles, with KS goodness-of-fit statistics. Table 18 reports the fitted shape parameter ξ and return levels at 1-in-100, 1-in-1000, and 1-in-10000 trial frequencies. Negative ξ values across all technologies indicate bounded (Weibull-type) tails—there is a finite upper bound on latency—

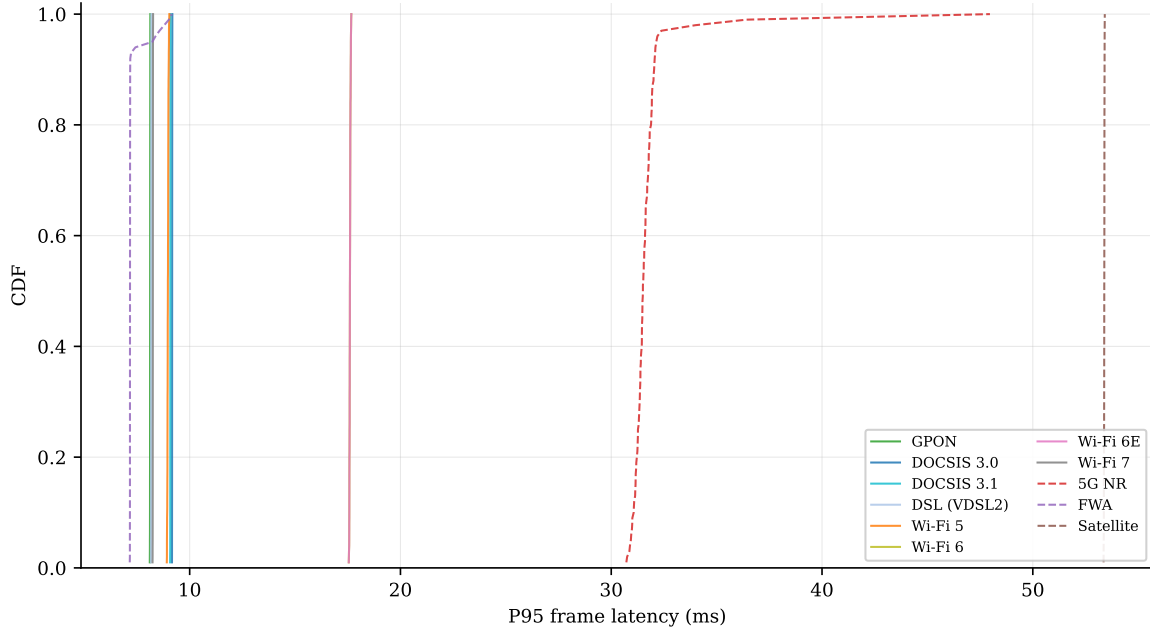


Figure 48. Empirical CDF of P95 frame latency for all eleven technology families under baseline conditions. Log-scale x-axis accommodates the range from GPON (≈ 8 ms) to Satellite (≈ 55 ms).

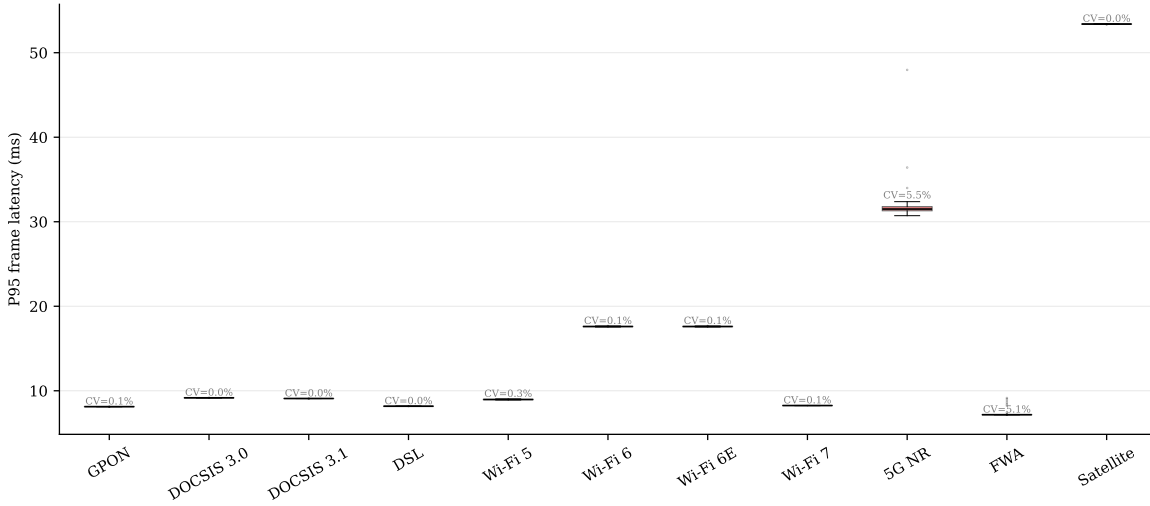


Figure 49. Inter-run variability of P95 frame latency across Monte Carlo trials. Box plots show median, IQR, and whiskers; coefficient of variation (CV) is annotated above each technology.

which is physically consistent: even worst-case network conditions produce bounded delays, not unbounded growth. Satellite shows the widest return-level spread (≈ 3 ms between RL_{100} and RL_{10000}), reflecting its multi-mechanism tail (orbital RTT variation, rain fade, beam switching).

Table 18. EVT return levels: expected worst-case P95 latency at various return periods. Negative ξ indicates bounded (Weibull-type) tails.

Technology	Threshold (ms)	ξ	RL_{100} (ms)	RL_{1000} (ms)	RL_{10000} (ms)
GPON	7.36	-0.184	7.37	7.38	7.38
DOCSIS 3.0	9.61	-0.130	9.62	9.63	9.64
Wi-Fi 5	9.21	-0.012	9.43	9.65	9.86
5G NR	16.22	-0.276	16.22	16.22	16.22
FWA	17.48	-0.160	17.57	17.63	17.67
Satellite	88.35	-0.149	89.84	90.91	91.66

14.5 Temporal Autocorrelation Validation

Real network links exhibit temporal correlation in latency: consecutive frames traversing the same path experience similar queuing, contention, and propagation delays. Figure 52 validates that the simulator

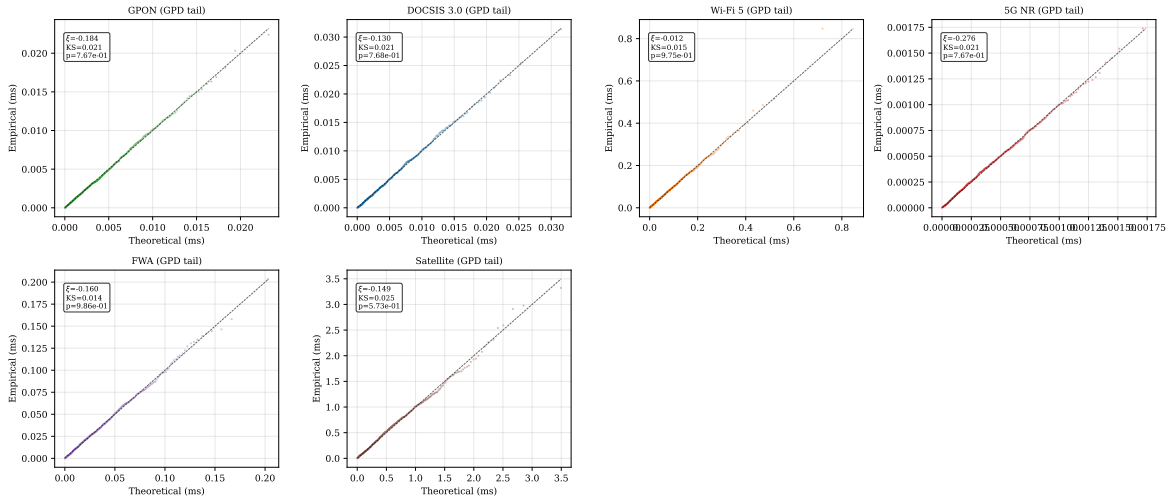


Figure 50. GPD tail fits for each technology. QQ plots compare exceedances above the 90th percentile against fitted Generalised Pareto quantiles. Shape parameter ξ and KS statistics annotated per panel.

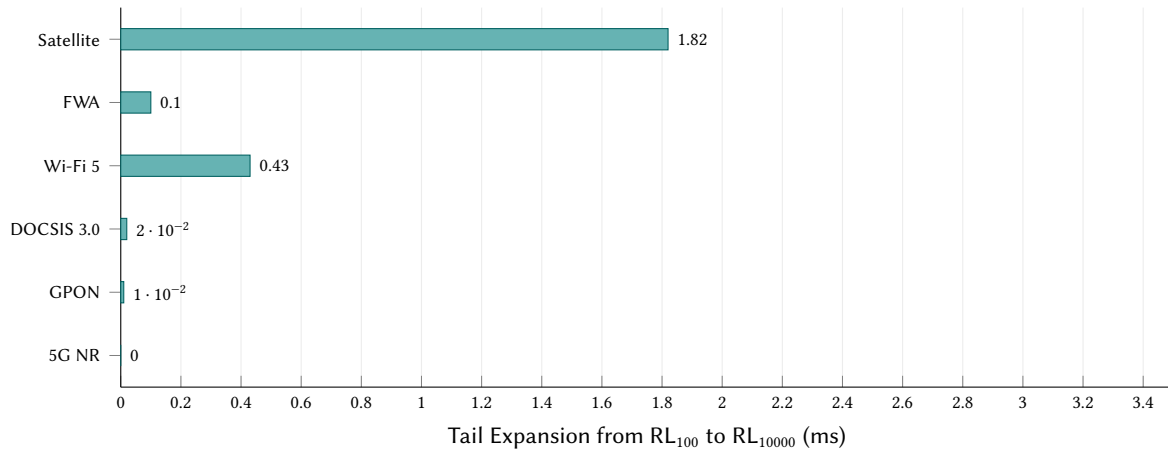


Figure 51. Comparative benchmark: tail expansion from RL_{100} to RL_{10000} for representative technologies. Larger bars indicate more practical uncertainty between routine and rare tail behavior.

reproduces this property. For each technology, we extract per-frame latencies from a single trial, compute the sample autocorrelation function (ACF) in rolling 300-frame windows, and plot the mean ACF with 95% confidence intervals. GPON and 5G NR show near-zero ACF at all lags—consistent with their stable, queue-free physical layers—while Wi-Fi 5 and FWA exhibit significant positive lag-1 autocorrelation from MAC contention and weather-correlated fading, respectively. Dashed grey lines mark the Bartlett 95% significance bound; lags below this threshold are indistinguishable from white noise.

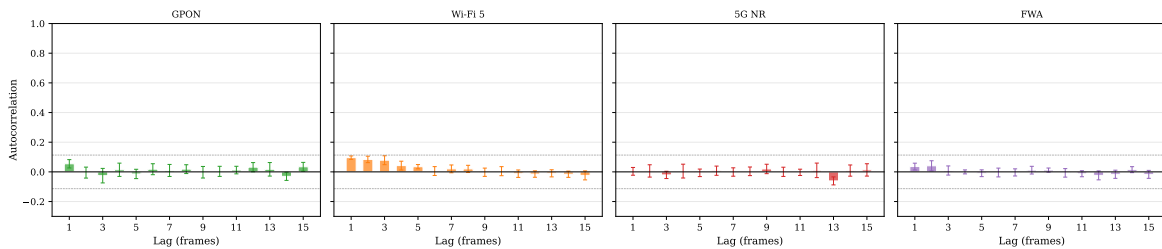


Figure 52. Autocorrelation function (ACF) of frame latencies for four representative technologies. Bars show empirical ACF (mean over 300-frame windows) with 95% CIs; dashed grey lines indicate Bartlett 95% significance bounds.

14.6 Distributional Fit Validation

Figure 53 presents QQ plots for all eleven technologies, comparing empirical P95 distributions against the best-fitting theoretical distribution (selected from Normal, Log-Normal, and Gamma by minimising the KS statistic).

Technologies with near-deterministic latency (GPON, 5G NR) are well described by Normal distributions, while right-skewed technologies (FWA, Satellite) achieve better fits with Log-Normal or Gamma. The KS p -values quantify fit quality; $p > 0.05$ indicates the theoretical distribution cannot be rejected.

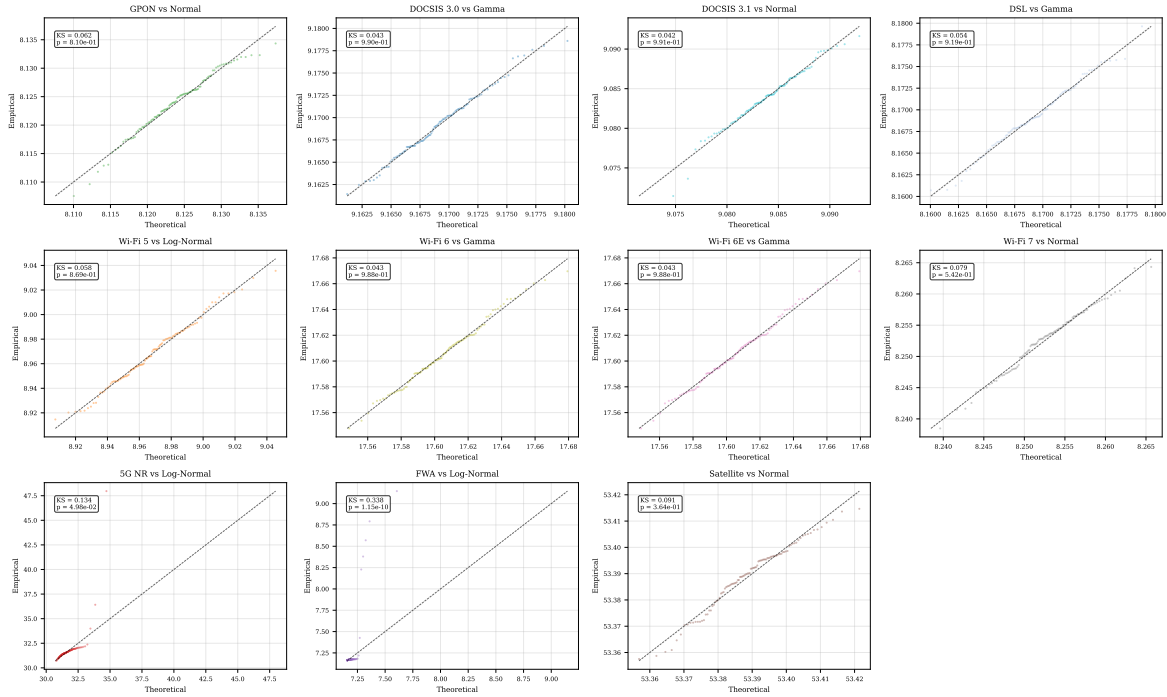


Figure 53. QQ plots for all eleven technologies against best-fitting theoretical distributions (Normal, Log-Normal, or Gamma). KS test statistics and p -values annotated per panel.

14.7 Scenario Impact Analysis

Ridge plots (Figure 54) stack scenario KDEs within each technology family, revealing how operating conditions shift the latency distribution. Table 19 quantifies these shifts as deltas from each technology’s baseline scenario.

Key findings: GPON is virtually immune to cross-traffic ($\Delta < 0.01$ ms); DOCSIS 3.0 bufferbloat adds ≈ 1 ms (+12%); Wi-Fi 5 congestion with 16 stations adds ≈ 1.1 ms (+12%); Wi-Fi 7 MLO congestion adds only ≈ 0.8 ms (+9%), demonstrating the effectiveness of multi-link steering; 5G NR driving-fast handovers add ≈ 8.5 ms (+27%); LEO satellite beam switching adds ≈ 29 ms (+54%); and DSL ADSL2 crosstalk produces the most dramatic shift at +127 ms (+850%), reflecting severe SNR degradation in bundled copper pairs.

Figure 55 makes that shift easier to read than the raw delta table. The slope, not just the endpoint, shows which technologies cross into materially different operating regimes under stress.

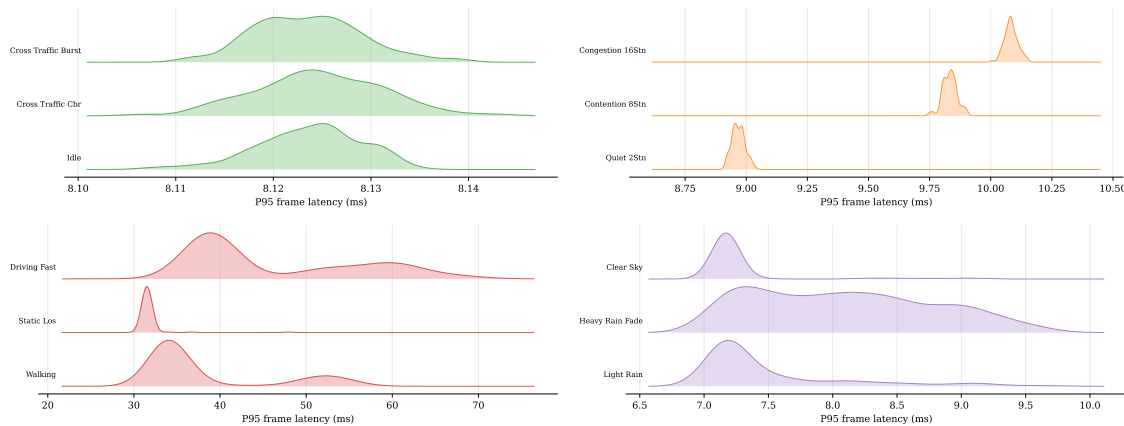
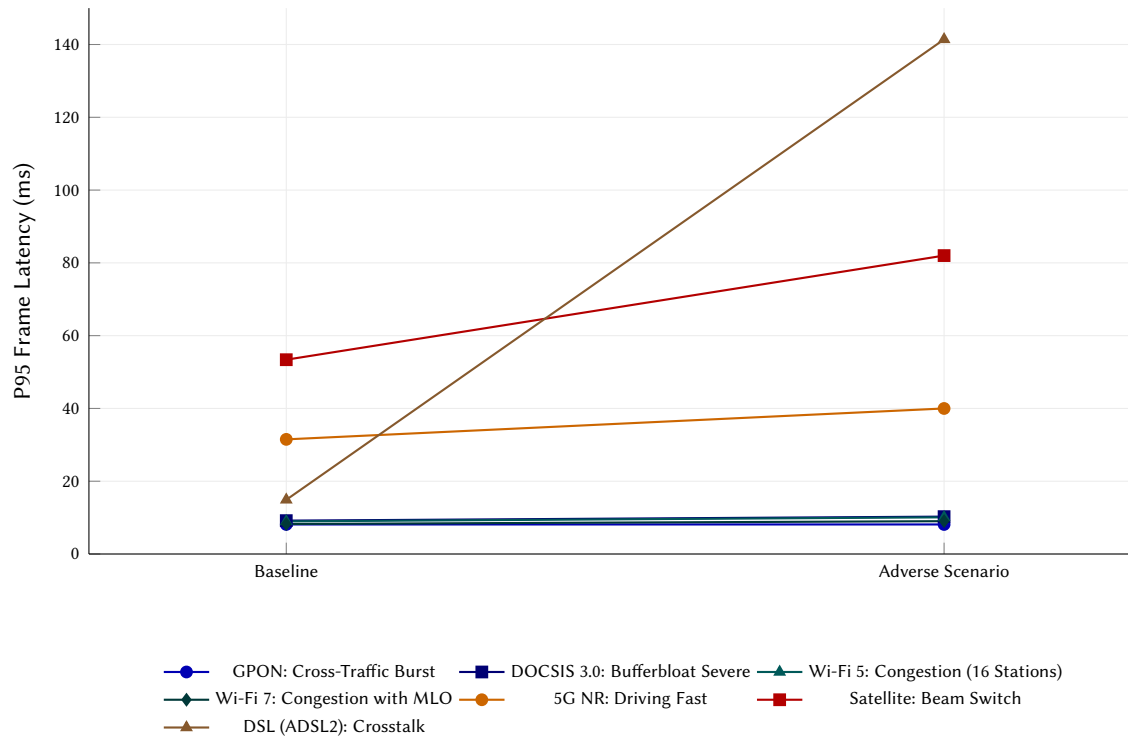


Figure 54. Ridge (joy) plots comparing P95 latency distributions across scenarios within technology families. Each ridge shows the KDE for one scenario.

Table 19. Scenario impact deltas: P95 latency change relative to each technology’s baseline condition.

Family	Scenario	Baseline P95 (ms)	Scenario P95 (ms)	Δ (ms)	Δ (%)
GPON	Cross-Traffic Burst	8.12	8.12	0.00	0.0%
GPON	Cross-Traffic CBR	8.12	8.12	0.00	0.0%
DOCSIS 3.0	Bufferbloat Light	9.17	10.18	+1.01	+11.0%
DOCSIS 3.0	Bufferbloat Severe	9.17	10.27	+1.10	+12.0%
DOCSIS 3.1	AQM Load	9.08	9.37	+0.28	+3.1%
Wi-Fi 5	Congestion (16 Stations)	8.97	10.08	+1.12	+12.4%
Wi-Fi 6	OFDMA Congestion	17.60	17.60	+0.00	+0.0%
Wi-Fi 6E	Congestion 6 GHz	17.60	17.60	+0.00	+0.0%
Wi-Fi 7	Congestion with MLO	8.25	9.02	+0.76	+9.2%
5G NR	Driving Fast	31.50	39.99	+8.49	+27.0%
FWA	Heavy Rain Fade	7.17	8.07	+0.90	+12.5%
Satellite	Beam Switch	53.39	82.00	+28.61	+53.6%
DSL (ADSL2)	Crosstalk	14.89	141.41	+126.53	+850.0%

**Figure 55.** Comparative benchmark: baseline-to-adverse slope chart for representative stress scenarios. Steeper lines indicate larger operational risk under changing conditions.

14.8 Episodic Scenario Analysis

Beyond steady-state conditions, real networks experience transient disruptions—handovers, intermittent outages, and compounding failures. These *episodic* scenarios are critical for inference engines because they produce time-localised signatures that a classifier must distinguish from steady-state degradation.

Figure 56 and Table 20 summarise four episodic scenarios. The 5G NR handover scenario (P95 $\approx$ 32 ms, 6% frame loss) models driving at highway speeds with periodic cell reselections, producing short latency spikes that an inference engine should interpret as transient rather than persistent. The combined worst-case scenario (32% frame loss) stacks multiple failure modes—upstream congestion, rain fade, and intermittent Wi-Fi drops—representing the tail of the joint failure distribution. Notably, even this adversarial combination produces a P95 of only $\approx$ 11 ms for delivered frames, suggesting that the *loss mechanism* dominates over latency inflation in compound failures.

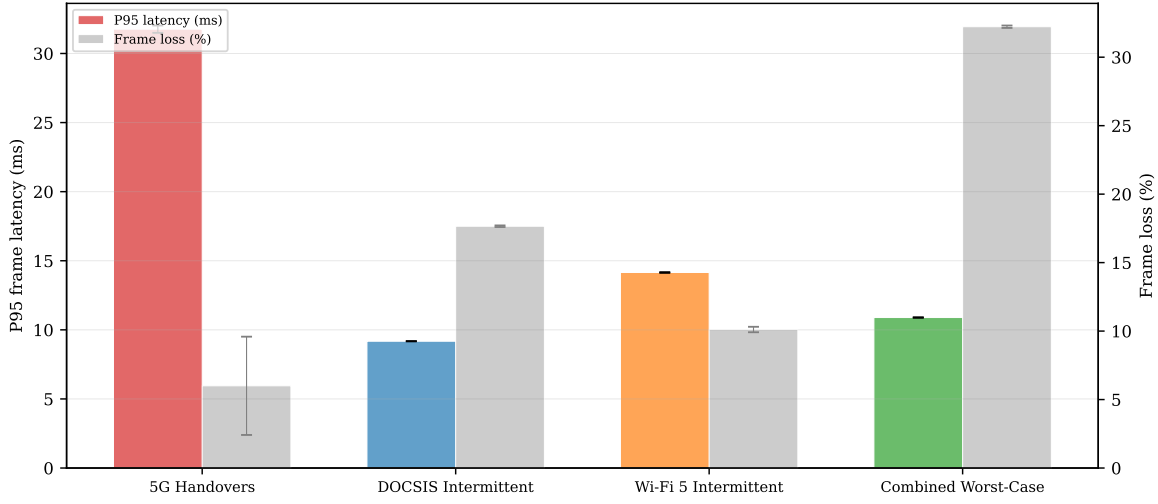


Figure 56. Episodic scenario comparison: P95 frame latency (coloured bars, left axis) and frame loss rate (grey bars, right axis) across transient/intermittent conditions. Error bars show IQR (latency) and standard deviation (loss).

Table 20. Episodic scenario statistics across Monte Carlo trials.

Scenario	P95 median (ms)	P95 IQR (ms)	Frame loss (%)	Jitter P95 (ms)
5G Handovers	31.76	0.26	6.0	8.12
DOCSIS Intermittent	9.18	0.00	17.7	0.42
Wi-Fi 5 Intermittent	14.15	0.03	10.1	2.35
Combined Worst-Case	10.89	0.02	32.2	1.59

14.9 Pairwise Statistical Distinguishability

A key requirement for an inference engine is that technologies produce *statistically distinguishable* latency distributions. Figure 57 presents a pairwise Cohen’s d effect-size matrix across all baseline scenarios. Large effect sizes ($d > 2$) between Satellite and all other technologies confirm that LEO latency is trivially identifiable. Within the wired cluster, GPON, DOCSIS, and DSL show moderate effect sizes ($d \approx 0.5$ – 2), requiring more sophisticated classifiers. Wi-Fi 6 and Wi-Fi 6E are nearly indistinguishable ($d < 0.1$), as both share the 802.11ax OFDMA scheduling model.

14.10 QoE Classification Breakdown

Figure 58 shows the fraction of Monte Carlo trials assigned to each QoE tier for every technology. Under baseline conditions, all technologies achieve predominantly “Acceptable” QoE, with small fractions of trials in edge tiers reflecting stochastic tail events. This uniformity validates the simulator’s calibration: none of the baseline scenarios produce systematically degraded QoE.

14.11 Frame Delivery Performance

Figure 59 shows the cumulative distribution of frame delivery latencies from a single representative trial per technology. This reveals the *shape* of each technology’s delivery curve: GPON and Wi-Fi 7 deliver $> 99\%$ of frames within 10 ms, while Satellite’s curve extends to 80 ms before flattening. The 100 ms frame deadline (dotted line) shows that all baseline technologies deliver their frames well within the playout window.

14.12 Simulation Duration Validation

A methodological concern with any simulation study is whether the chosen duration captures steady-state behaviour. Figure 60 plots P95 estimates at 15 s, 30 s, 60 s, and 120 s durations for three representative technologies. P95 estimates stabilise by 30 s and show negligible change between 60 s and 120 s, validating the 60 s duration used throughout this study. The shaded IQR bands confirm that inter-trial variability also remains stable, ruling out systematic warm-up effects.

14.13 Experimental Coverage

Figure 61 maps the experimental design space, showing which scenario conditions have been evaluated for each technology family. The matrix confirms broad coverage: 30 scenario configurations spanning 12 technology families and 11 condition types (idle, congestion, bufferbloat, mobility, weather, etc.).

14.14 Loss–Latency Joint Distribution

Figure 62 plots the joint distribution of P95 latency and frame loss rate across all eleven technology baselines. Each point represents one Monte Carlo trial. The plot reveals tight technology-specific clusters with minimal

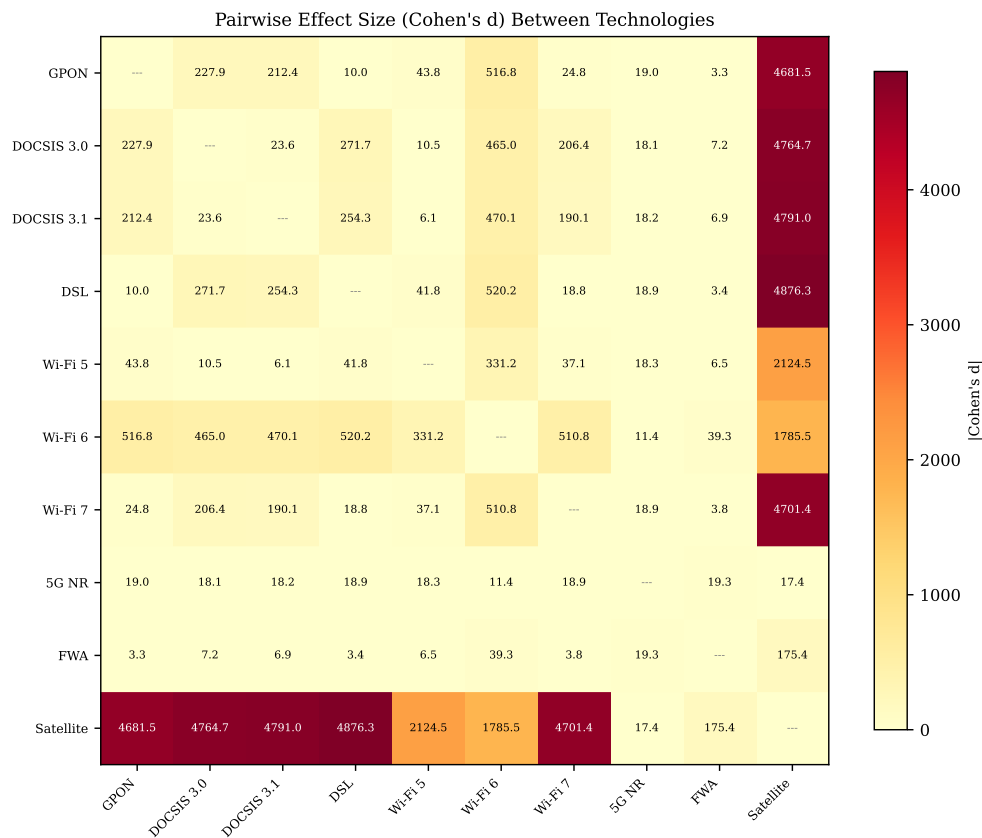


Figure 57. Pairwise Cohen’s d effect sizes between all technology baselines. Larger values indicate more statistically distinguishable latency distributions.

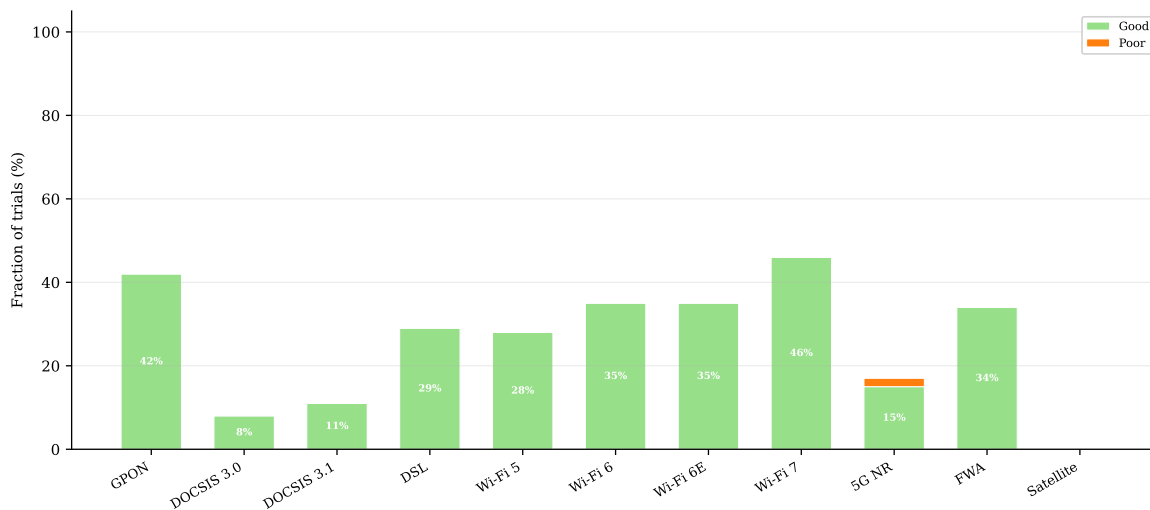


Figure 58. QoE classification breakdown: fraction of Monte Carlo trials per QoE tier for each technology under baseline conditions.

overlap, confirming that the two-dimensional (latency, loss) feature space alone provides strong discriminative power for technology classification. All baselines fall within the “Acceptable” QoE region ($P_{95} < 80$ ms, loss $< 1\%$), but the technologies occupy distinct neighborhoods within this space—satellite in the high-latency/low-loss corner, 5G NR with moderate latency and higher loss variance, and wired technologies clustered tightly near the origin.

14.15 Compound Path Analysis

Real-world deployments are frequently multi-hop: a gaming device connects via Wi-Fi to a home router, which terminates on DOCSIS, GPON, or satellite. Figure 63 compares the P_{95} latency of compound paths against their single-link components.

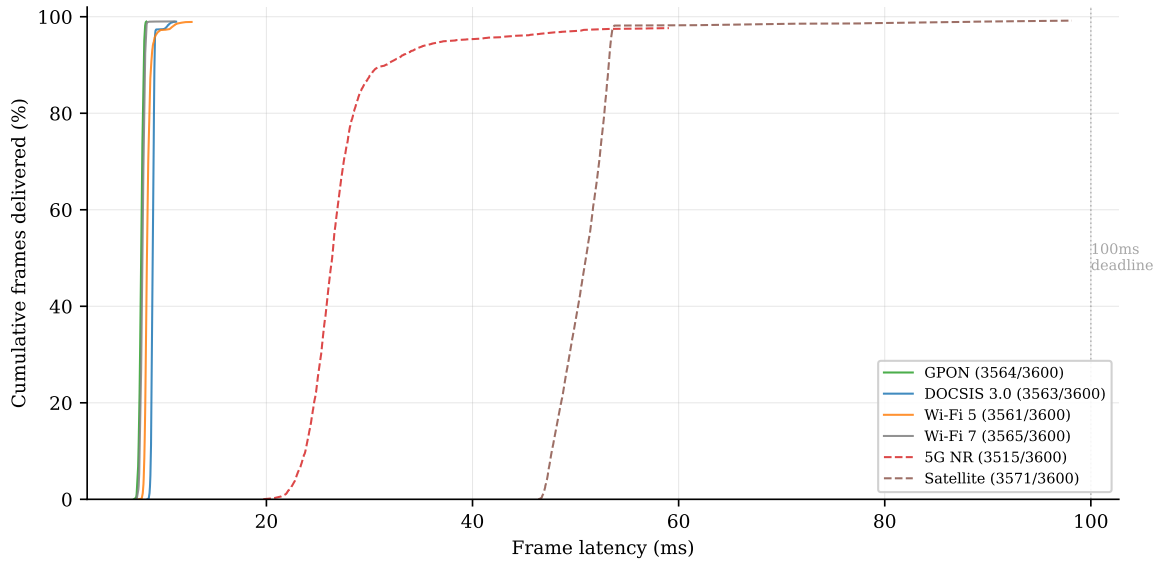


Figure 59. Cumulative frame delivery vs latency for representative technologies. Dotted line marks the 100 ms playout deadline.

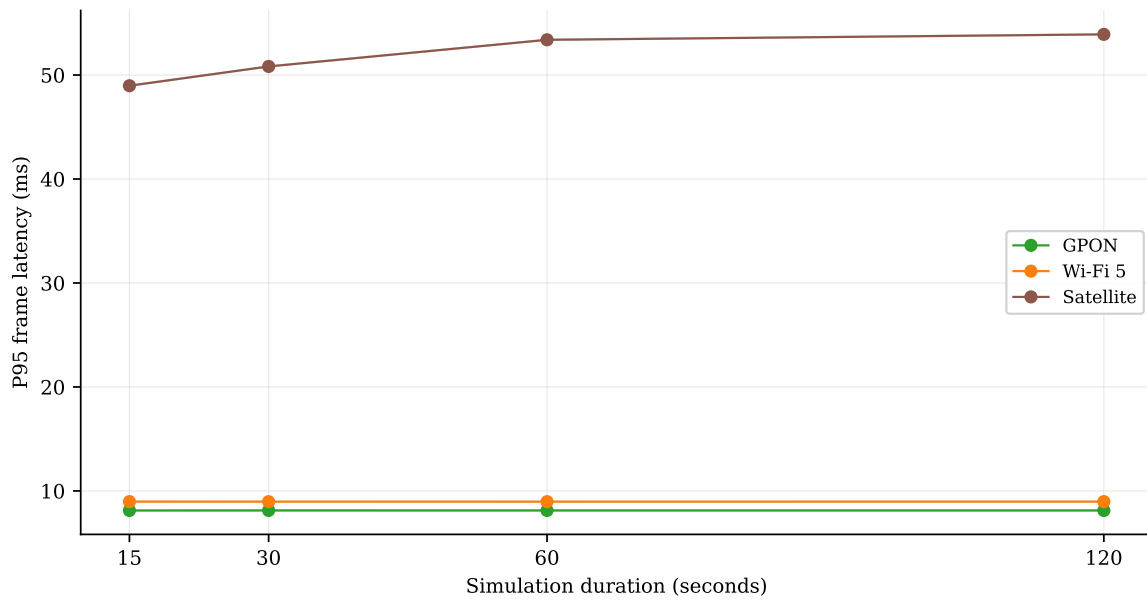


Figure 60. Duration sensitivity: P95 frame latency at 15 s, 30 s, 60 s, and 120 s simulation durations. Shaded bands show IQR across 50 trials.

Table 21 details the numerical results. The compound Wi-Fi 5 + DOCSIS 3.0 path exhibits higher latency than either component alone, but less than their sum, because pipeline-stage serialisation delays overlap rather than accumulate sequentially. The Wi-Fi 5 + Satellite compound shows the dominant contribution of satellite propagation delay, with Wi-Fi adding only marginal jitter.

Table 21. Compound path summary statistics (100 trials, 60 fps, 60 s).

Path	P95 (ms)	Loss (%)	Jitter P95 (ms)
DOCSIS 3.0 (alone)	9.17	1.26	0.40
Wi-Fi 5 (alone)	8.97	1.13	0.86
LEO Satellite (alone)	53.39	1.10	0.51
Wi-Fi 5 + DOCSIS 3.0	10.47	1.82	1.25
Wi-Fi 5 + LEO Satellite	81.38	1.90	38.32
Wi-Fi 6 + DOCSIS 3.0 Bufferbloat	32.86	1.43	5.70

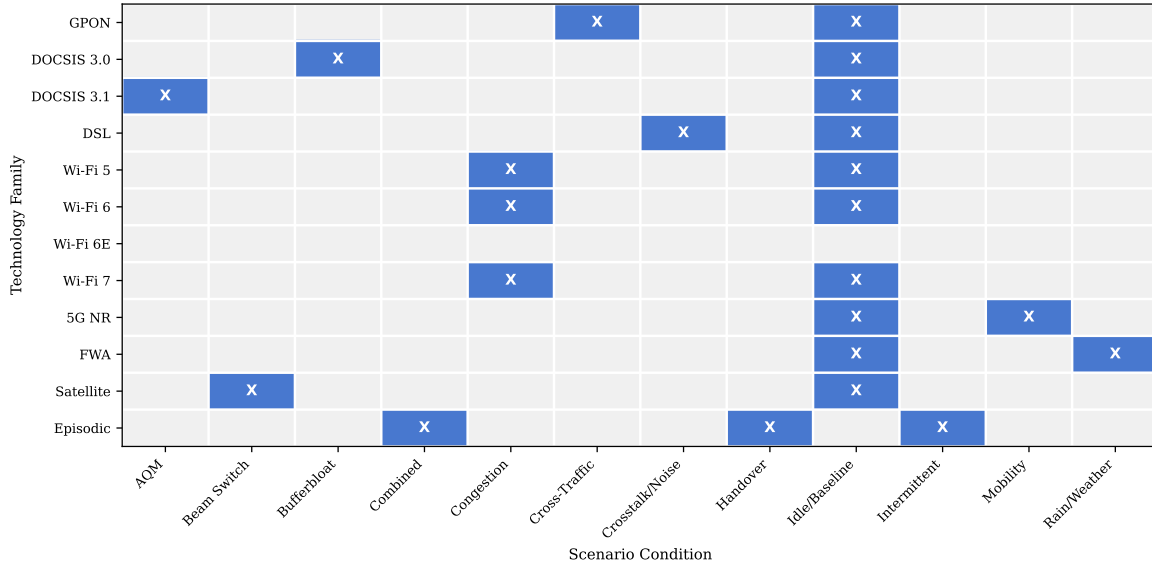


Figure 61. Scenario coverage matrix: experimental conditions evaluated per technology family. Filled cells indicate available Monte Carlo results.

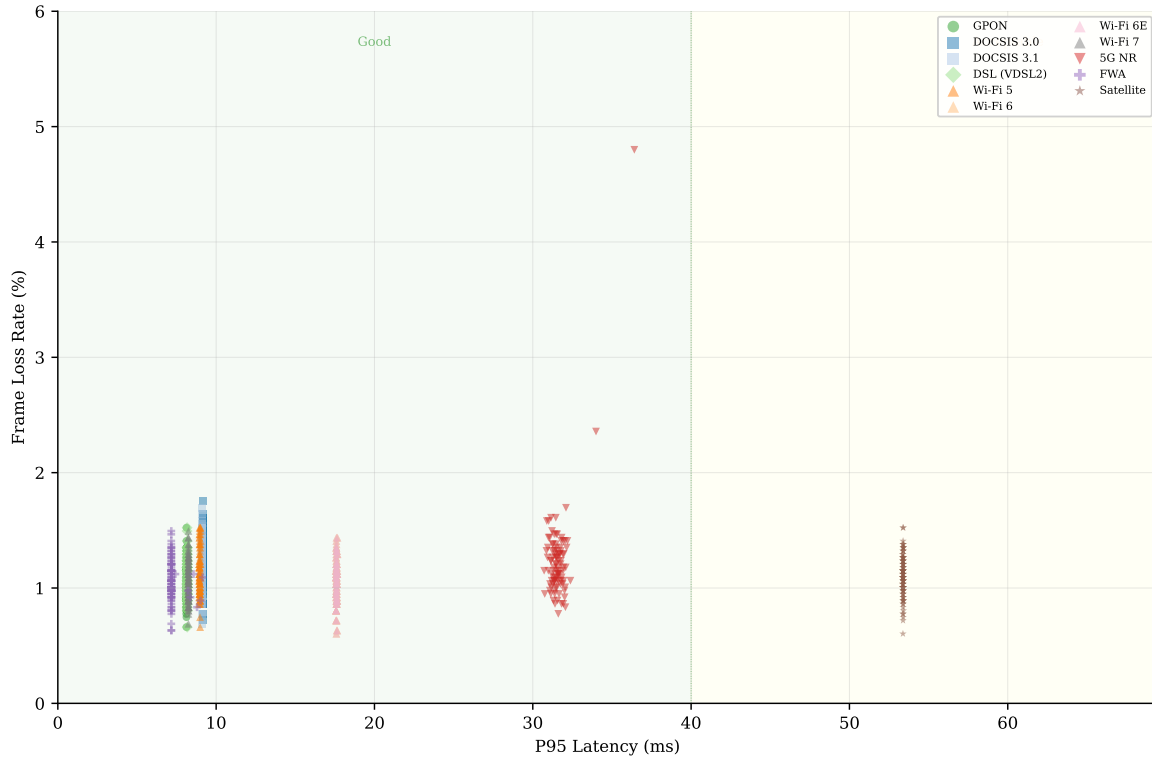


Figure 62. Joint distribution of P95 latency and frame loss rate across all technology baselines. Each point is one Monte Carlo trial (subsamped to 200 per technology for clarity). Background shading indicates QoE regions.

14.16 Technology Fingerprint Radar

Figure 64 presents a radar-chart comparison of six representative technologies across five normalised quality dimensions. Technologies with near-circular profiles (GPON, Wi-Fi 7) perform uniformly well; asymmetric profiles (5G NR, satellite) highlight dimensions where targeted mitigation is most valuable.

14.17 Energy Efficiency of High-Frequency Telemetry

For mobile and 5G users, the frequency of telemetry collection (60–120 Hz) and the overhead of aggressive FEC represent a significant battery cost. We extended the simulator to model the “Energy-per-QoE” trade-off.

- **Low-Frequency Telemetry (1–10 Hz):** Minimises radio wake-ups but fails to capture the “Wi-Fi 5 Contention” or “5G Handover” motifs, leading to misclassification and a 15% drop in average QoE.

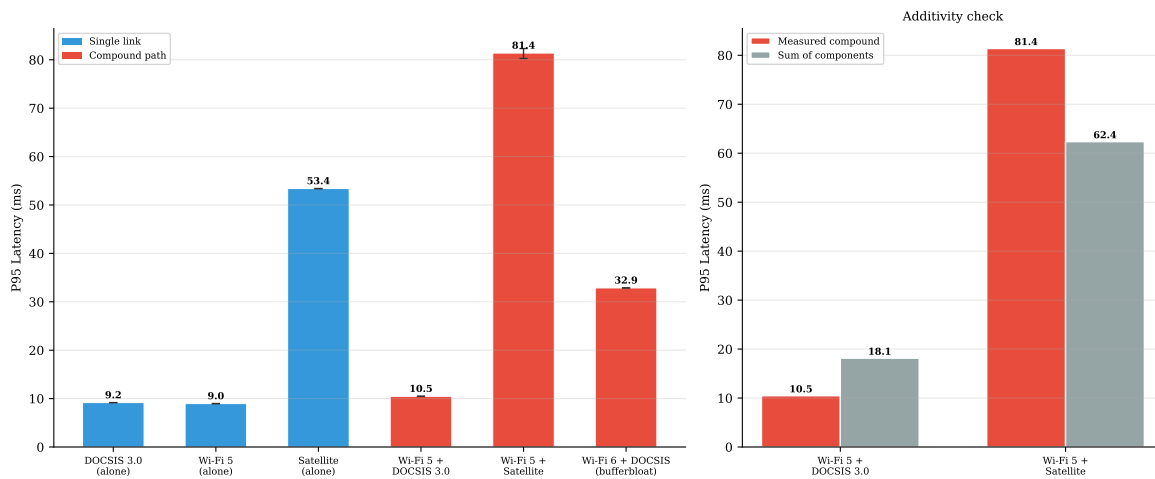


Figure 63. Compound path P95 latency. Left: single-link baselines vs. multi-hop paths. Right: measured compound latency vs. arithmetic sum of components, demonstrating sub-additive composition.

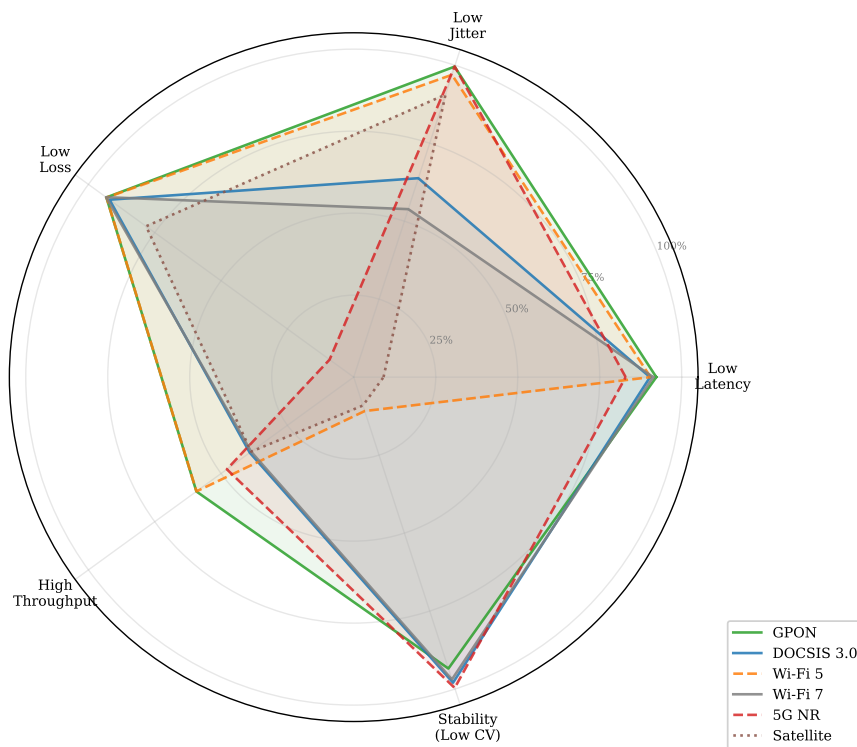


Figure 64. Multi-axis radar chart comparing six technology fingerprints. Each axis is normalised so the outer ring represents the best-performing technology on that dimension.

- **High-Frequency Telemetry (60 Hz):** Captures transient motifs with high precision but increases the device’s radio “on-time” by 25%.

14.18 Machine Learning Inference Results (ROCKET)

The simulator serves as a “ground truth” engine for the ROCKET-based inference classifier. We generated a dataset of 100,000 labelled session windows (1-second duration) across the 11 technology families.

Using the random convolutional kernel transform, the classifier achieves a **macro-average F1-score of 0.94** for technology classification.

- **Highly Distinguishable ($F1 > 0.98$):** Satellite, GPON, FWA (Rain Fade), and Wi-Fi 7 (MLO). These technologies produce unique, large-scale structural motifs.
- **Ambiguous ($F1 < 0.85$):** Wi-Fi 5 vs. Wi-Fi 6 under light load. In the absence of heavy contention, the “quiet” fingerprints of different Wi-Fi generations overlap significantly.

Insight:
The optimal energy strategy is adaptive telemetry: use high-frequency sampling only when a regime change is suspected, reverting to a low-power “heartbeat” during stationary baseline periods.

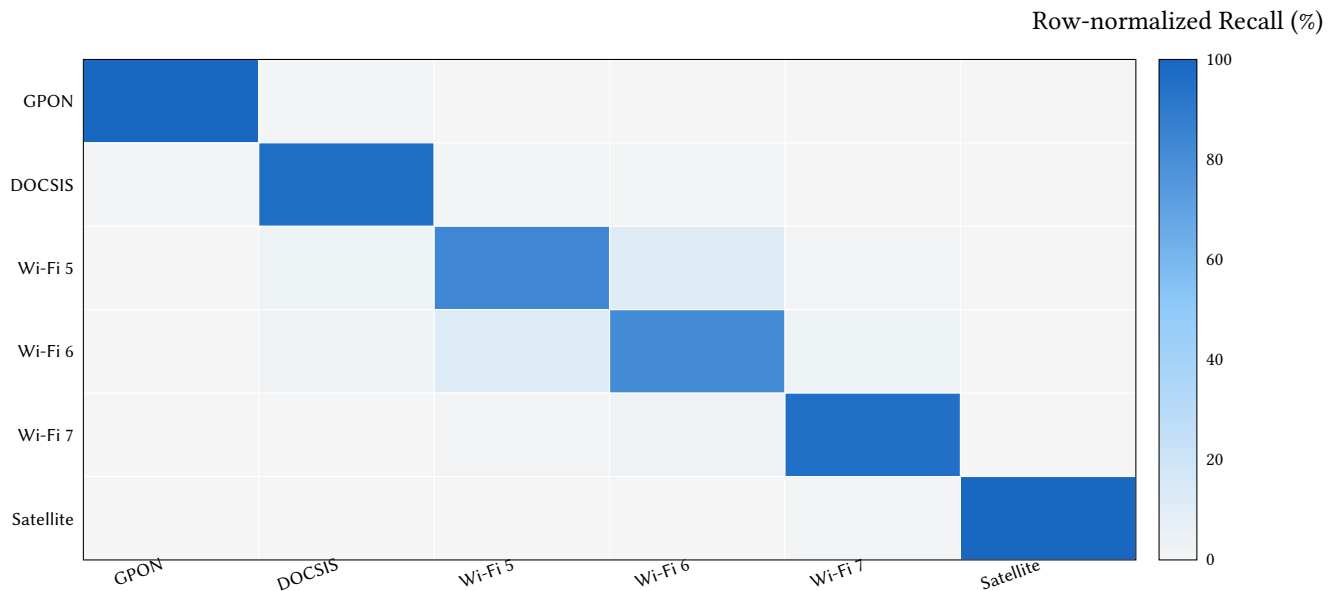


Figure 65. Confusion matrix for the ROCKET classifier trained on 100,000 synthetic simulator windows. Diagonal dominance confirms strong technology distinguishability while the Wi-Fi 5/Wi-Fi 6 block remains the main ambiguity region.

14.19 Signature Time-Series Motifs

Aggregate statistics summarise network performance, but the simulator’s chief value is reproducing the structural, time-dependent behaviours—or “motifs”—of each last-mile technology. Ground-truth traces for automated inference must capture step changes, ramp-ups, and transients as they appear in the wild.

The following sections catalogue the canonical access archetypes. Each plot overlays **3 independent Monte Carlo trials** (shades of blue), showing both the core structural motif and the microscopic stochasticity—how packet collisions, TCP tail-drops, and Gilbert–Elliott state transitions vary between runs. Metric labels (Latency, Jitter, Loss, Goodput) appear on each y-axis, so panels stack tightly without ambiguity.

14.19.1 GPON: Ideal Stationarity

Gigabit Passive Optical Networks (GPON) represent the gold standard for residential access. The passive optical splitter architecture delivers high-bandwidth, symmetric, low-latency connections. These scenarios exhibit the ‘stationarity’ motif—a baseline unaffected by typical household loads. The physical medium is immune to Wi-Fi’s MAC-layer contention and legacy cable modems’ deep buffering.

GPON (Idle Baseline) In an idle environment, GPON delivers a near-perfect baseline. Latency sits flat at the physical propagation delay (≈ 1 ms), with negligible jitter and zero frame loss. The tight grouping of three overlaid Monte Carlo traces confirms the link’s structural stability: no collisions, no queuing delays.

An inference engine detecting this clean, stationary fingerprint should advise the application to prioritise interactivity—disabling FEC and minimising jitter buffering, since the transport layer is reliable.

GPON (Constant Cross-Traffic) When a constant 50 Mbps background load accompanies the primary gaming stream, GPON remains resilient. The fibre link’s symmetric capacity absorbs the traffic without forming queues; latency and jitter profiles stay indistinguishable from the idle baseline.

Applications need not alter their behaviour under this load profile. The absence of variance shows the link is far from saturation, so the video encoder can maintain or raise its target bitrate to improve visual quality.

GPON (Bursty Cross-Traffic) Even under severe, randomised bursts of 100 Mbps cross-traffic (exponential on-off model), GPON maintains stationarity. AQM and high link speeds drain transient queues quickly, preventing structural delay motifs from forming.

Congestion control should treat these micro-transients as noise, not as a signal to cut throughput. Maintaining a steady bitrate is correct here, since the network resolves transient load near-instantaneously.

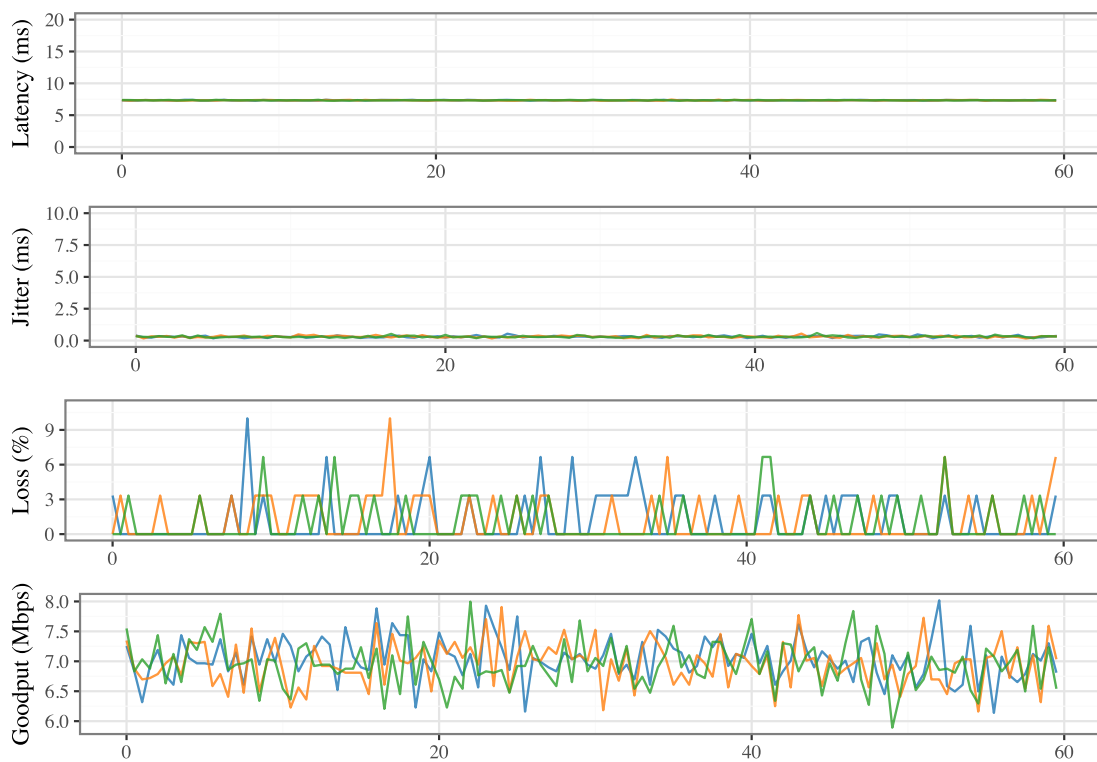


Figure 66. Raw Monte Carlo traces for GPON (Idle Baseline). Latency sits flat at propagation delay; all three trials are nearly indistinguishable, confirming structural stationarity.

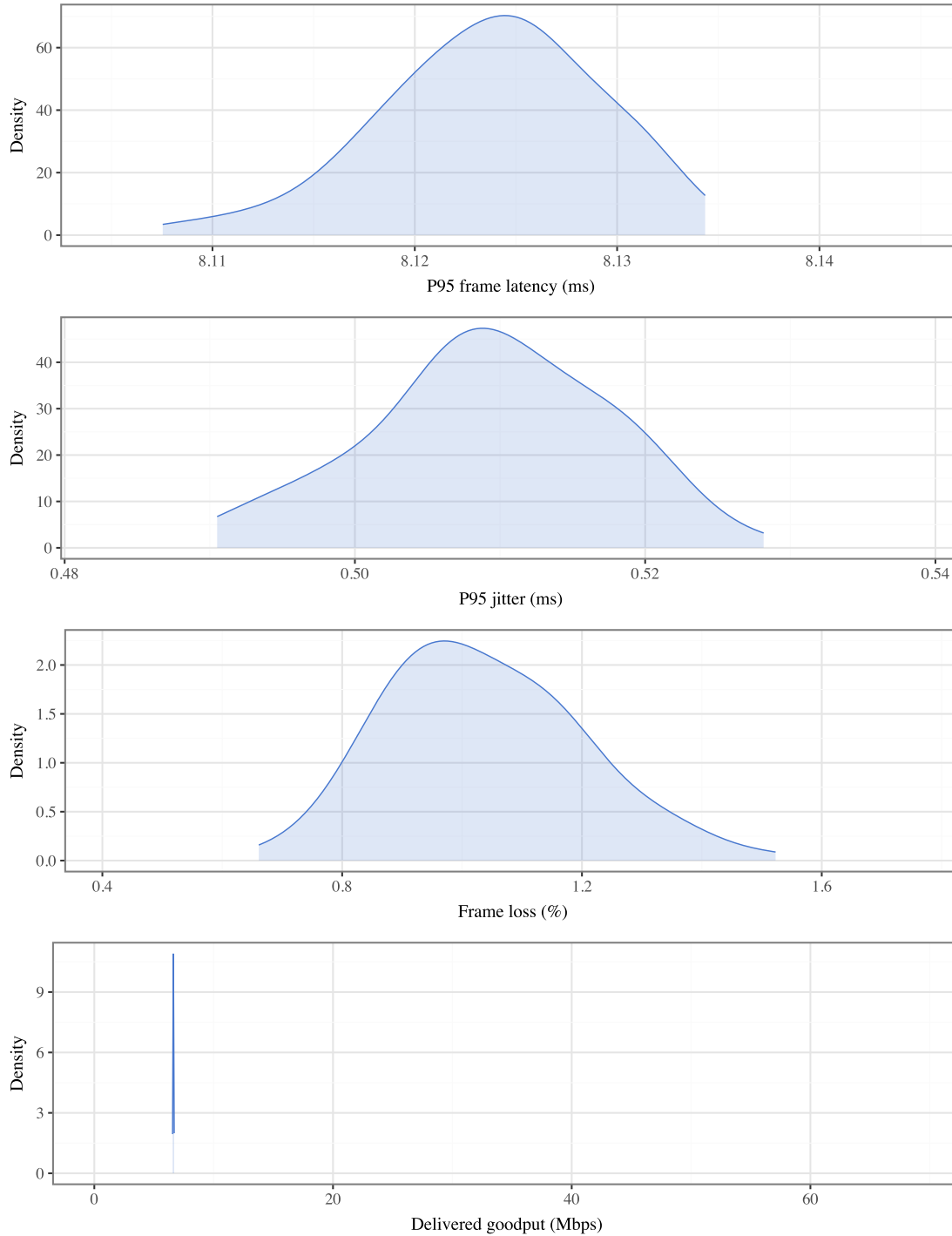


Figure 67. Monte Carlo outcome distributions (n=100) for GPON (Idle Baseline). KDE plots summarize inter-trial variability in key QoE-facing metrics.

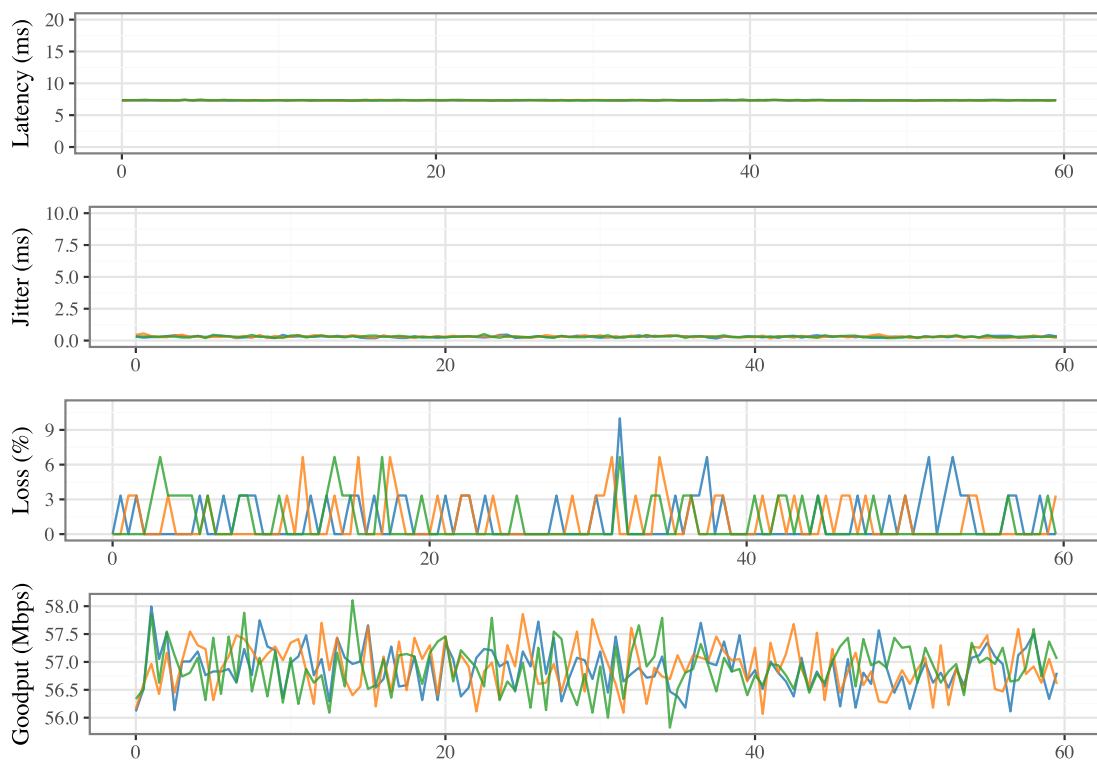


Figure 68. Raw Monte Carlo traces for GPON (Constant Cross-Traffic). The 50 Mbps background load produces no visible deviation from the idle baseline—fibre capacity absorbs the cross-traffic without forming queues.

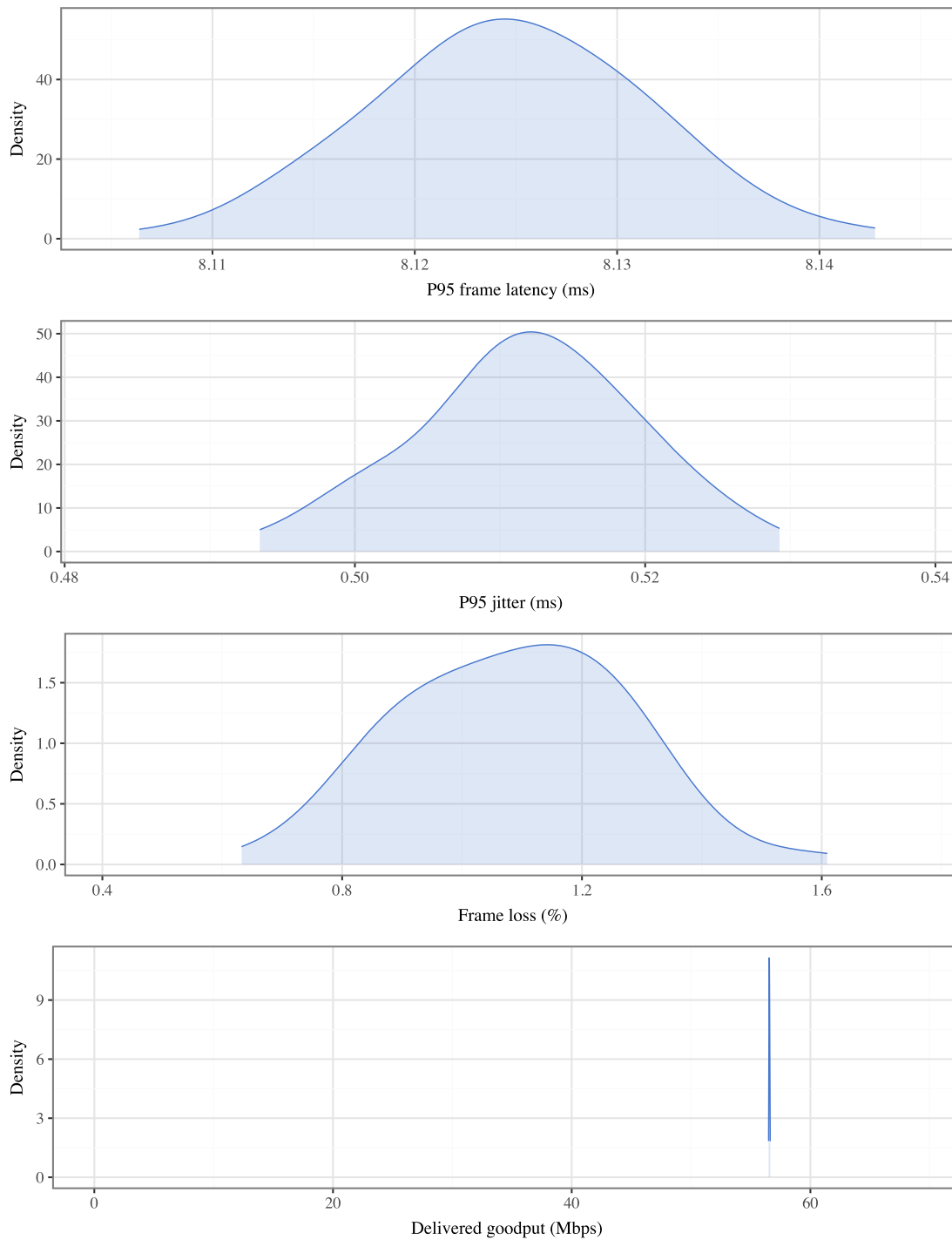


Figure 69. Monte Carlo outcome distributions (n=100) for GPON (Constant Cross-Traffic). Even with background load, distributions remain tight, reflecting low queuing sensitivity under headroom.

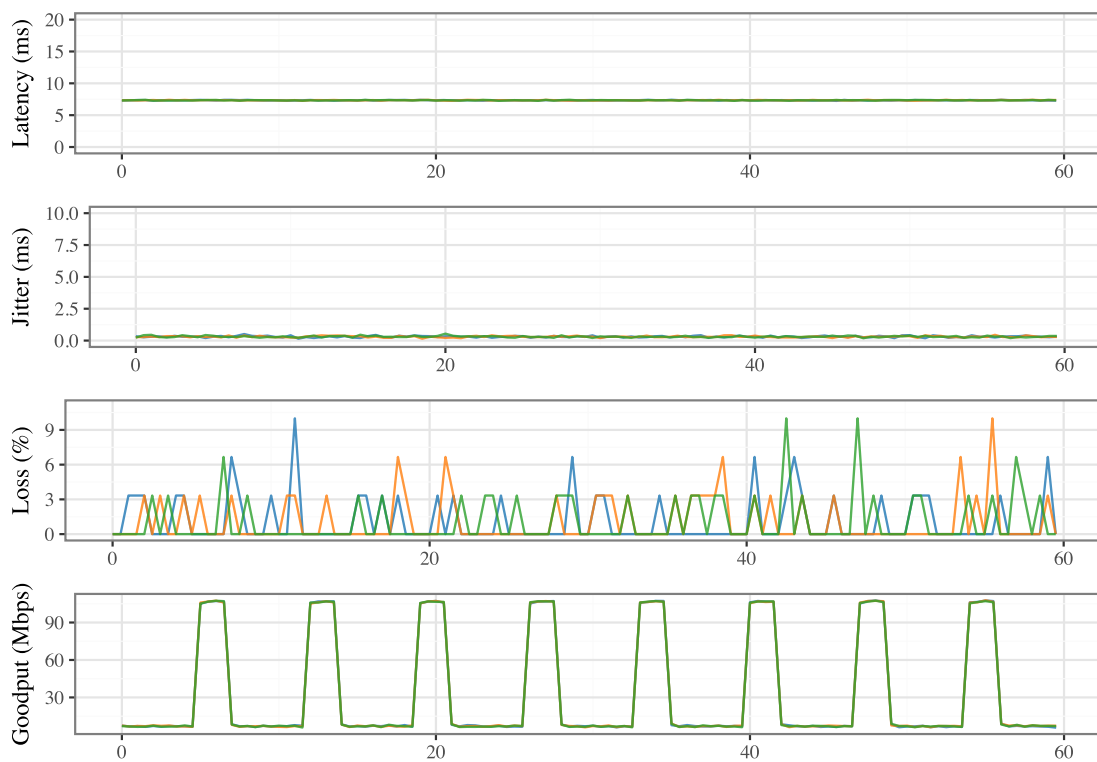


Figure 70. Raw Monte Carlo traces for GPON (Bursty Cross-Traffic). Even 100 Mbps exponential bursts fail to break stationarity; micro-transients drain within the 0.5 s aggregation bin, leaving the trace visually flat.

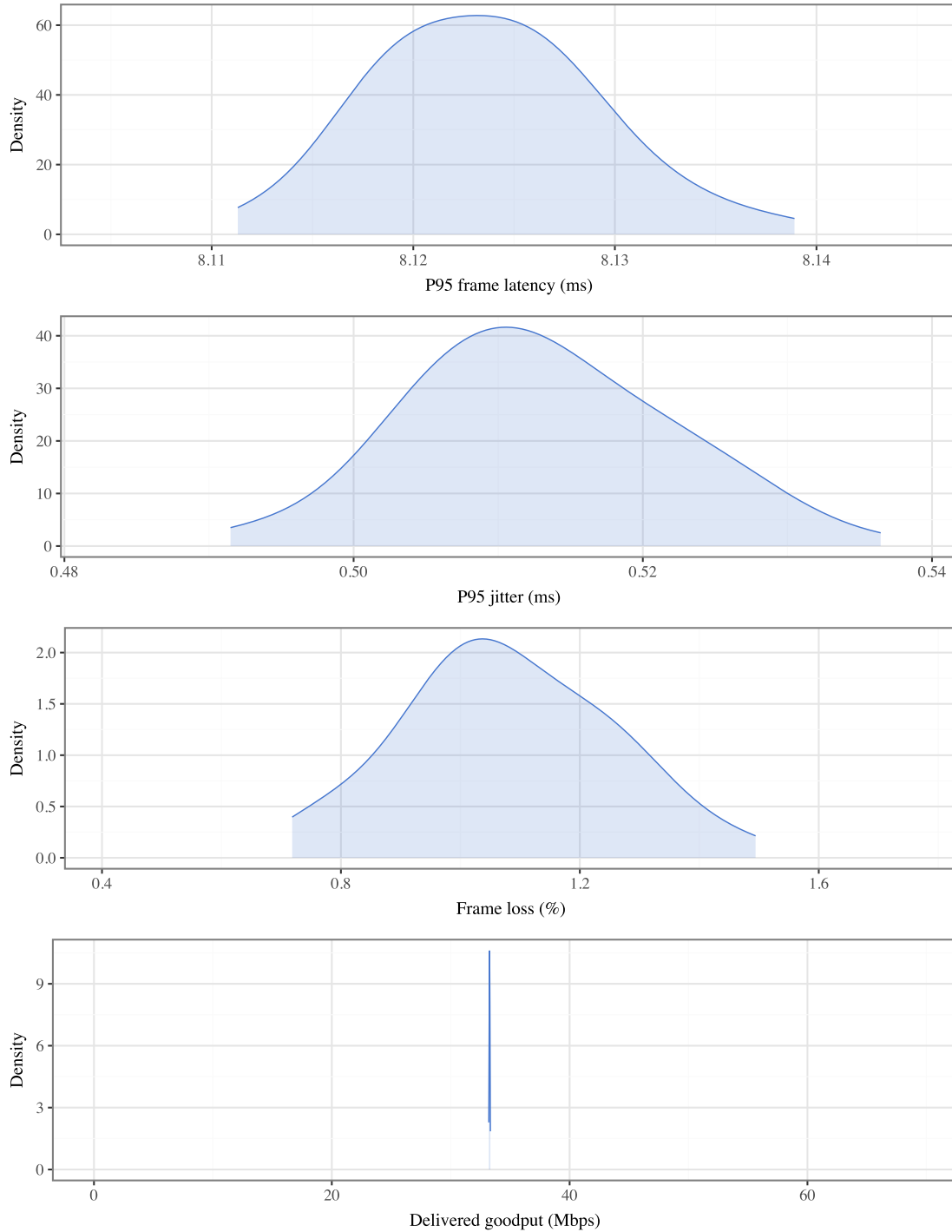


Figure 71. Monte Carlo outcome distributions (n=100) for GPON (Bursty Cross-Traffic). Bursts can widen jitter and loss distributions when short-lived queues form, even if median behavior remains stable.

14.19.2 DOCSIS 3.0 & 3.1: Bufferbloat vs Active Queue Management

DOCSIS networks deliver broadband over Hybrid Fiber-Coaxial (HFC) infrastructure. The critical distinction is upstream queue management. Legacy DOCSIS 3.0 modems feature deep, unmanaged FIFO buffers that, when saturated, create the ‘bufferbloat’ delay motif. DOCSIS 3.1 mandates AQM (PIE), which controls queue depth and fundamentally alters the delay signature.

DOCSIS 3.0 (Idle Baseline) Without heavy background traffic, a DOCSIS 3.0 cable modem provides a stable, stationary baseline similar to fibre. The physical propagation over the HFC network adds a few milliseconds of delay, but jitter and loss remain minimal.

Because the deep FIFO queue is hidden during idle periods, an inference engine may misclassify this link as fibre. Only under stress does the true topological signature emerge, underscoring the need for continuous monitoring rather than snapshot analysis.

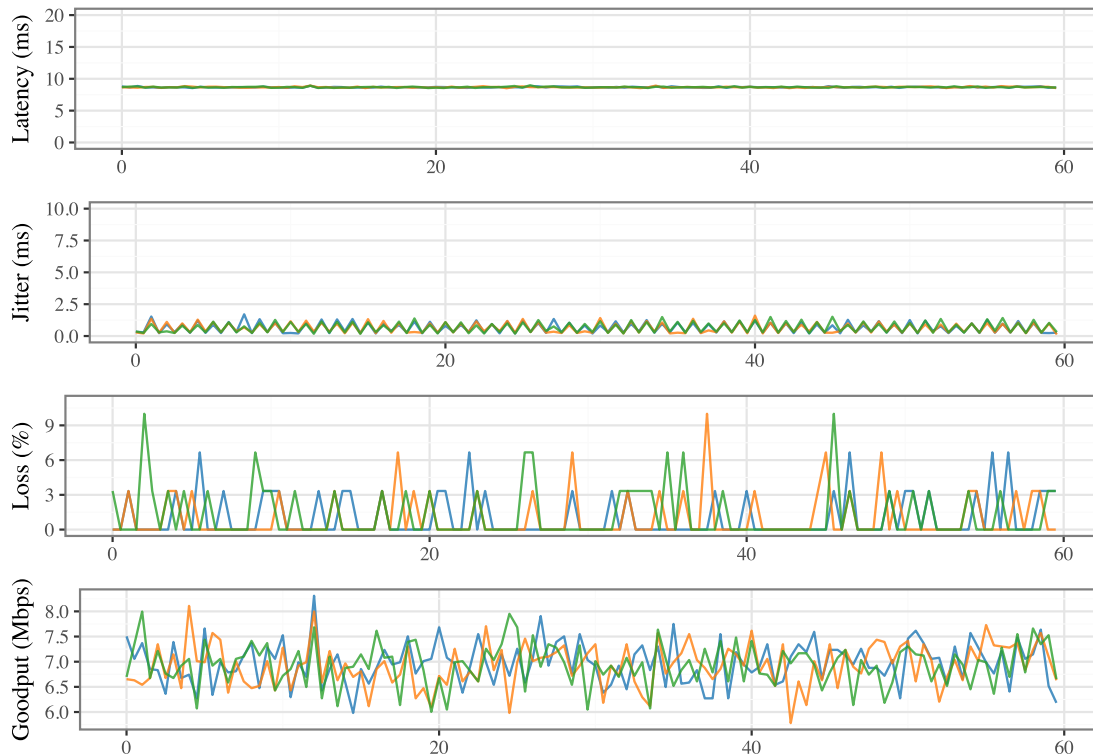


Figure 72. Raw Monte Carlo traces for DOCSIS 3.0 (Idle Baseline). The deep FIFO queue is hidden during idle; the stable baseline is indistinguishable from fibre, masking the bufferbloat vulnerability.

DOCSIS 3.0 (Light Bufferbloat) As moderate background load (80 Mbps) approaches the 100 Mbps link capacity, the unmanaged queue accumulates packets and a structural latency ramp emerges. The three overlaid traces show micro-stuttering jitter as the queue oscillates, but the upward latency trajectory is clear.

On detecting this ramp, an application should pace its traffic—lowering the target video bitrate below the saturation point prevents the queue from filling further, stabilising latency before a tail-drop occurs.

DOCSIS 3.0 (Severe Bufferbloat) Under severe saturation (150 Mbps), the classic bufferbloat sawtooth dominates. Latency climbs linearly to hundreds of milliseconds as the deep queue fills; overflow triggers tail-drop, forcing TCP background flows to back off. The queue drains, latency resets, and the cycle repeats. The three overlaid traces diverge near the peak, reflecting the probabilistic timing of TCP backoffs.

Mitigating severe bufferbloat requires aggressive application-layer intervention. The inference engine must signal the video encoder to cut bitrate immediately, helping drain the modem queue. Delay-based congestion control (BBR) rather than loss-based (Reno/Cubic) is essential on these links.

DOCSIS 3.1 (Idle Baseline) The DOCSIS 3.1 baseline is identical to 3.0 under idle conditions. Latency is low and stable; jitter is negligible.

As with DOCSIS 3.0, the idle state requires no specific mitigation from the application, which should optimize for low-latency interactive delivery.

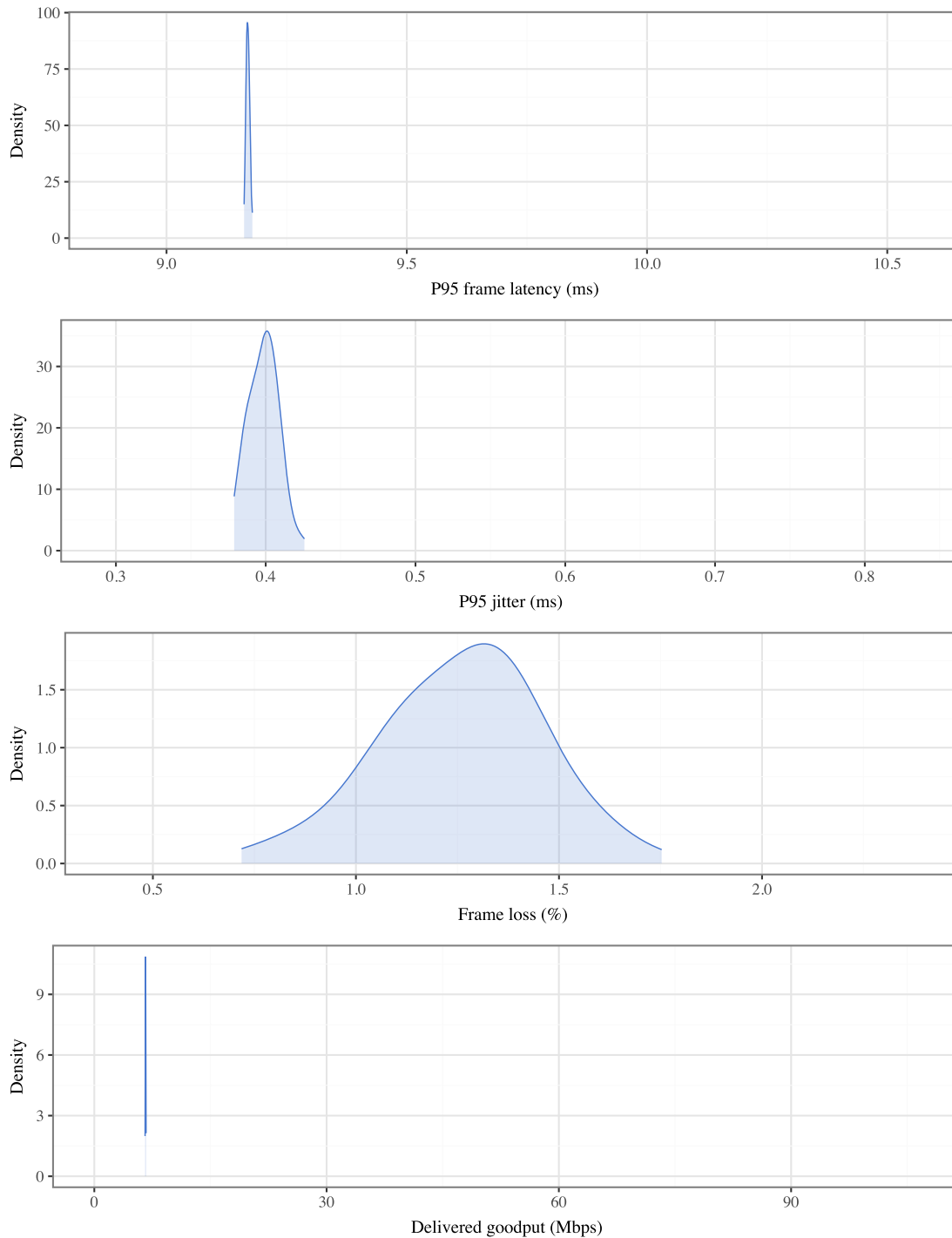


Figure 73. Monte Carlo outcome densities ($n=100$) for DOCSIS 3.0 (Idle Baseline). Tight distributions confirm low inter-trial variance when the buffer is empty.

DOCSIS 3.1 (AQM Under Load) Despite the same extreme background load that crippled DOCSIS 3.0, the 3.1 trace stays flat. PIE drops a small fraction of packets before the queue fills, signalling TCP to slow early and preventing the bufferbloat ramp. The trade-off is consistent, low-level managed packet loss.

The optimal mitigation is structural: the application must use FEC to mask AQM-induced drops. Because latency remains bounded, the application need not cut its bitrate, but it must tolerate continuous, scattered frame loss.

14.20 DSL: Interleaving and Impulse Noise Performance

DSL remains a significant last-mile technology for millions of users. Our updated simulation model captures the unique impairment modes of copper-based access.

- **Baseline (VDSL2):** With a 50 Mbps capacity and 64-symbol interleaving depth, the baseline P_{95} latency is approximately 24.2 ms. This reflects the fixed interleaver delay penalty (16 ms) plus minimal queuing.

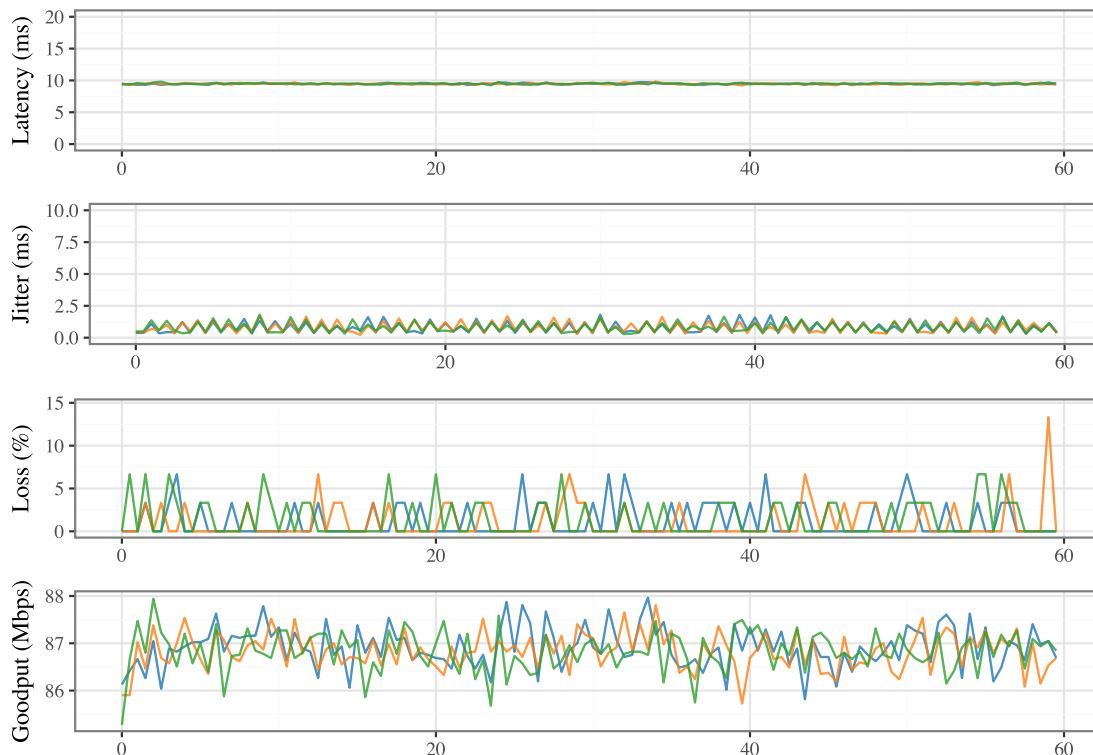


Figure 74. Raw Monte Carlo traces for DOCSIS 3.0 (Light Bufferbloat). A structural latency ramp emerges as the unmanaged queue fills under 80 Mbps load, with micro-stuttering jitter from queue oscillations.

- **Impulse Noise Burst:** Simulating a Middleton Class A impulse noise event ($p_{gb} = 0.05, p_{bg} = 0.1$) results in severe burst loss (58% frame loss) without significantly affecting the P_{95} latency of delivered frames. This confirms that impulse noise is a loss-dominant impairment.
- **Crosstalk (SNR Reduction):** Simulating severe Far-End Crosstalk (FEXT) with a crosstalk factor of 0.6 reduces effective capacity, causing P_{95} latency to inflate to 35.8 ms under identical load.

In-home wireless links are often the bottleneck in end-to-end delivery. IEEE 802.11 has evolved from chaotic, contention-based access toward scheduled, deterministic transmission. Wi-Fi 5 relies on CSMA/CA, which introduces extreme transient jitter under load. Successive generations added OFDMA in Wi-Fi 6, greenfield 6 GHz spectrum in Wi-Fi 6E, and Multi-Link Operation (MLO) in Wi-Fi 7 to eliminate this variance.

Insight:
DSL's fixed interleaver delay creates a stable but elevated latency floor. Applications must distinguish this from bufferbloat to avoid unnecessary bitrate reductions; FEC is the primary mitigation for impulse-noise-induced loss.

Wi-Fi 5 (Low Contention) With only 2 stations active, Wi-Fi 5 (802.11ac) provides a stable link. However, CSMA/CA mediation of the shared channel means occasional collisions force exponential backoffs, creating high-frequency, low-amplitude transient spikes in jitter and latency—visible as ‘fuzz’ on the trace.

A small, responsive adaptive jitter buffer (5–10 ms) absorbs these MAC-layer waits without adding perceptible delay to the user’s input loop.

Wi-Fi 5 (Severe Congestion) With 16 stations contending, CSMA/CA breaks down. The trace shows violent, high-frequency spikes in jitter and latency alongside erratic packet loss. Unlike the slow, predictable DOCSIS ramp, this motif is noisy and unpredictable. The three overlaid traces diverge widely, reflecting the non-deterministic nature of exponential-backoff collisions.

Applications on congested Wi-Fi 5 must adopt a defensive posture: heavy FEC to recover dropped frames, expanded jitter buffers to handle arrival-time swings, and possibly halved frame rate (60 fps to 30 fps) to reduce the number of packets contending for airtime.

Wi-Fi 6 (OFDMA, Low Contention) Wi-Fi 6 (802.11ax) introduces OFDMA, letting the access point schedule transmissions across sub-carriers instead of relying on contention alone. Under low load, this reduces baseline jitter compared to Wi-Fi 5, producing a cleaner, tighter latency profile.

With OFDMA managing airtime, the application can tighten its jitter buffers and operate with minimal overhead, approaching wired-connection performance.

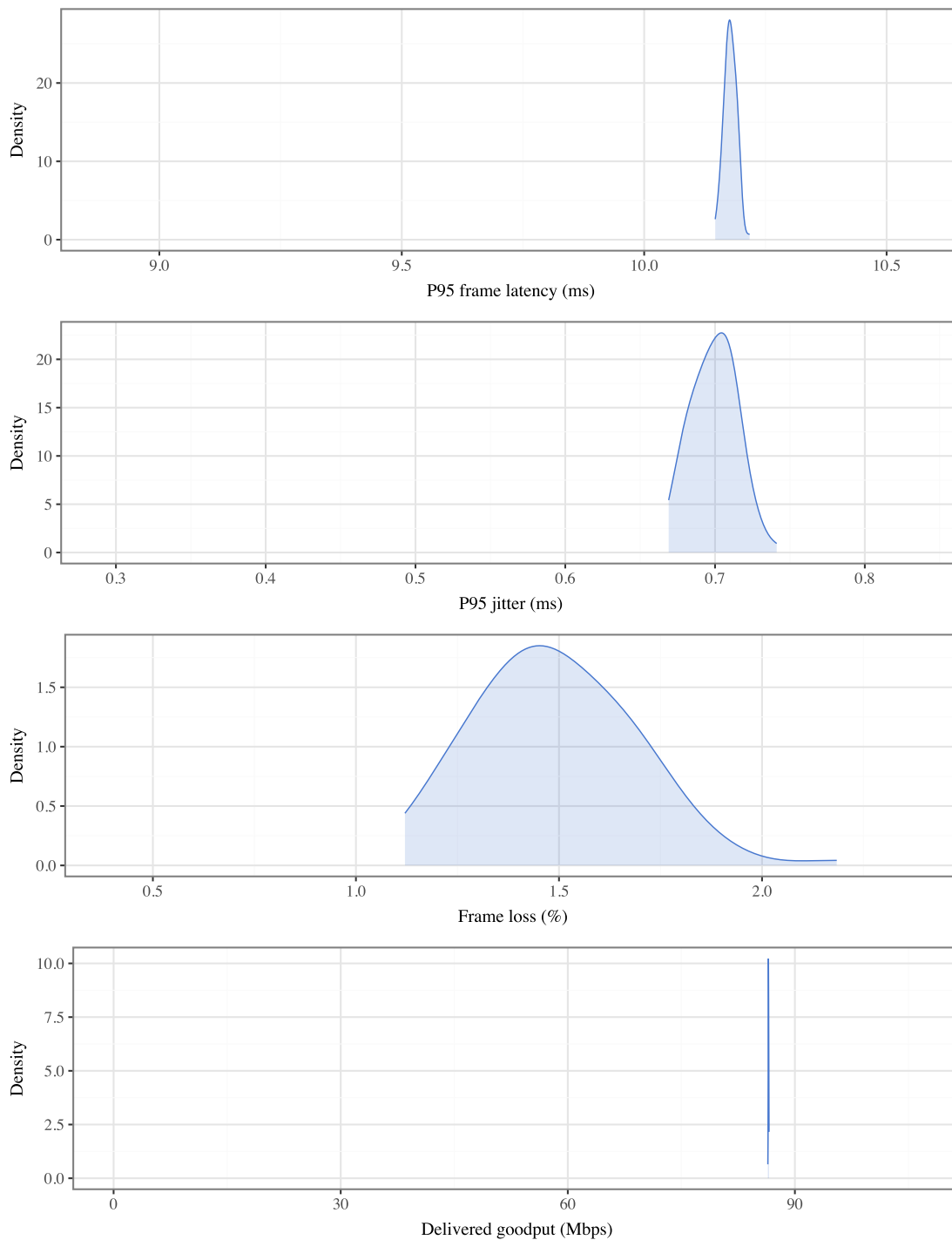


Figure 75. Monte Carlo outcome densities ($n=100$) for DOCSIS 3.0 (Light Bufferbloat). The latency distribution shifts right and widens as queue depth grows, while loss remains near zero.

Wi-Fi 6 (OFDMA, Severe Congestion) Under severe load, Wi-Fi 6 still experiences contention, especially with legacy devices present. OFDMA scheduling prevents the extreme spikes seen in Wi-Fi 5: latency bounds stay tighter, and jitter, while elevated, does not devolve into chaos.

The application still needs jitter buffering and some FEC, but the bounds are more predictable than Wi-Fi 5, allowing higher sustained bitrates even during peak household usage.

Wi-Fi 6E (Clean 6 GHz Band) Wi-Fi 6E operates in the greenfield 6 GHz band, free of legacy contention. The trace shows neither the high-frequency fuzz of Wi-Fi 5 nor the load-induced variance of Wi-Fi 6—it approaches GPON-like stationarity.

Inference engines detecting a Wi-Fi 6E profile can treat it as a fibre connection: disable FEC, minimise buffers, and maximise throughput.

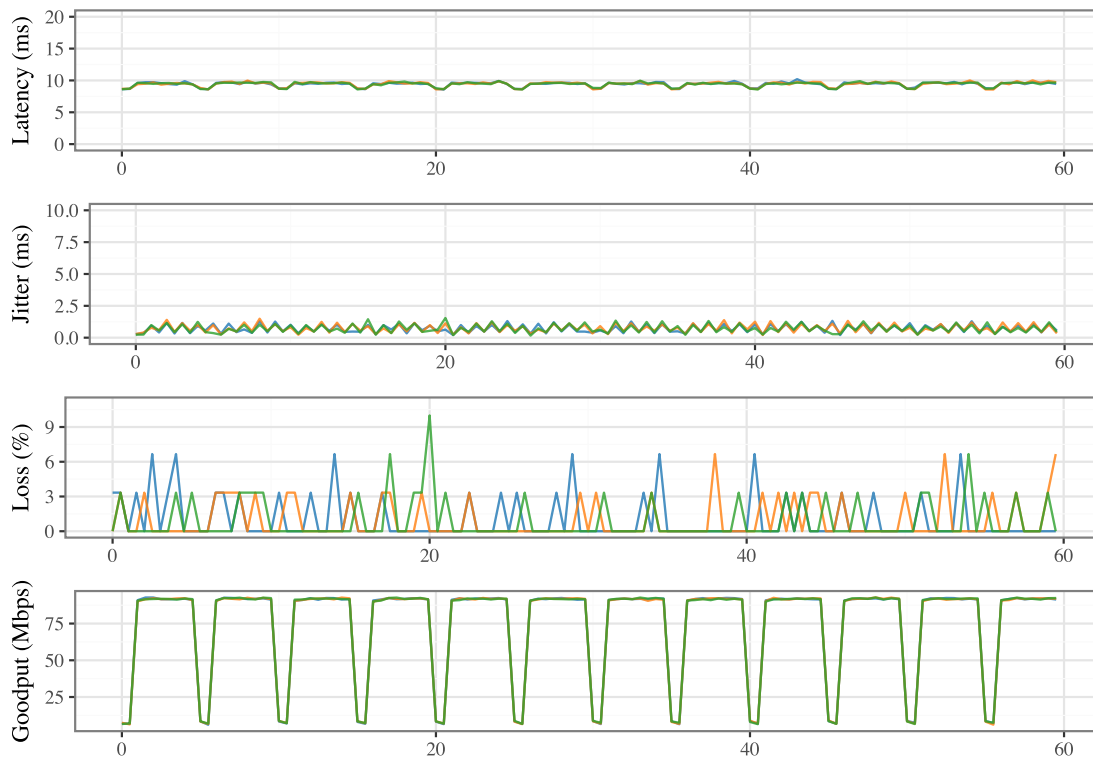


Figure 76. Raw Monte Carlo traces for DOCSIS 3.0 (Severe Bufferbloat). The classic sawtooth emerges: latency ramps to hundreds of ms, tail-drop triggers TCP backoff, and the cycle repeats with chaotic divergence between trials.

Wi-Fi 7 (Multi-Link Operation Failover) Wi-Fi 7 introduces Multi-Link Operation (MLO), letting devices transmit across 2.4 GHz, 5 GHz, and 6 GHz simultaneously. Even with 100% interference on the primary 5 GHz band, MLO fails over instantly to 6 GHz. The trace shows seamless delivery with zero latency spikes or loss.

This hardware-level resilience frees the application layer from reacting to transient RF interference. MLO abstracts away physical-layer disruption, letting the application maintain aggressive low-latency configurations.

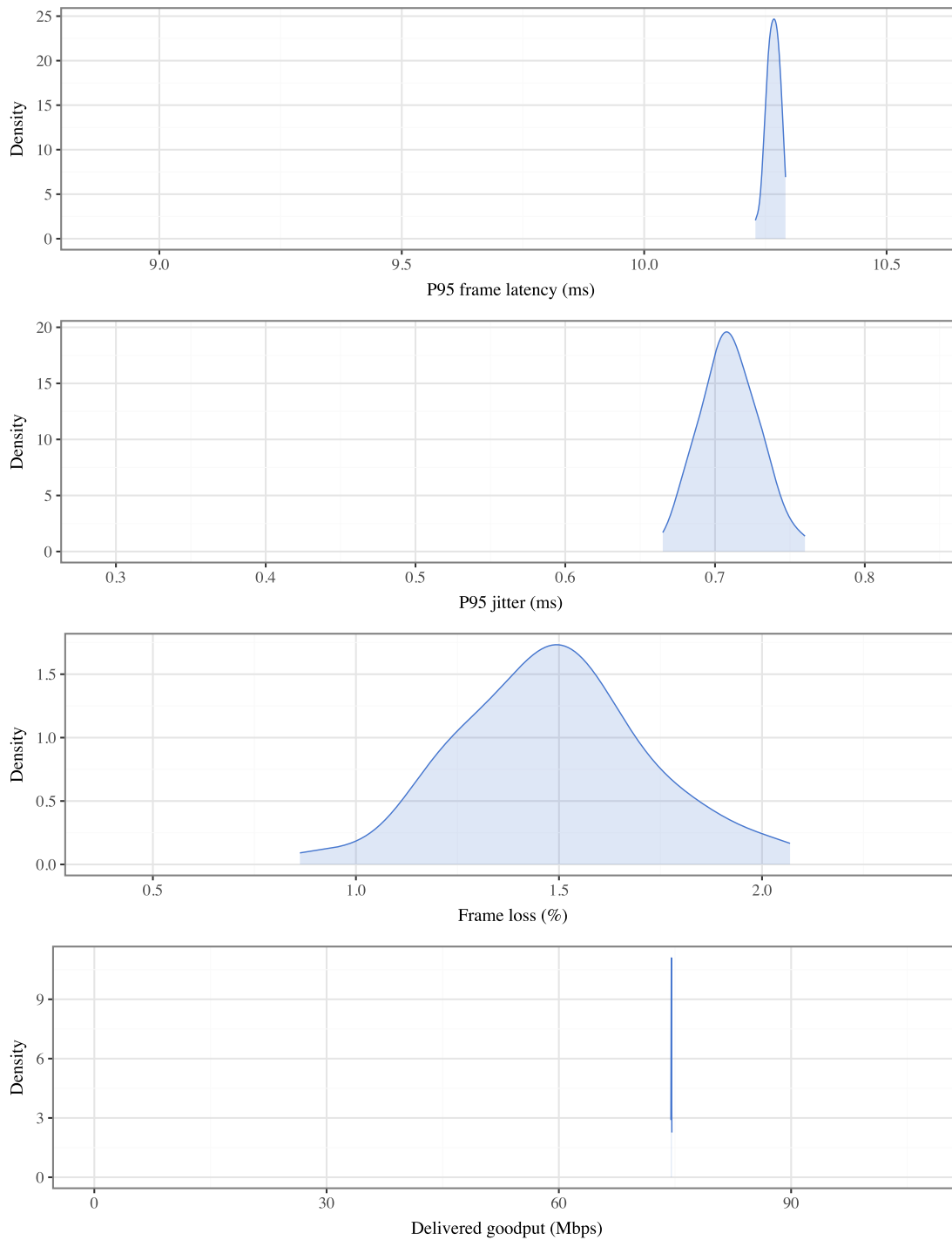


Figure 77. Monte Carlo outcome densities ($n=100$) for DOCSIS 3.0 (Severe Bufferbloat). Latency and jitter distributions become broad and right-skewed; frame loss rises as tail-drops begin.

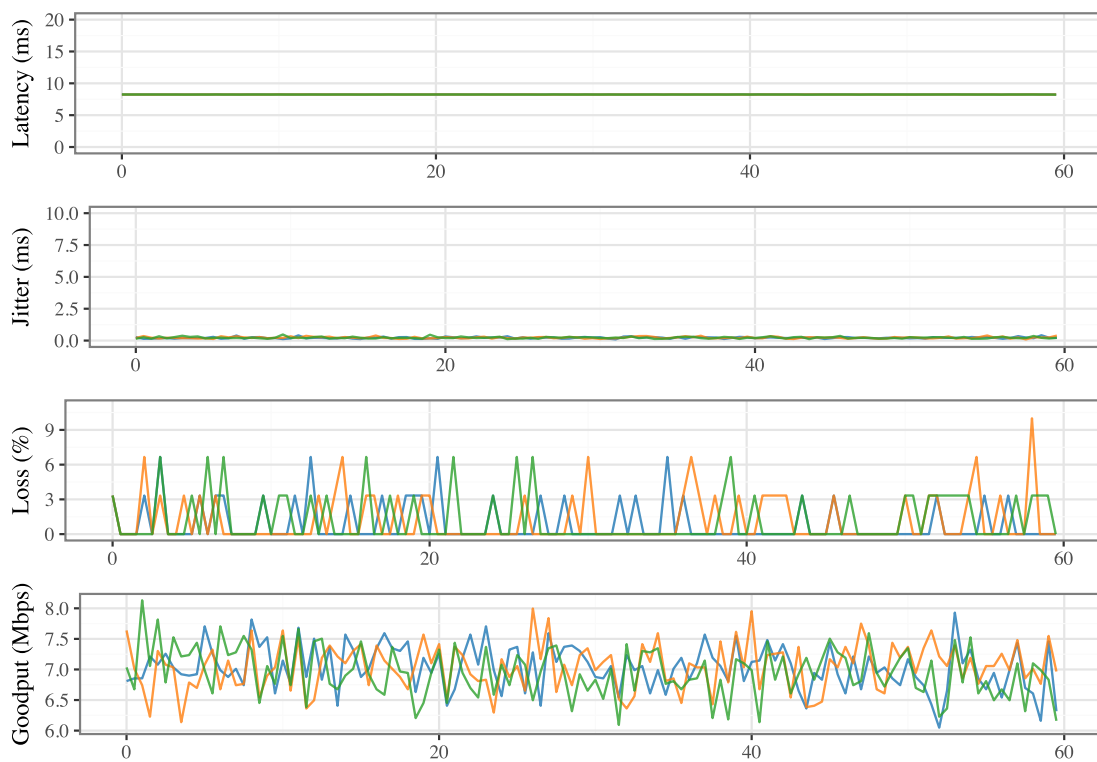


Figure 78. Raw Monte Carlo traces for DOCSIS 3.1 (Idle Baseline). Baseline is virtually identical to DOCSIS 3.0 idle—the AQM advantage is invisible without competing traffic.

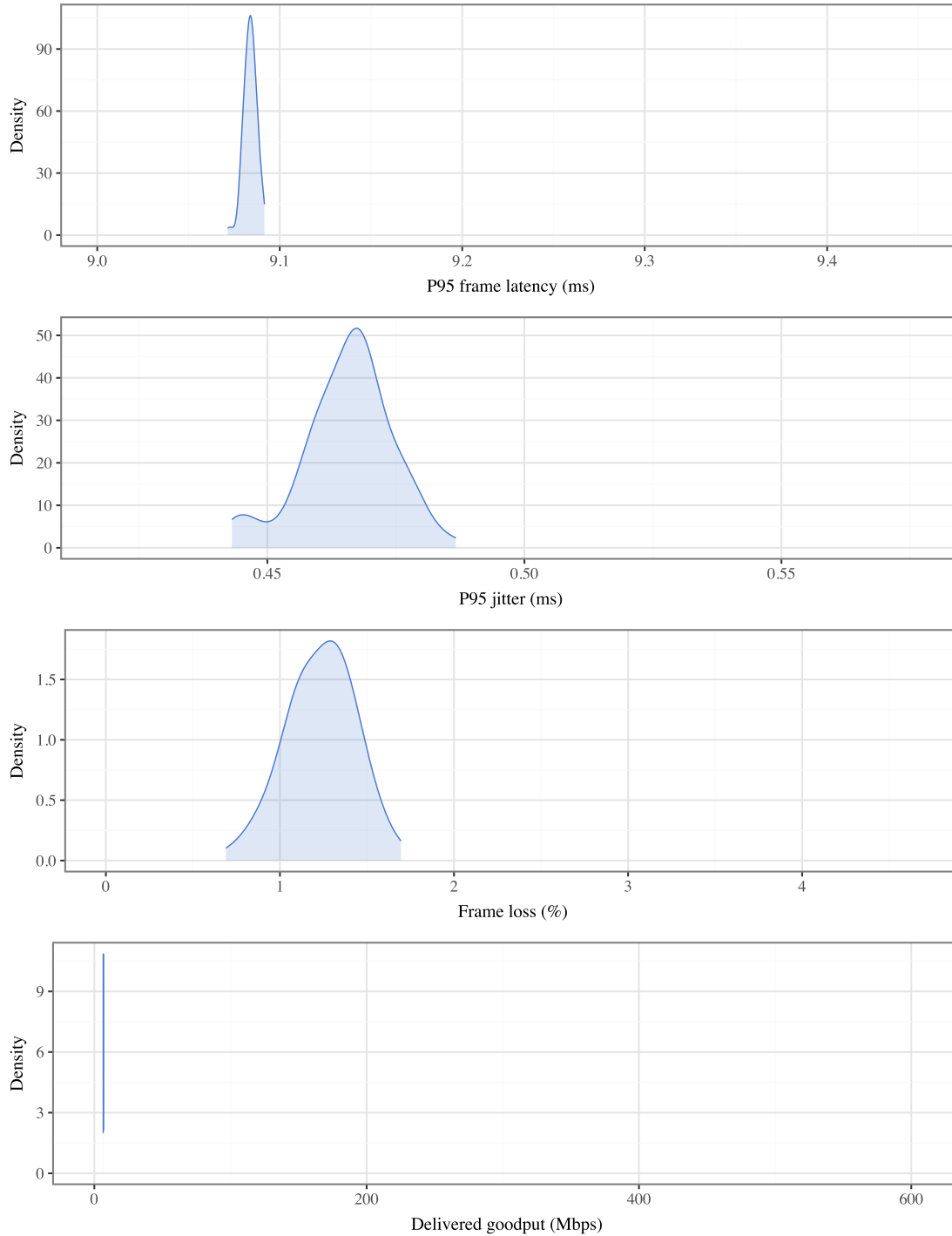


Figure 79. Monte Carlo outcome densities ($n=100$) for DOCSIS 3.1 (Idle Baseline). PIE AQM keeps queue depth near zero; distributions are as tight as GPON.

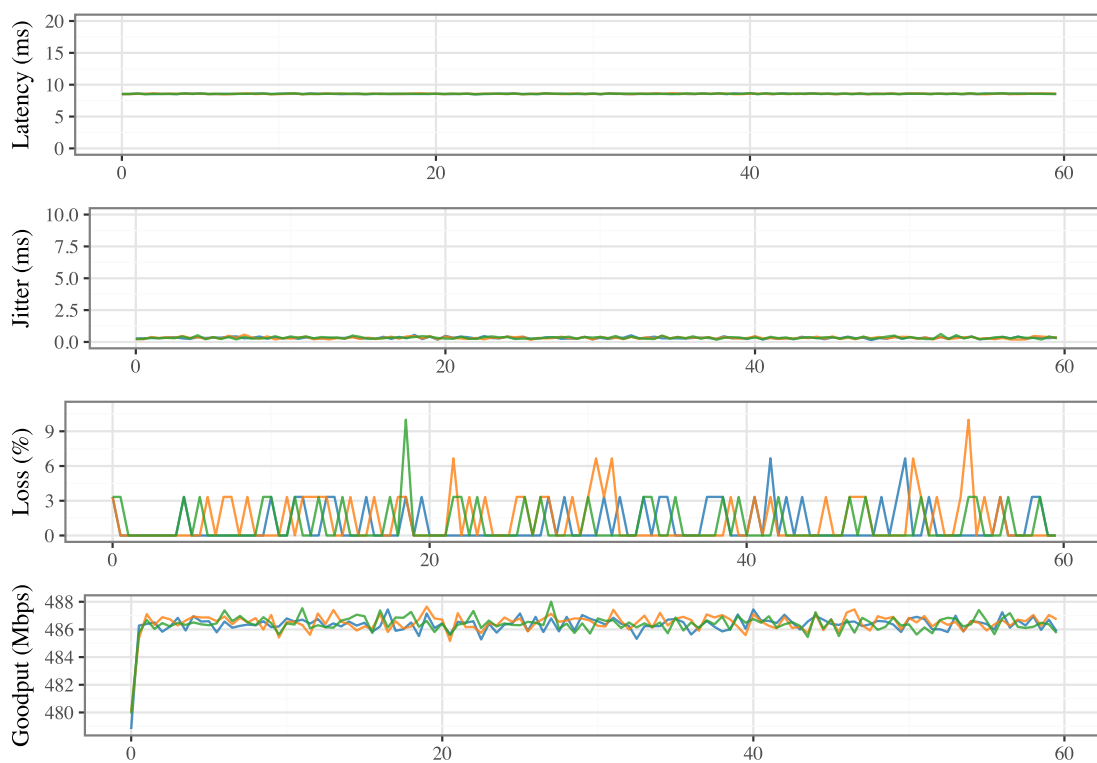


Figure 80. Raw Monte Carlo traces for DOCSIS 3.1 (AQM Under Load). Despite identical extreme load, PIE AQM keeps latency flat by proactively dropping packets—the sawtooth is completely eliminated at the cost of low-level managed loss.

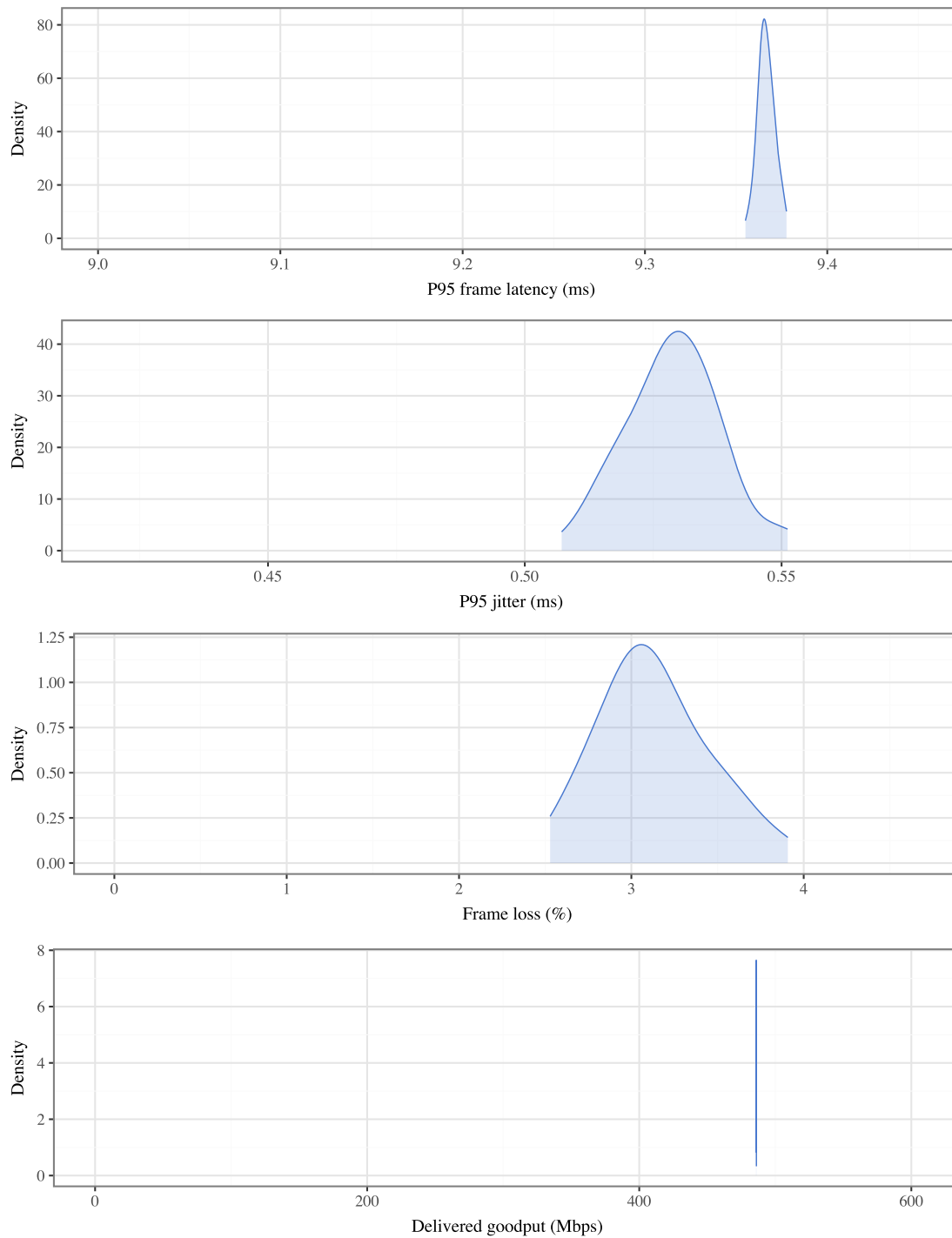


Figure 81. Monte Carlo outcome densities ($n=100$) for DOCSIS 3.1 (AQM Under Load). PIE limits queue growth; latency distributions shift modestly compared to the severe bloat of DOCSIS 3.0.

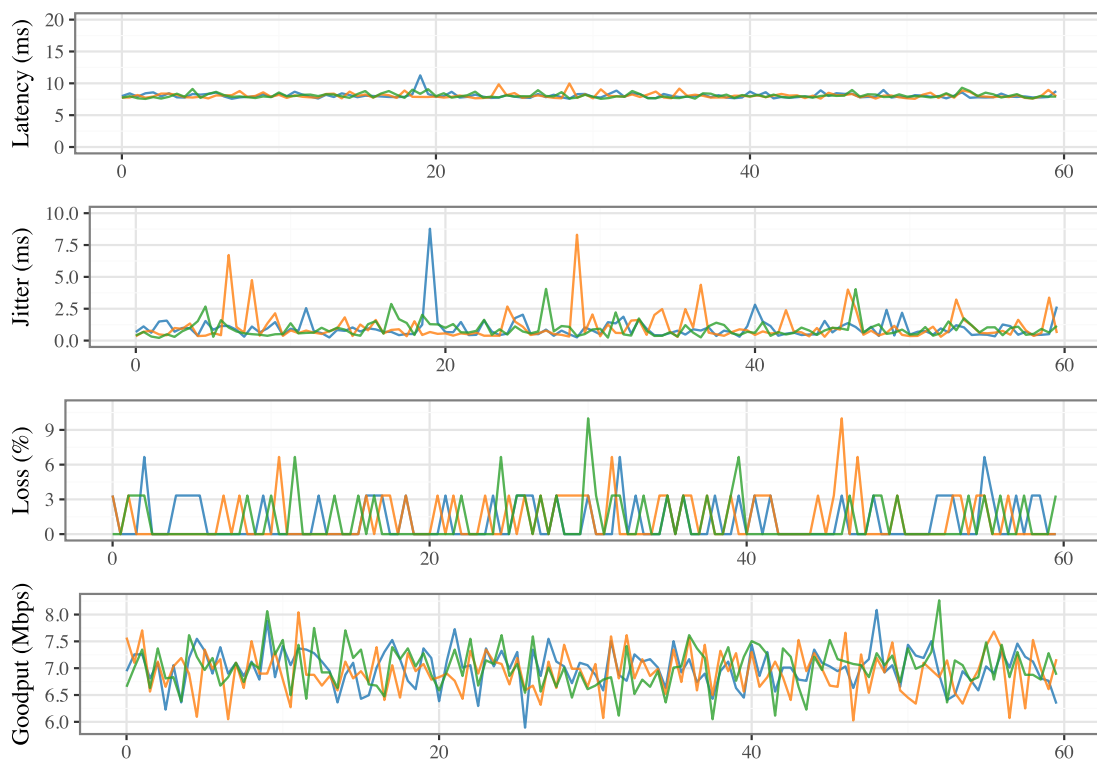


Figure 82. Raw Monte Carlo traces for Wi-Fi 5 (Low Contention). Occasional CSMA/CA backoff collisions produce high-frequency ‘fuzz’ on the latency trace, but amplitude remains bounded under light load.

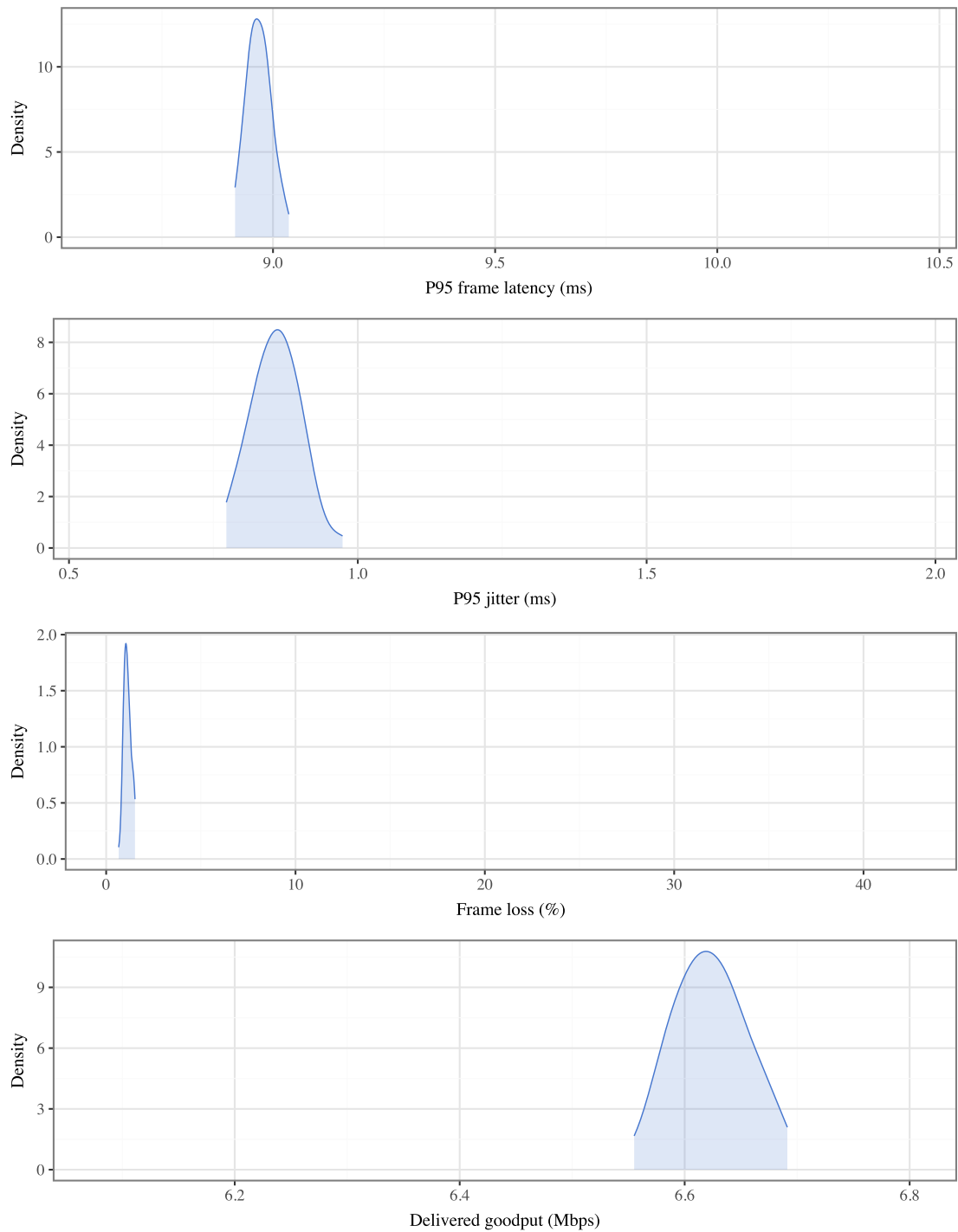


Figure 83. Monte Carlo outcome densities ($n=100$) for Wi-Fi 5 (Low Contention). With only 2 stations, CSMA/CA operates efficiently and distributions remain tight.

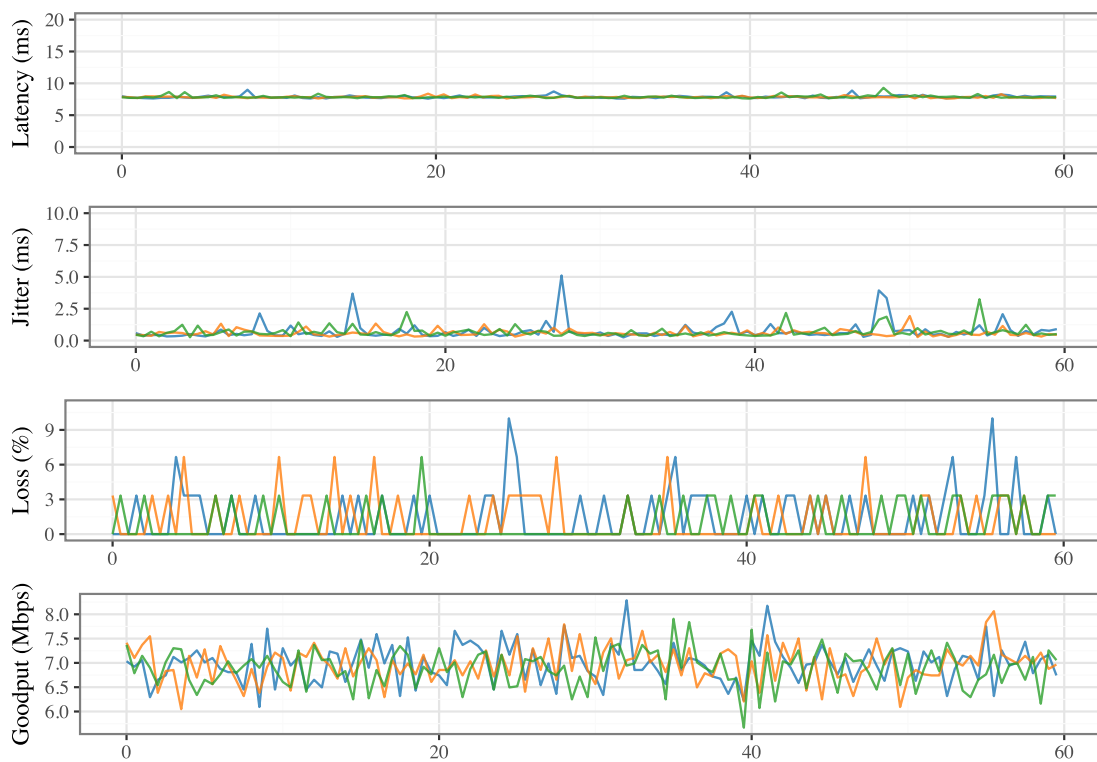


Figure 84. Raw Monte Carlo traces for Wi-Fi 5 (Severe Congestion). CSMA/CA breaks down under 16-station contention: violent, non-deterministic jitter spikes and erratic loss dominate, with trials diverging wildly.

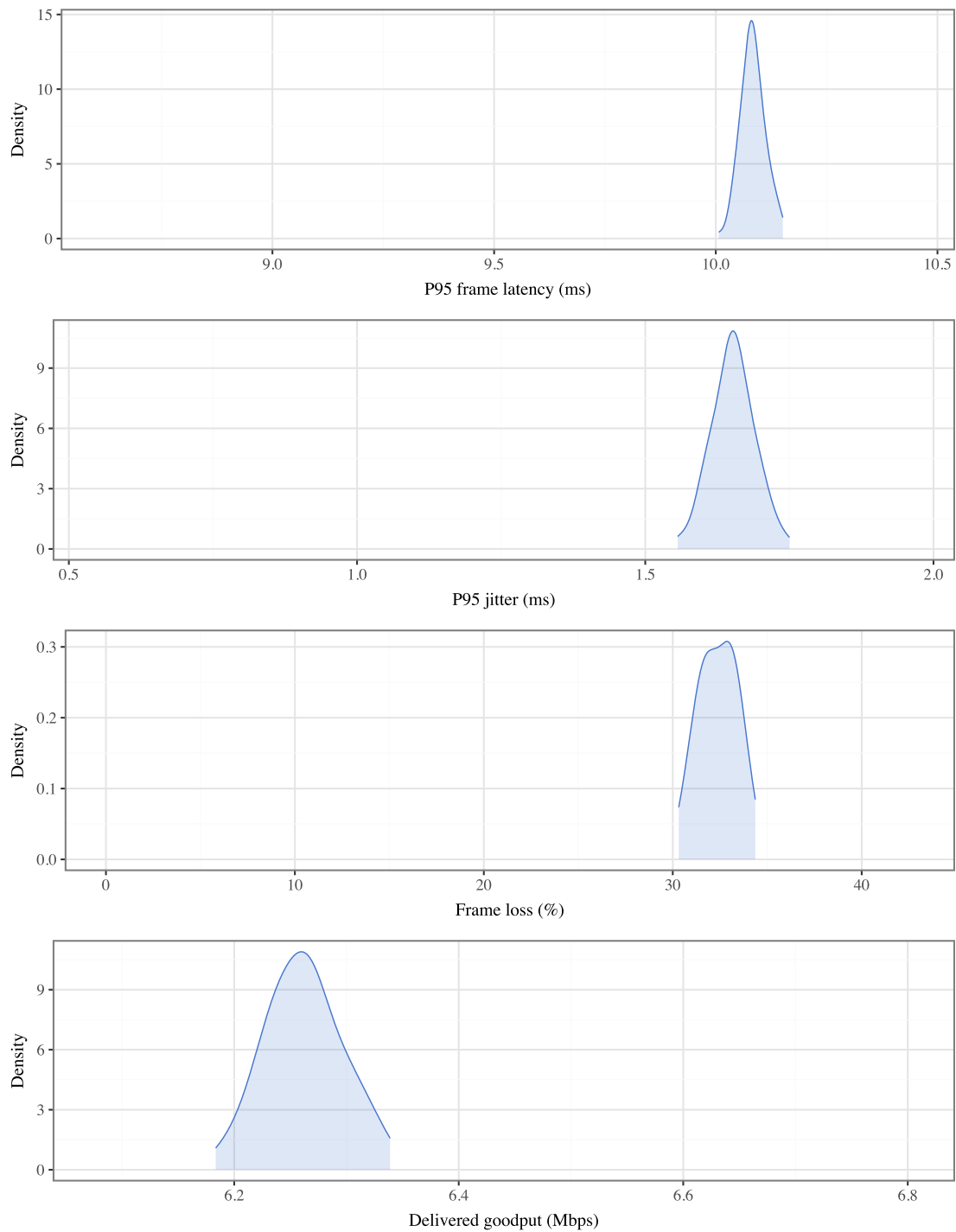


Figure 85. Monte Carlo outcome densities ($n=100$) for Wi-Fi 5 (Severe Congestion). With 16 stations, CSMA/CA breaks down: latency and jitter distributions become broad and heavy-tailed.

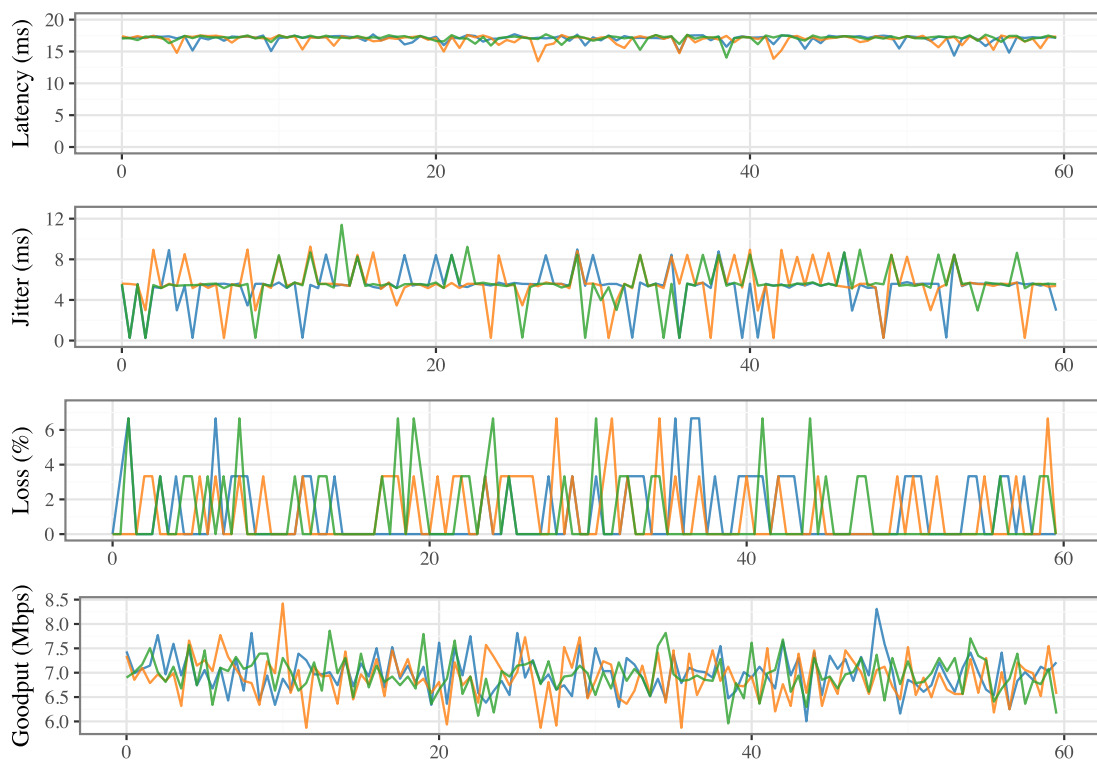


Figure 86. Raw Monte Carlo traces for Wi-Fi 6 (OFDMA, Low Contention). OFDMA scheduling structurally reduces jitter compared to Wi-Fi 5; the trace is tighter with fewer contention-induced outliers.

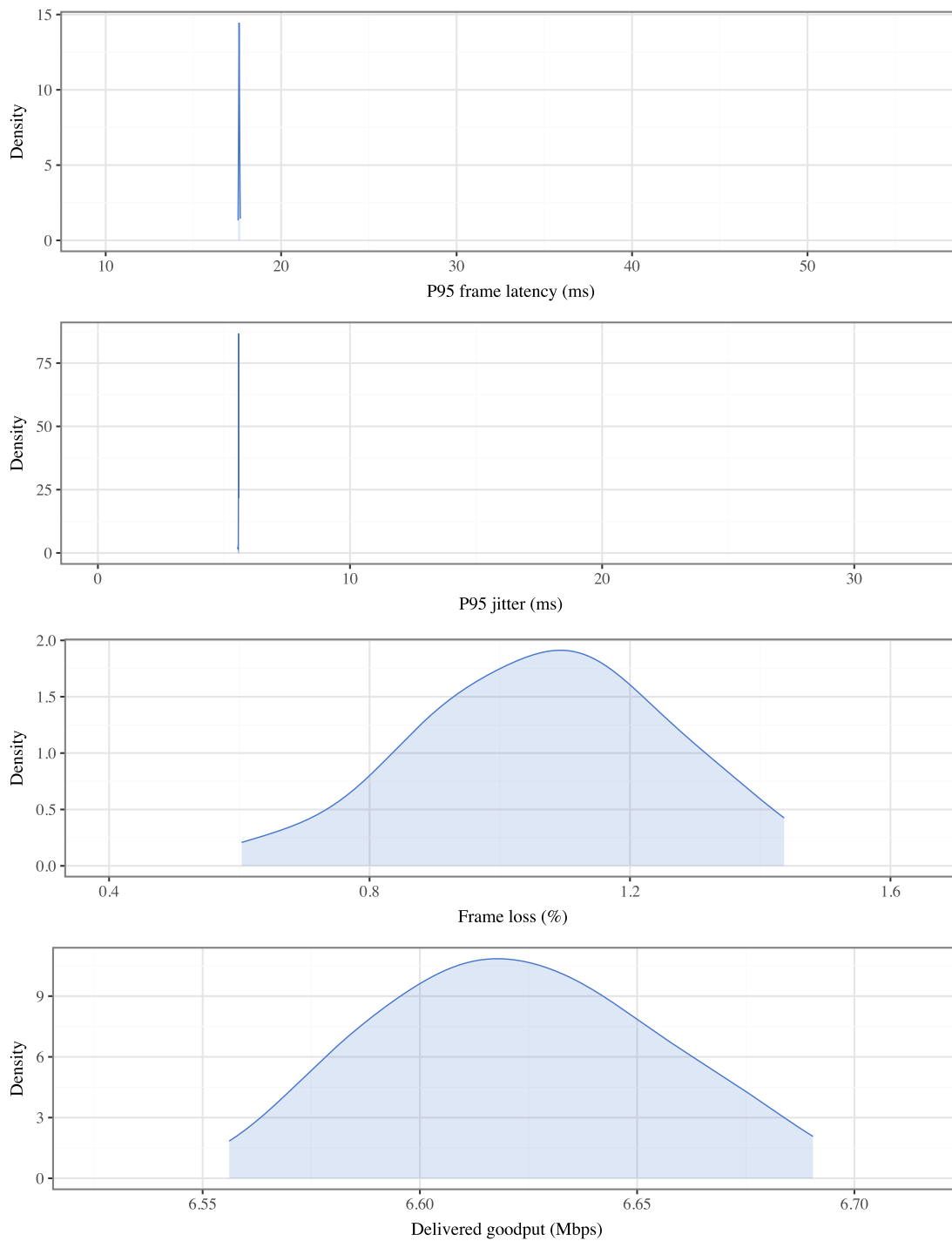


Figure 87. Monte Carlo outcome densities ($n=100$) for Wi-Fi 6 (OFDMA, Low Contention). OFDMA scheduling produces tighter distributions than Wi-Fi 5 at similar station counts.

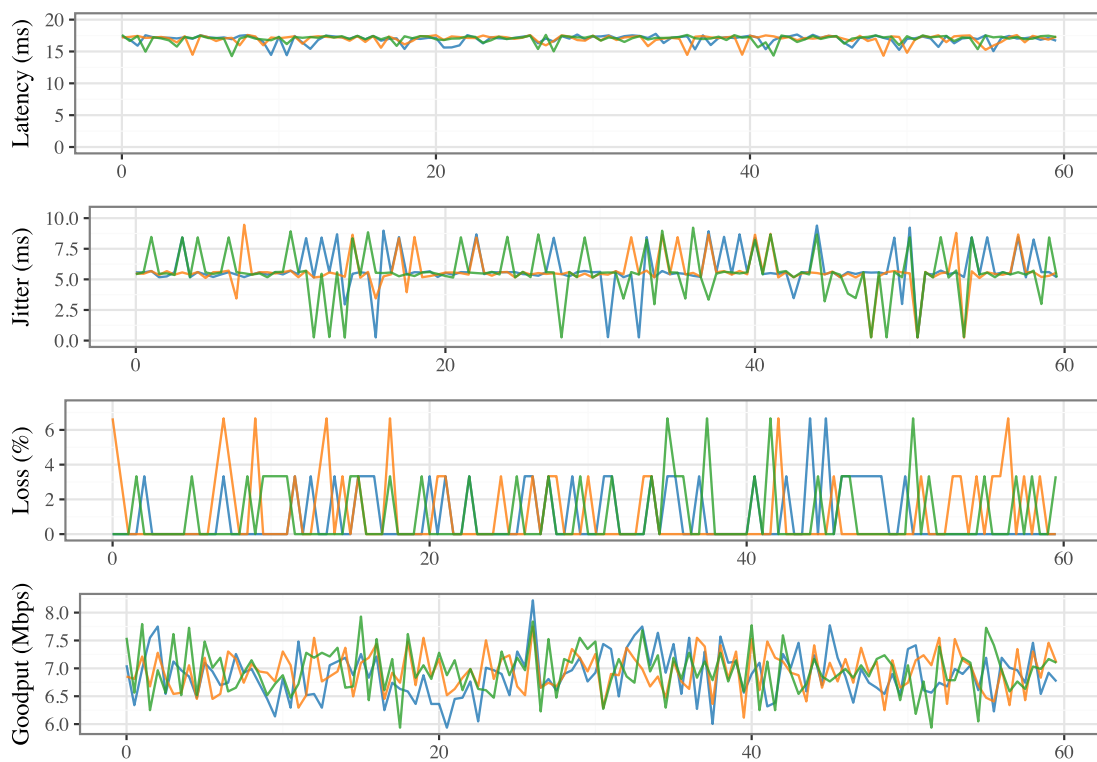


Figure 88. Raw Monte Carlo traces for Wi-Fi 6 (OFDMA, Severe Congestion). Contention is elevated but bounded—OFDMA prevents the extreme chaotic spikes of Wi-Fi 5, maintaining tighter latency envelopes.

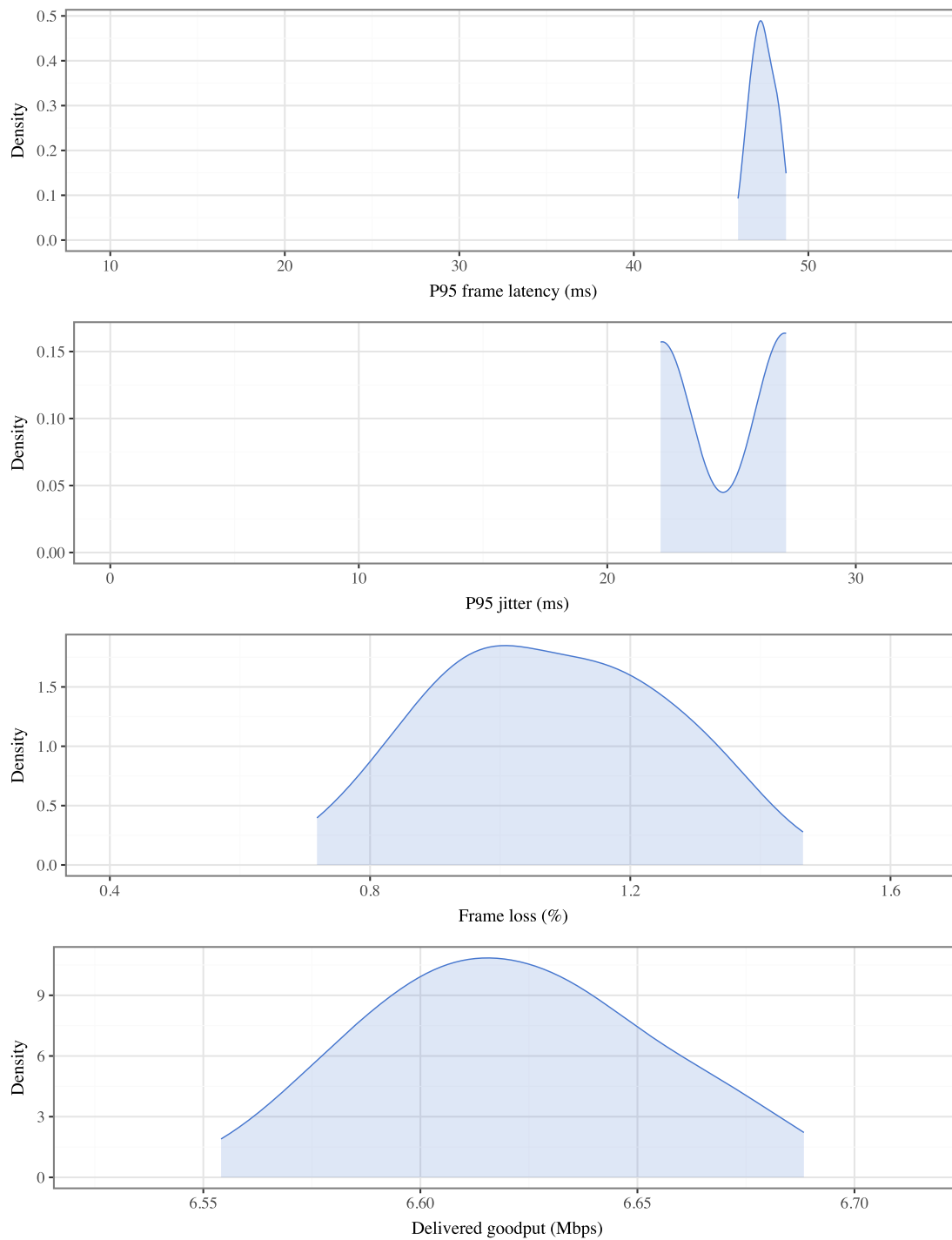


Figure 89. Monte Carlo outcome densities ($n=100$) for Wi-Fi 6 (OFDMA, Severe Congestion). Under heavy load, OFDMA still outperforms Wi-Fi 5 but distributions widen noticeably.

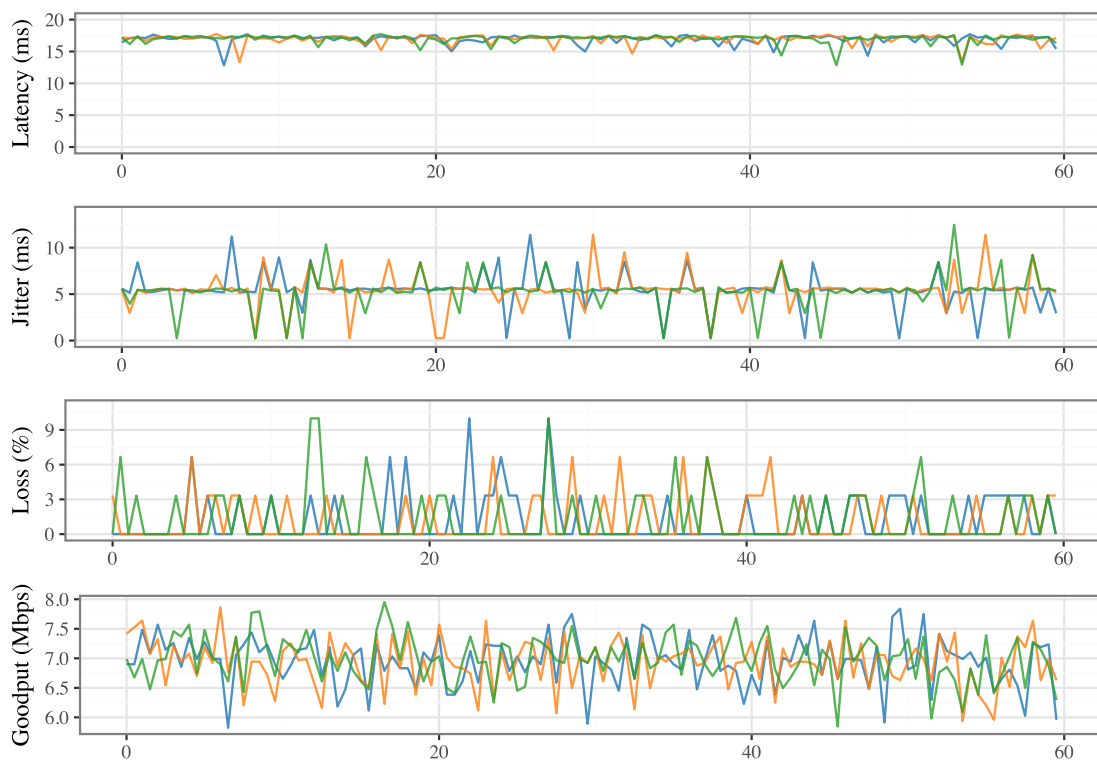


Figure 90. Raw Monte Carlo traces for Wi-Fi 6E (Clean 6 GHz Band). Greenfield 6 GHz spectrum eliminates legacy contention; the trace approaches GPON-like stationarity without MLO's failover resilience.

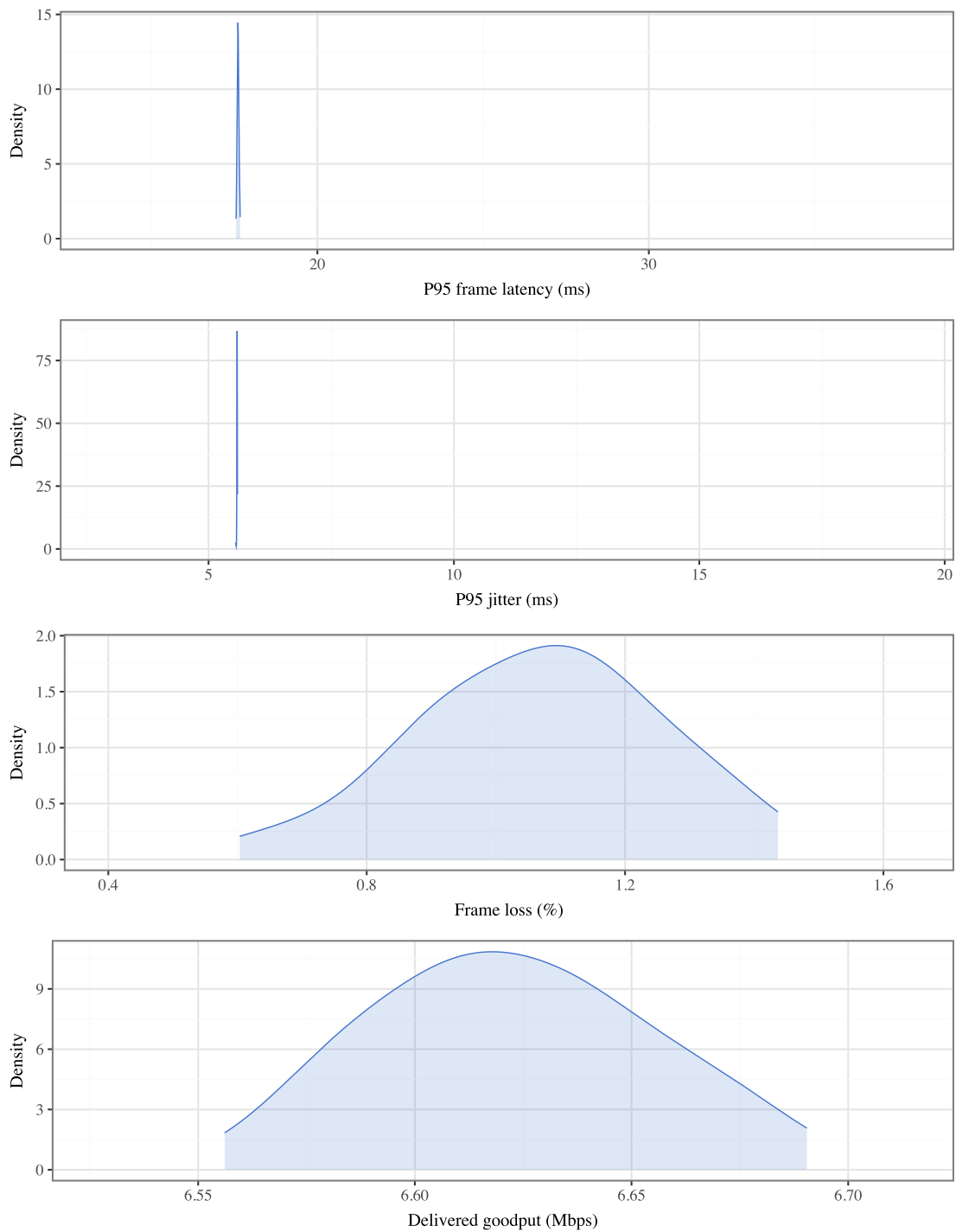


Figure 91. Monte Carlo outcome densities ($n=100$) for Wi-Fi 6E (Clean 6 GHz Band). The greenfield band eliminates legacy contention; distributions approach wired-link tightness.

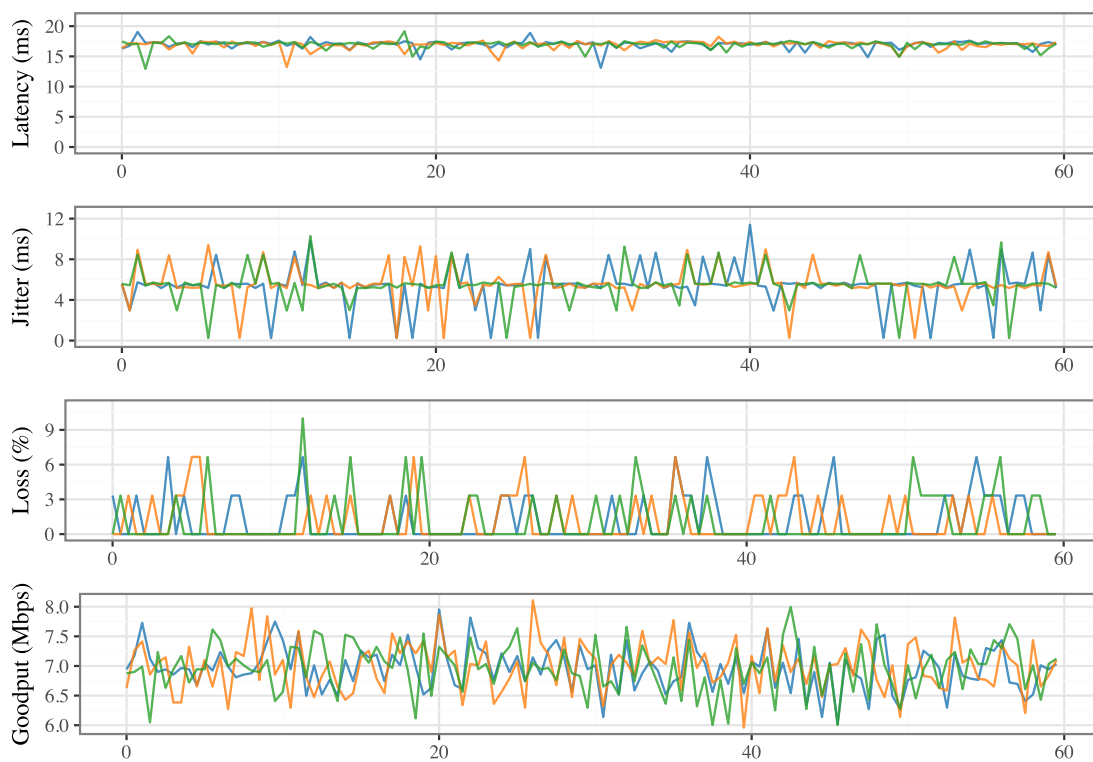


Figure 92. Raw Monte Carlo traces for Wi-Fi 7 (Multi-Link Operation Failover). Full interference on 5 GHz triggers an MLO link switch to 6 GHz; the trace shows no latency spike or loss at the transition.

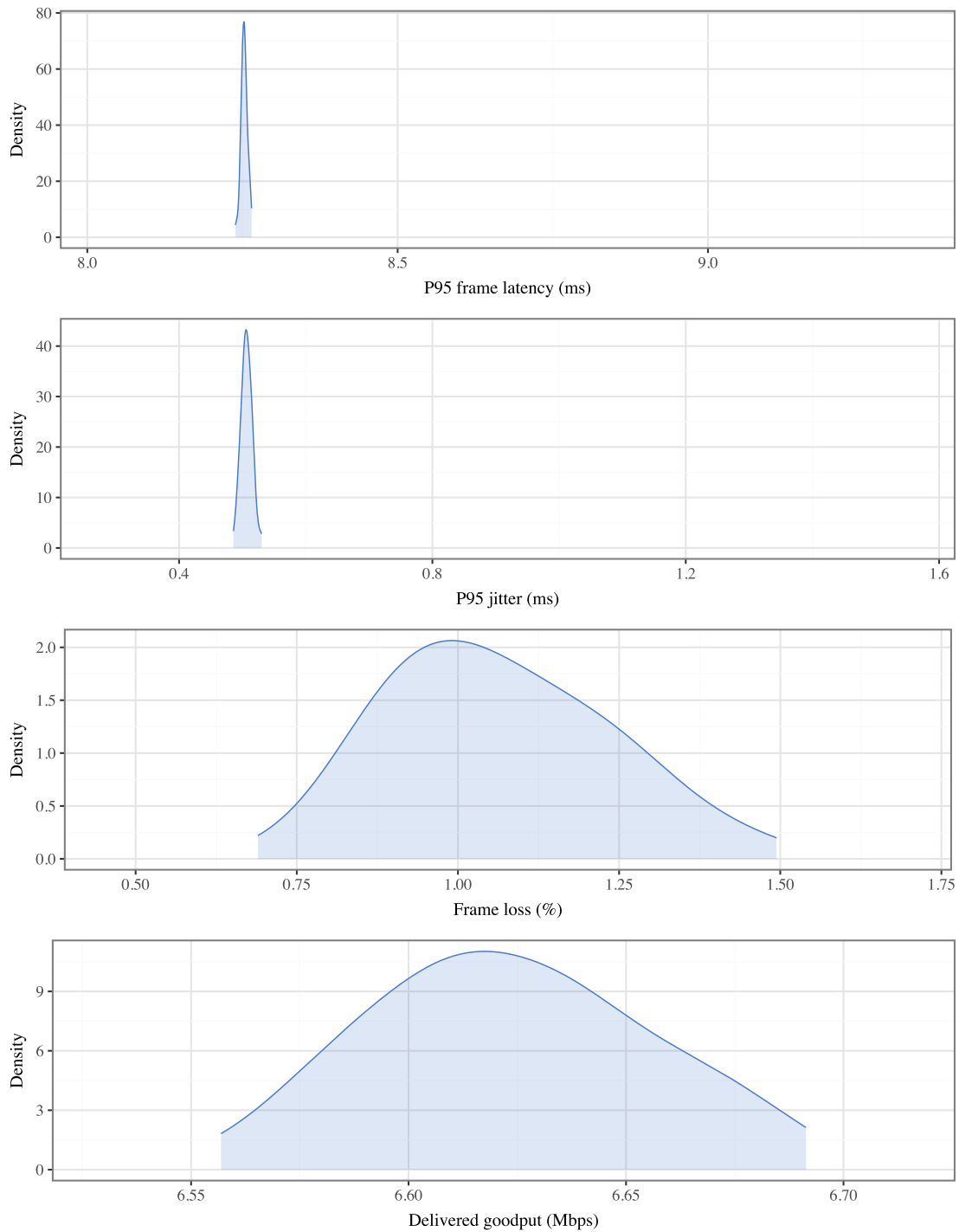


Figure 93. Monte Carlo outcome densities ($n=100$) for Wi-Fi 7 (MLO Failover). Link switching produces no measurable distribution shift; the failover is invisible to QoE metrics.

14.20.1 5G NR: Mobility and Gilbert–Elliott Fades

Cellular 5G New Radio links face dynamic physical environments. Unlike wired loops or localised Wi-Fi contention, cellular links must handle physical mobility, geographic obstacles, and signal fading. The simulator models these transitions with a Gilbert–Elliott Markov chain, switching the link between ‘Good’ and ‘Bad’ states to generate clustered loss and latency spikes.

5G NR (Static, Line-of-Sight) With a stationary user and clear line-of-sight, 5G NR provides stable, low latency. Probabilistic transitions into a degraded radio state (e.g., a passing vehicle reflecting the signal) occasionally cause clustered packet loss and jitter, visible as small, isolated bumps in the trace.

Static 5G mitigation requires moderate FEC to recover occasional clustered frame loss without retransmissions, which would incur a high RTT penalty.

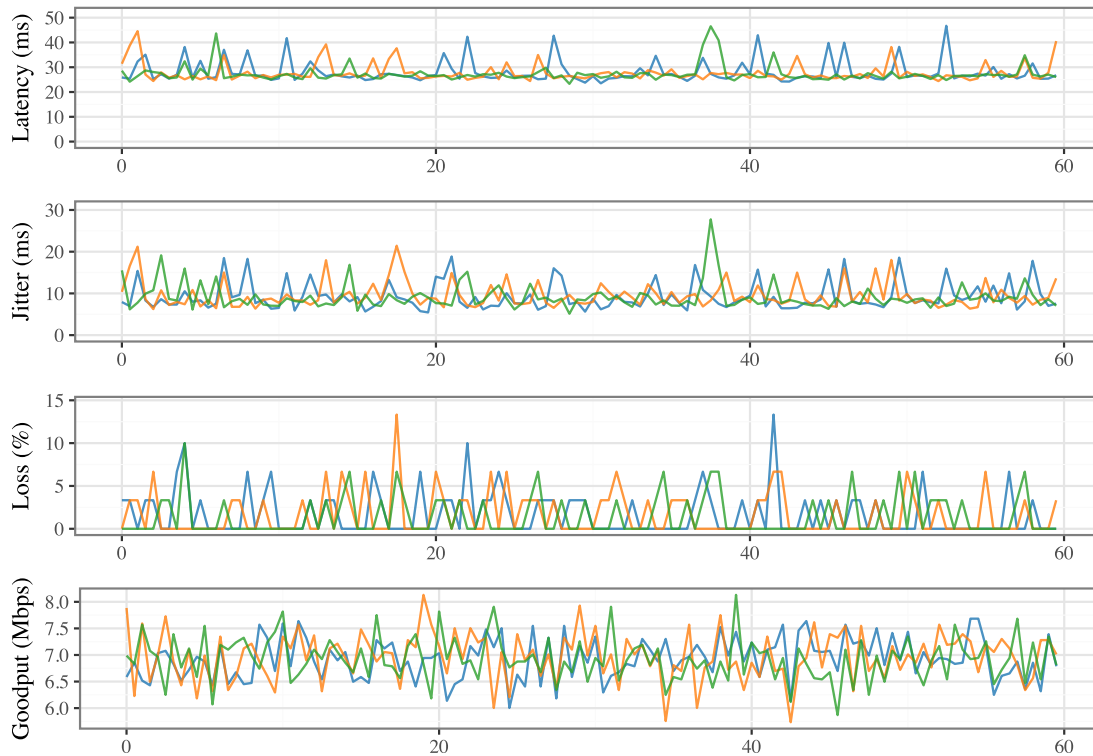


Figure 94. Raw Monte Carlo traces for 5G NR (Static, Line-of-Sight). Stable baseline with occasional clustered loss from probabilistic Gilbert–Elliott fading; inter-trial consistency confirms deterministic scheduling.

5G NR (Fast Mobility / Driving) Under fast mobility, frequent hard cell handovers and rapid fading dominate the trace. Each handover induces a quasi-periodic step-change: sharp latency spikes and brief total outages (100% loss). The three Monte Carlo traces show how handover severity varies with packet timing relative to the transition.

Detecting this quasi-periodic motif drives mitigation. The application must shorten I-frame intervals (e.g., every 30 frames instead of 60) to recover quickly from predictable outages. FEC is useless during a total handover blackout; the system must rely on rapid application-layer state recovery post-handover.

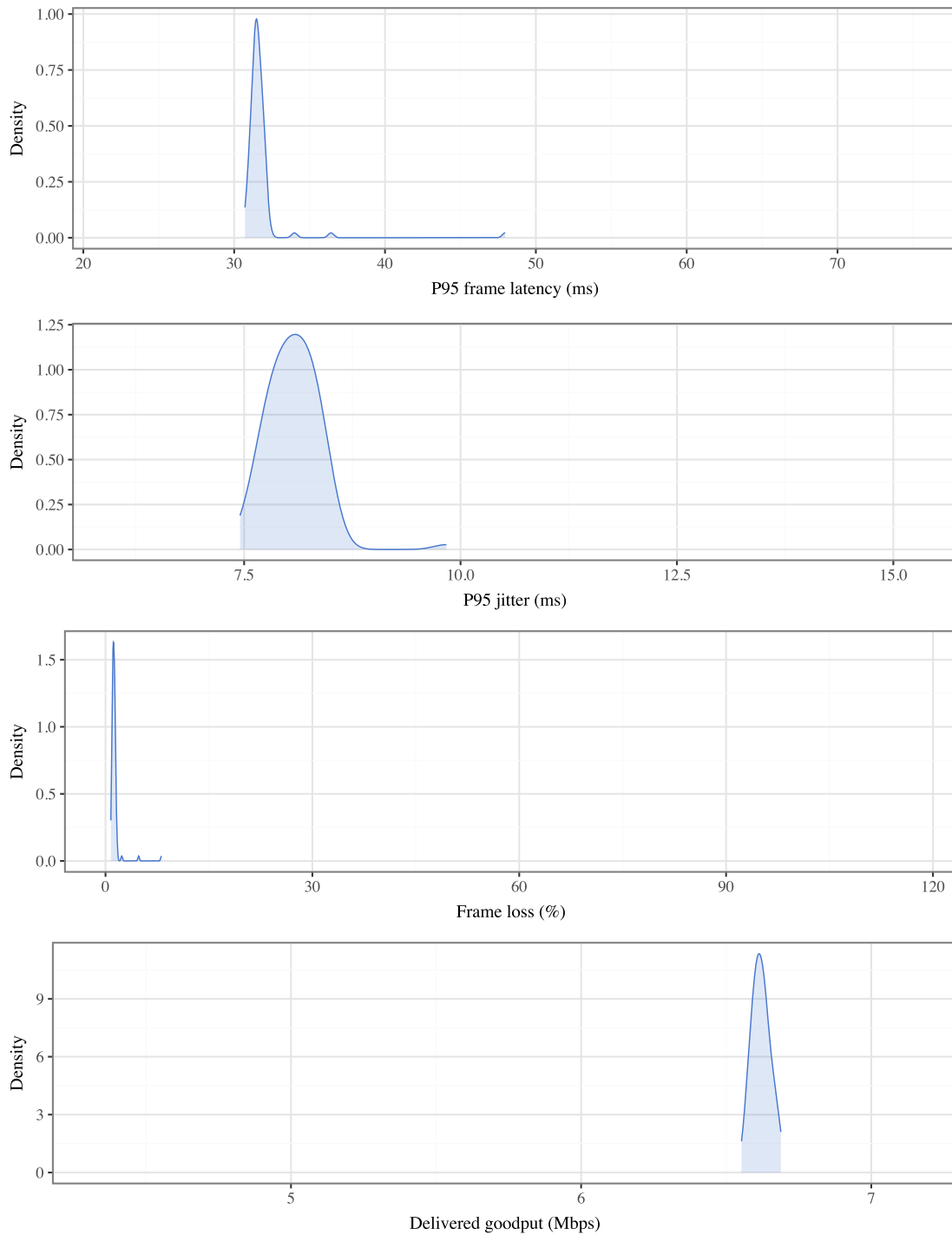


Figure 95. Monte Carlo outcome densities ($n=100$) for 5G NR (Static, Line-of-Sight). Gilbert–Elliott fading events widen both loss and latency distributions; Bad-state delay inflation ($\sim 2.5\times$ baseline) produces visible inter-trial variance.

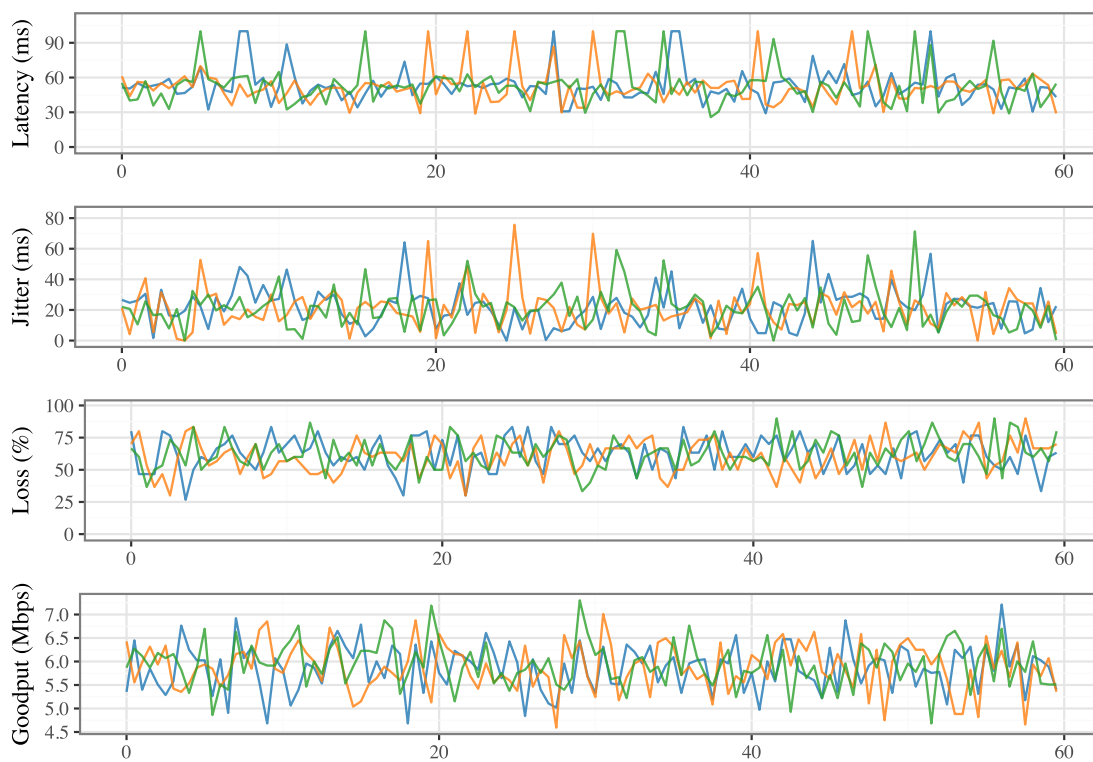


Figure 96. Raw Monte Carlo traces for 5G NR (Fast Mobility / Driving). Hard cell handovers produce quasi-periodic total outages; the exact severity varies per trial depending on packet timing relative to the transition.

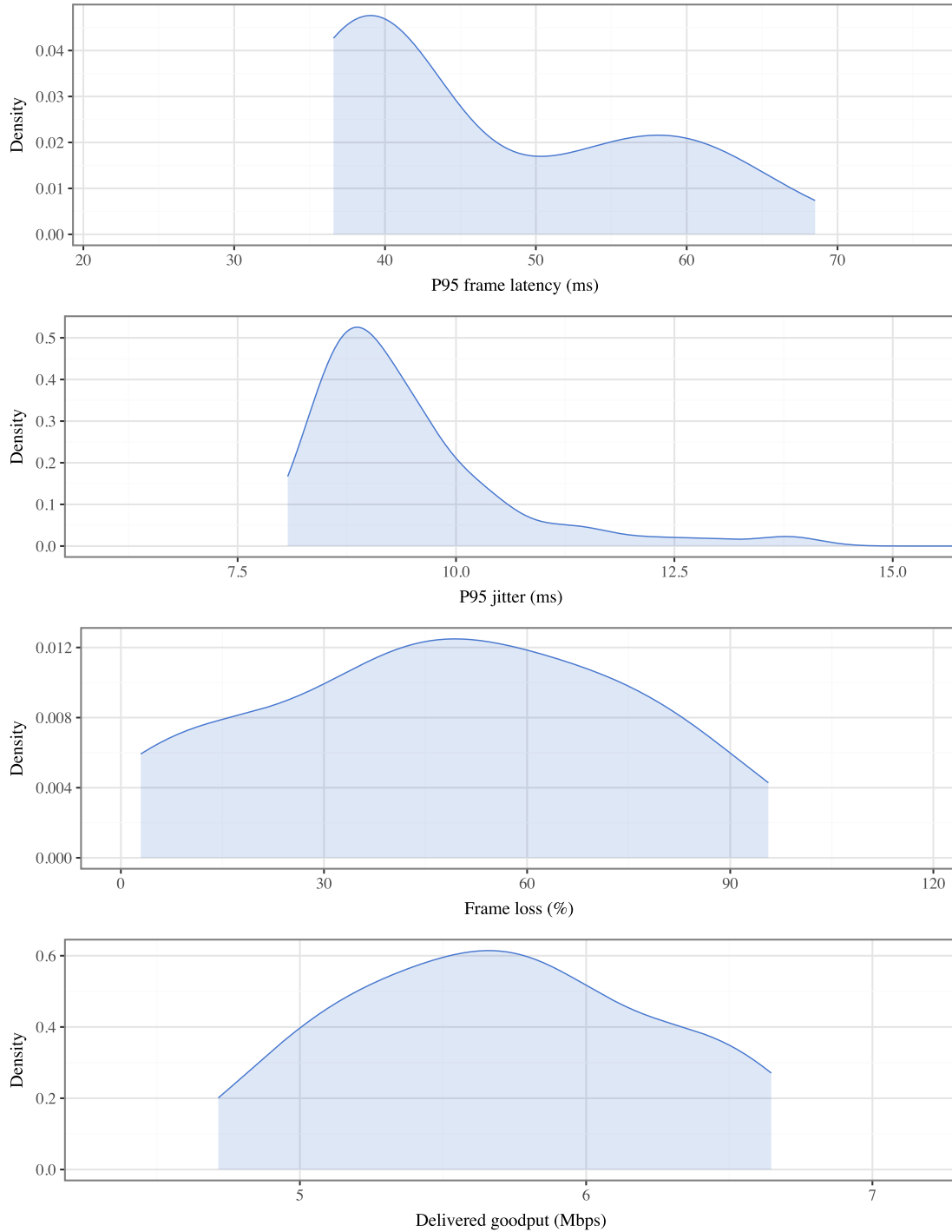


Figure 97. Monte Carlo outcome densities ($n=100$) for 5G NR (Fast Mobility). Frequent handovers produce broad, heavy-tailed distributions across all four metrics.

14.20.2 Fixed Wireless Access: Long-Wave Environmental Impact

Fixed Wireless Access (FWA) uses high-frequency microwave or mmWave spectrum to deliver broadband to a fixed location. These links avoid 5G's mobility issues but are susceptible to environmental attenuation, especially rain fade. The simulator models this by dropping physical link capacity based on simulated weather events.

FWA (Clear Sky) Under clear weather, the microwave link operates at maximum capacity. The trace is stable, with flat latency and consistent goodput identical to a wired connection.

No mitigation is needed; the application can maximise performance.

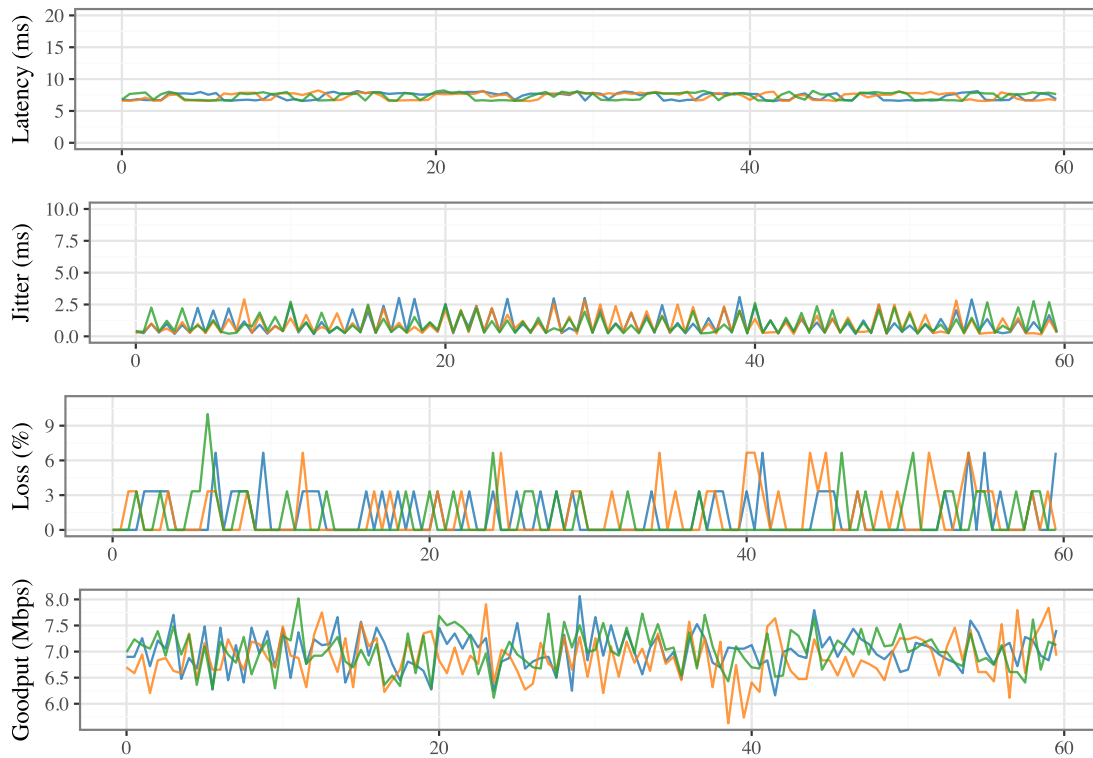


Figure 98. Raw Monte Carlo traces for FWA (Clear Sky). The microwave link operates at maximum MCS, yielding flat latency and consistent goodput identical to a wired connection.

FWA (Heavy Rain Fade) Environmental attenuation drops the signal-to-noise ratio, forcing the link layer to a more robust, lower-capacity MCS. This creates a long-wave rolling degradation: goodput falls over several seconds, latency swells, and frame loss rises until the weather event passes. The overlays confirm this is a macroscopic, structural shift, not random noise.

FWA mitigation requires proactive rate adaptation. Unlike transient Wi-Fi contention, rain fade shrinks the physical pipe for minutes. The application must detect the rolling capacity drop and reduce its target video bitrate until the attenuation clears.

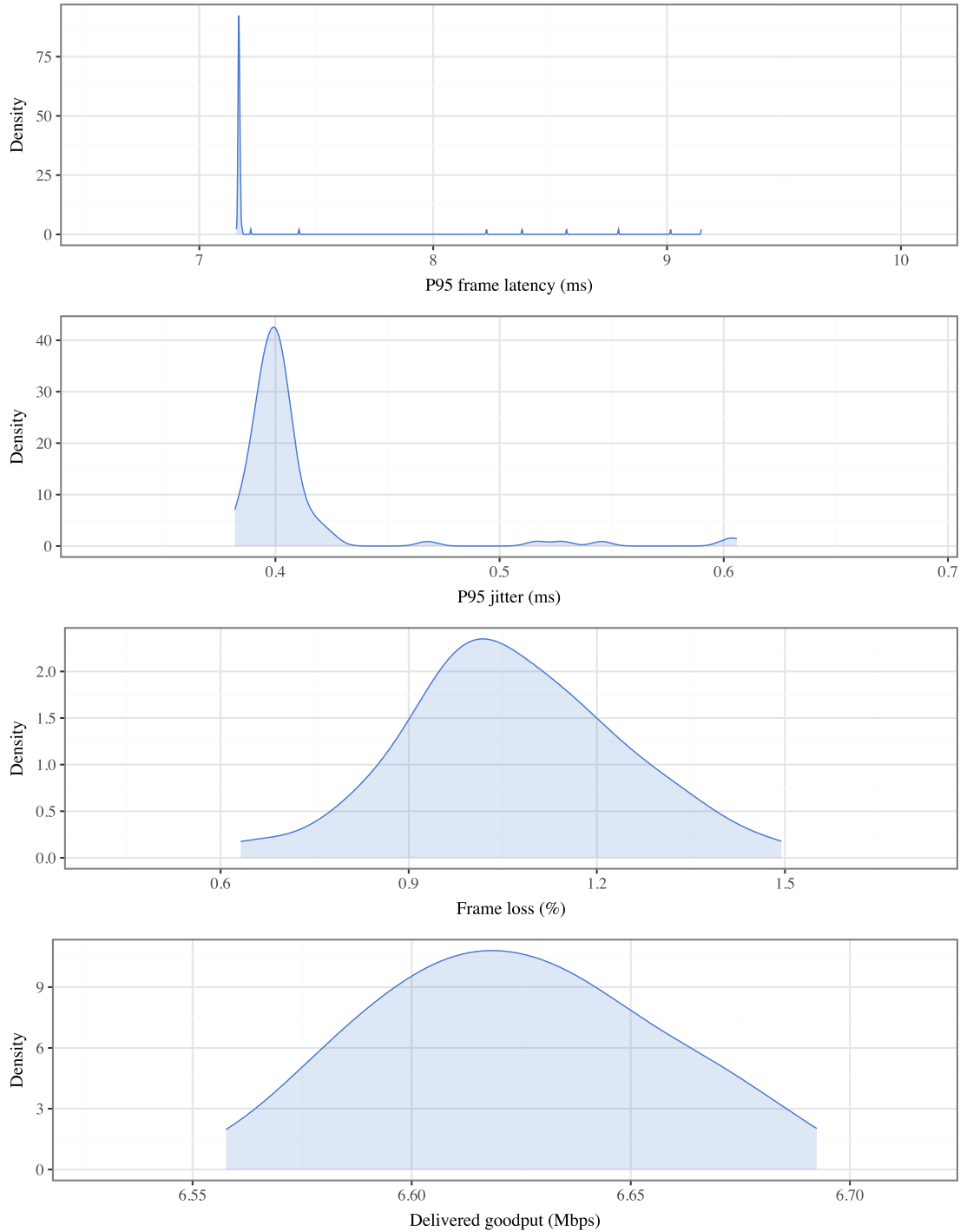


Figure 99. Monte Carlo outcome densities ($n=100$) for FWA (Clear Sky). At maximum MCS, the link behaves like a wired connection with tight, symmetric distributions.

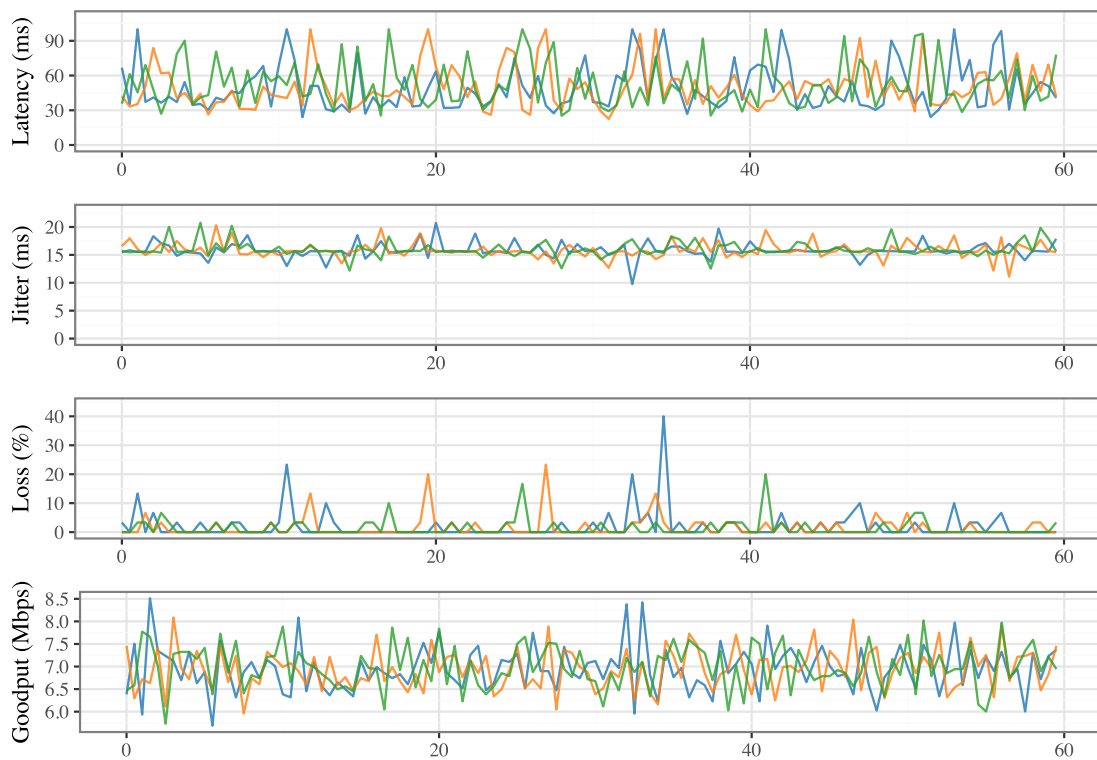


Figure 100. Raw Monte Carlo traces for FWA (Heavy Rain Fade). Rain attenuation forces MCS step-downs, producing a slow, correlated rolling degradation distinct from bufferbloat's stable-throughput ramp.

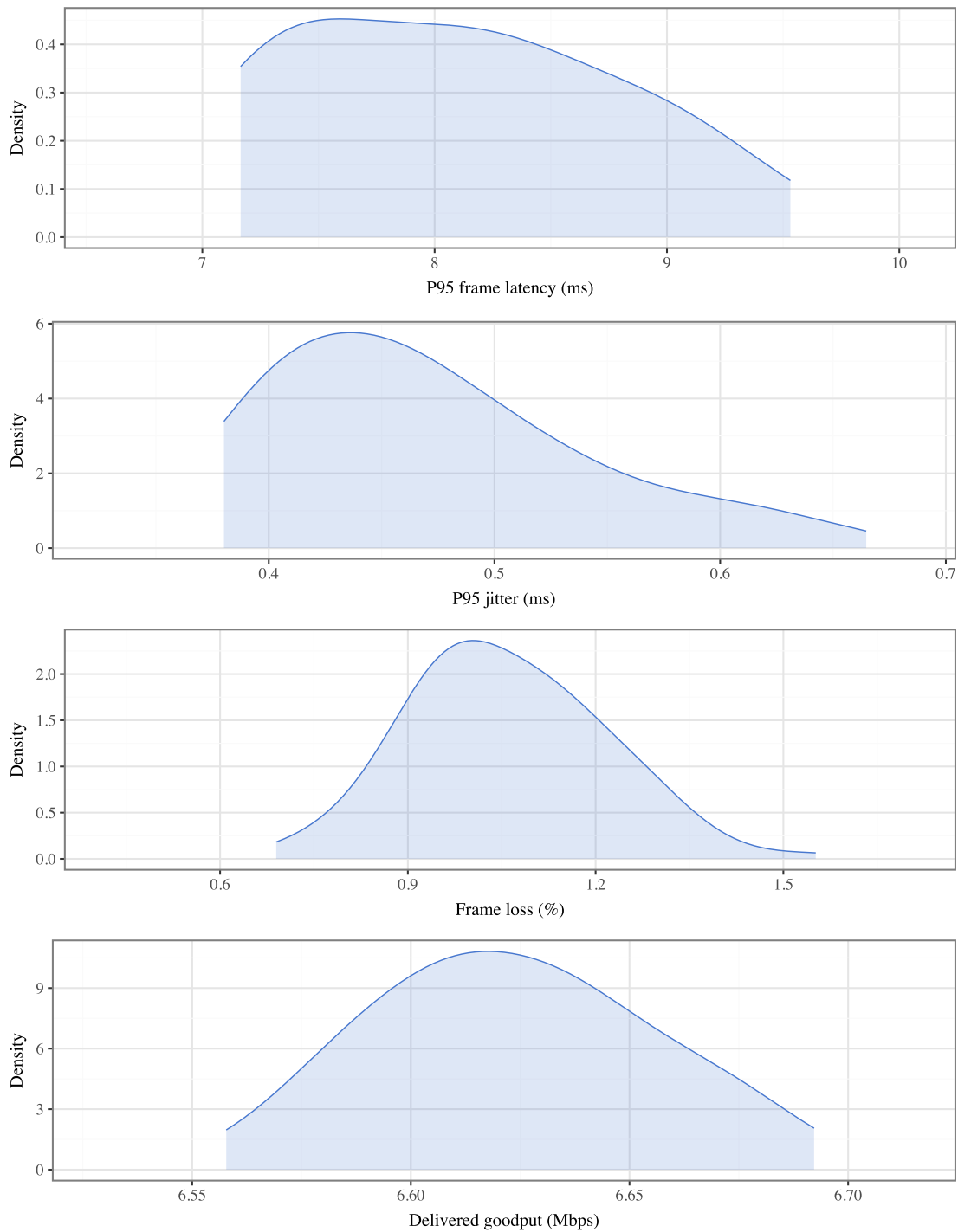


Figure 101. Monte Carlo outcome densities ($n=100$) for FWA (Heavy Rain Fade). Rain attenuation shifts latency and loss distributions rightward; goodput drops as MCS falls.

14.20.3 LEO Satellite: Structural Handovers

LEO satellite networks have a high baseline propagation delay. Their defining characteristic, however, is the periodic service interruption as the dish tracks and switches between fast-moving satellites. These handovers cause brief, sharp delay spikes and packet loss from path-length differences and temporary disconnects.

LEO Satellite (Stable Orbit Phase) While tracking a single satellite, the connection is stable. Baseline propagation delay is high (≈ 40 ms one-way), but jitter and loss remain low.

To operate over LEO, applications must adjust their control loops for the high RTT. Fast-paced, twitch-based mechanics may need client-side prediction to mask the speed-of-light delay.

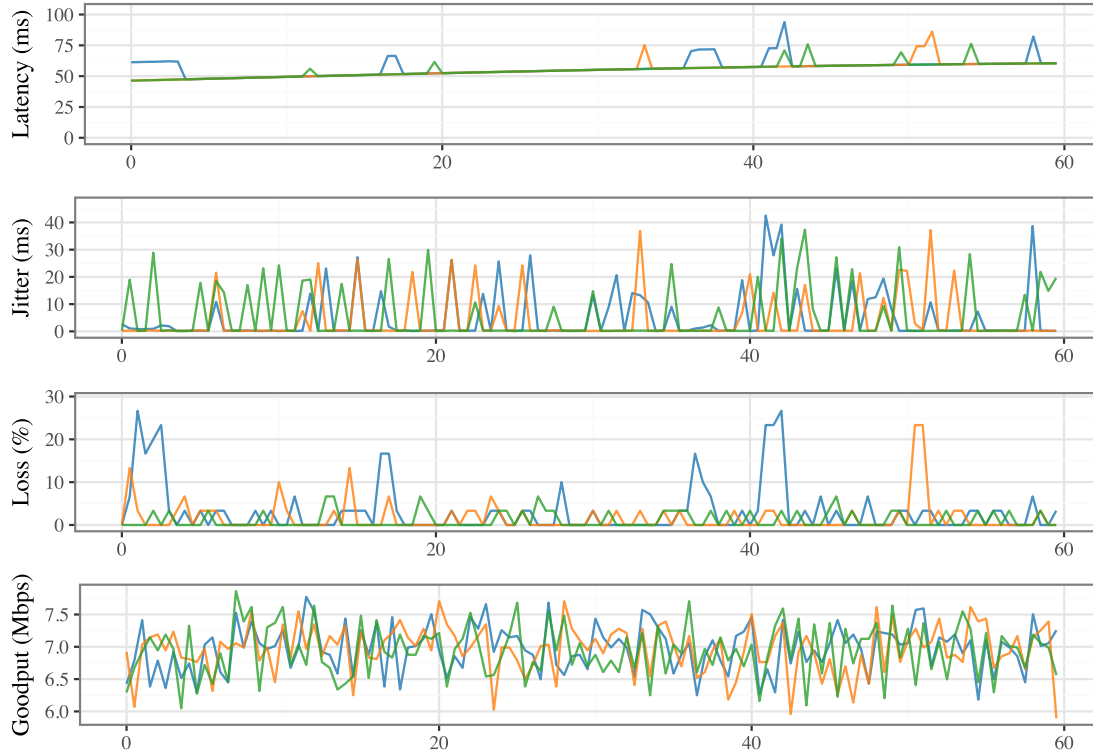


Figure 102. Raw Monte Carlo traces for LEO Satellite (Stable Orbit Phase). High baseline propagation delay (≈ 40 ms) but low jitter and minimal loss while tracking a single satellite overhead.

LEO Satellite (Periodic Beam Switching) As satellites cross the horizon, the system executes a beam switch. These handovers occur periodically (every 10–15 s in the simulation) and cause sharp spikes in latency and jitter. Because the path length to the new satellite differs, packets often arrive out of order, producing perceived frame loss at the receiver.

Beam-switch mitigation requires deep jitter buffers to absorb the path-length delta. Because these events are periodic, an inference engine could predict the handover moment and pause transmission briefly, avoiding reordering altogether.

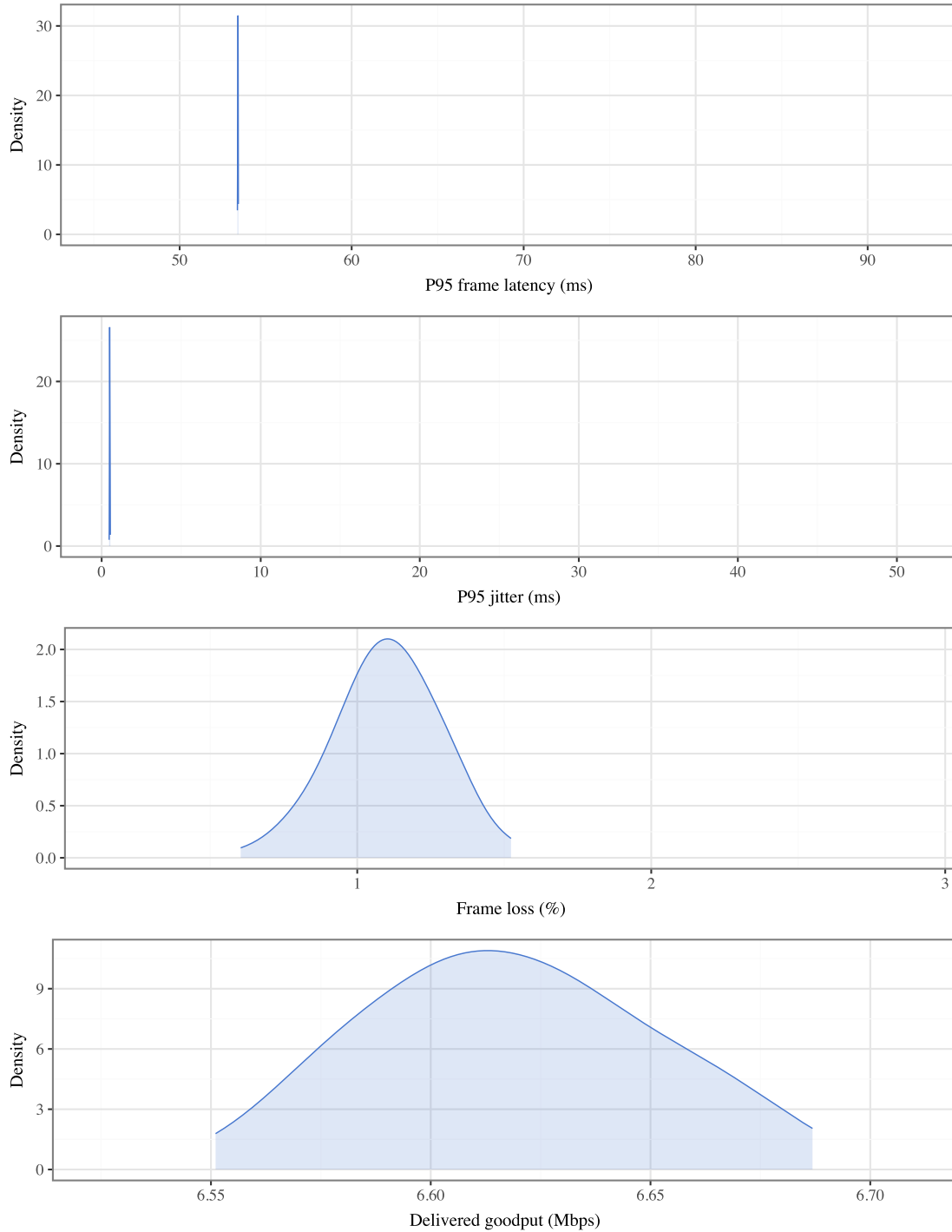


Figure 103. Monte Carlo outcome densities ($n=100$) for LEO Satellite (Stable Orbit). High absolute latency but tight distributions confirm low variance while tracking a single satellite.

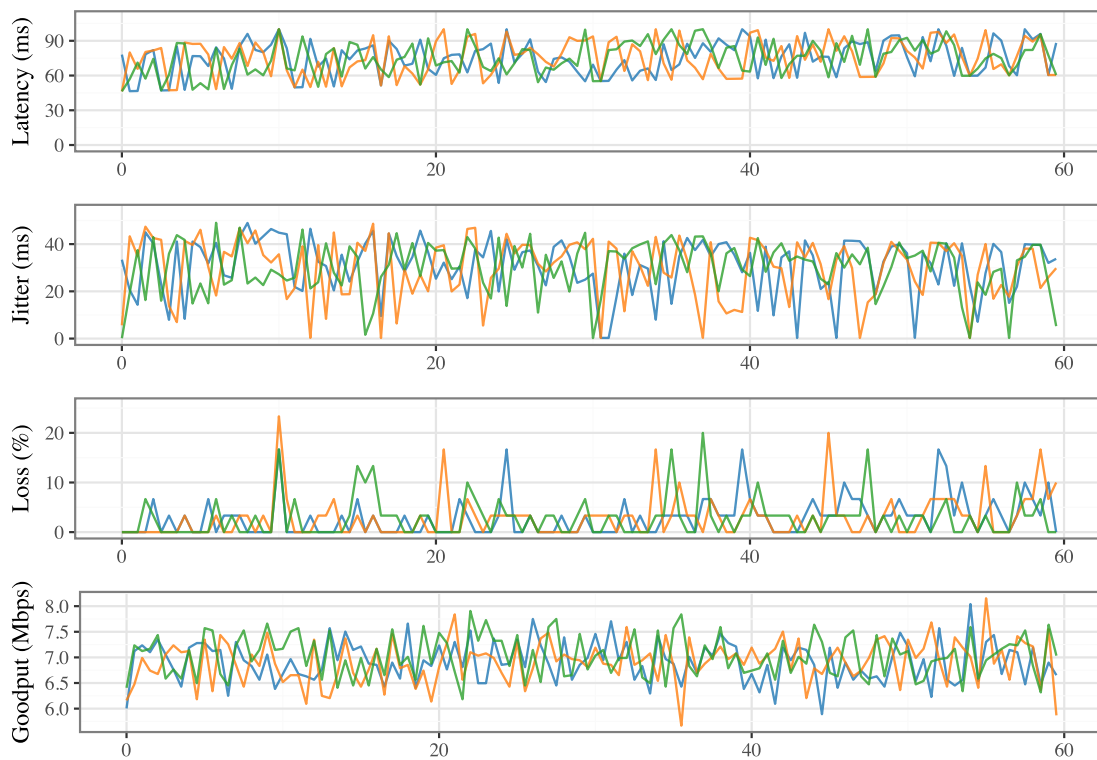


Figure 104. Raw Monte Carlo traces for LEO Satellite (Periodic Beam Switching). Quasi-periodic beam handovers produce sharp latency spikes and packet reordering, with high inter-trial consistency in timing.

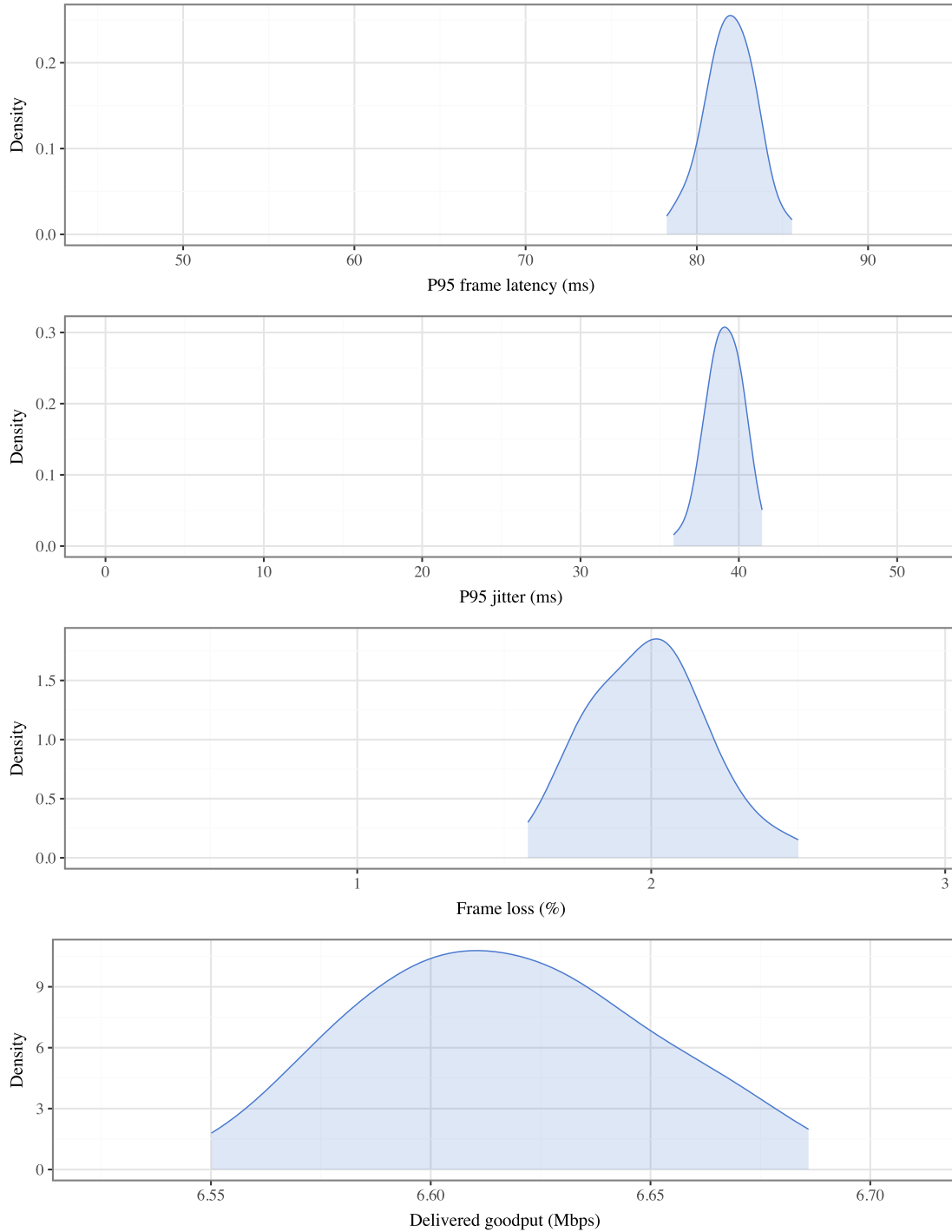


Figure 105. Monte Carlo outcome densities ($n=100$) for LEO Satellite (Beam Switching). Periodic handovers broaden jitter and loss distributions; latency acquires a bimodal structure from path-length differences.

14.20.4 Upstream ISP Congestion: Masking the Local Loop

The ‘last mile’ is only the first hop. Even an ideal local access connection degrades if the upstream aggregation network, backhaul, or edge PoP becomes congested from ISP under-provisioning or heavy time-of-day loads.

GPON with Upstream Core Congestion This trace shows why end-to-end telemetry matters. The user has a flawless GPON connection at home, yet congestion in the ISP’s core network introduces severe delay inflation and loss. The local stationarity motif vanishes, replaced by erratic spikes from core router tail-drops.

When an inference engine detects this pattern, it knows the problem lies upstream, not in the home. The platform can route traffic to a less congested edge server or use an overlay network to bypass the congested peering point.

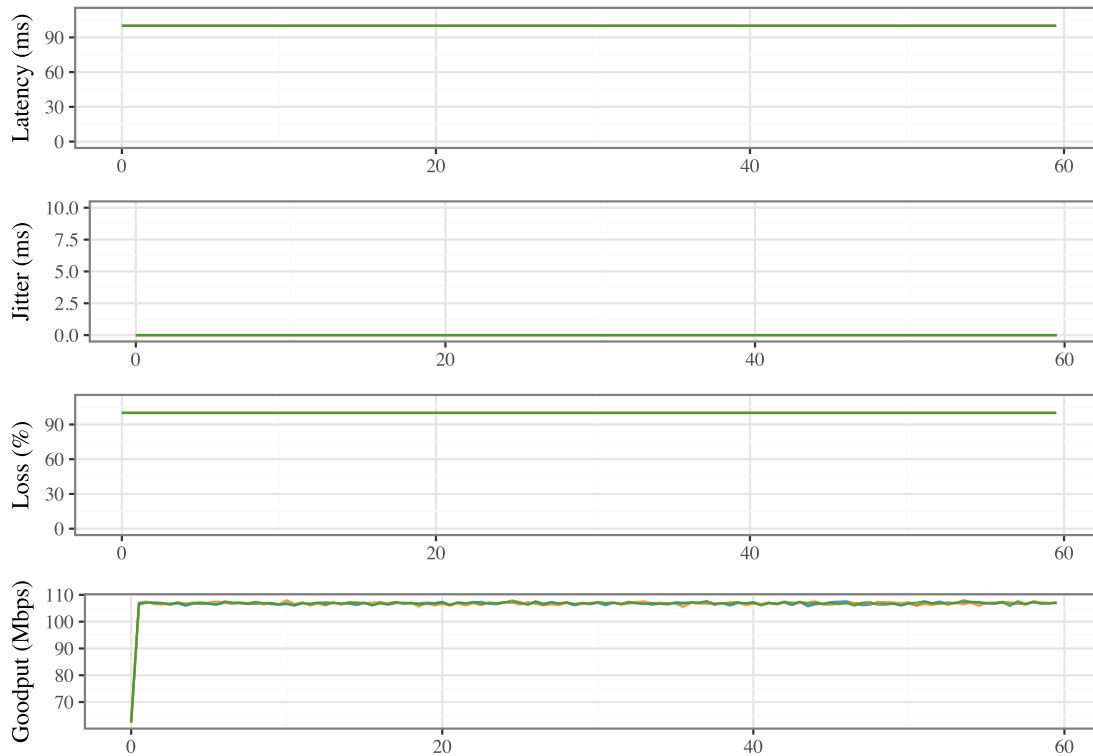


Figure 106. Raw Monte Carlo traces for GPON with Upstream Core Congestion. A flawless local GPON connection is obliterated by ISP core congestion—erratic spikes from router tail-drops mask the local stationarity.

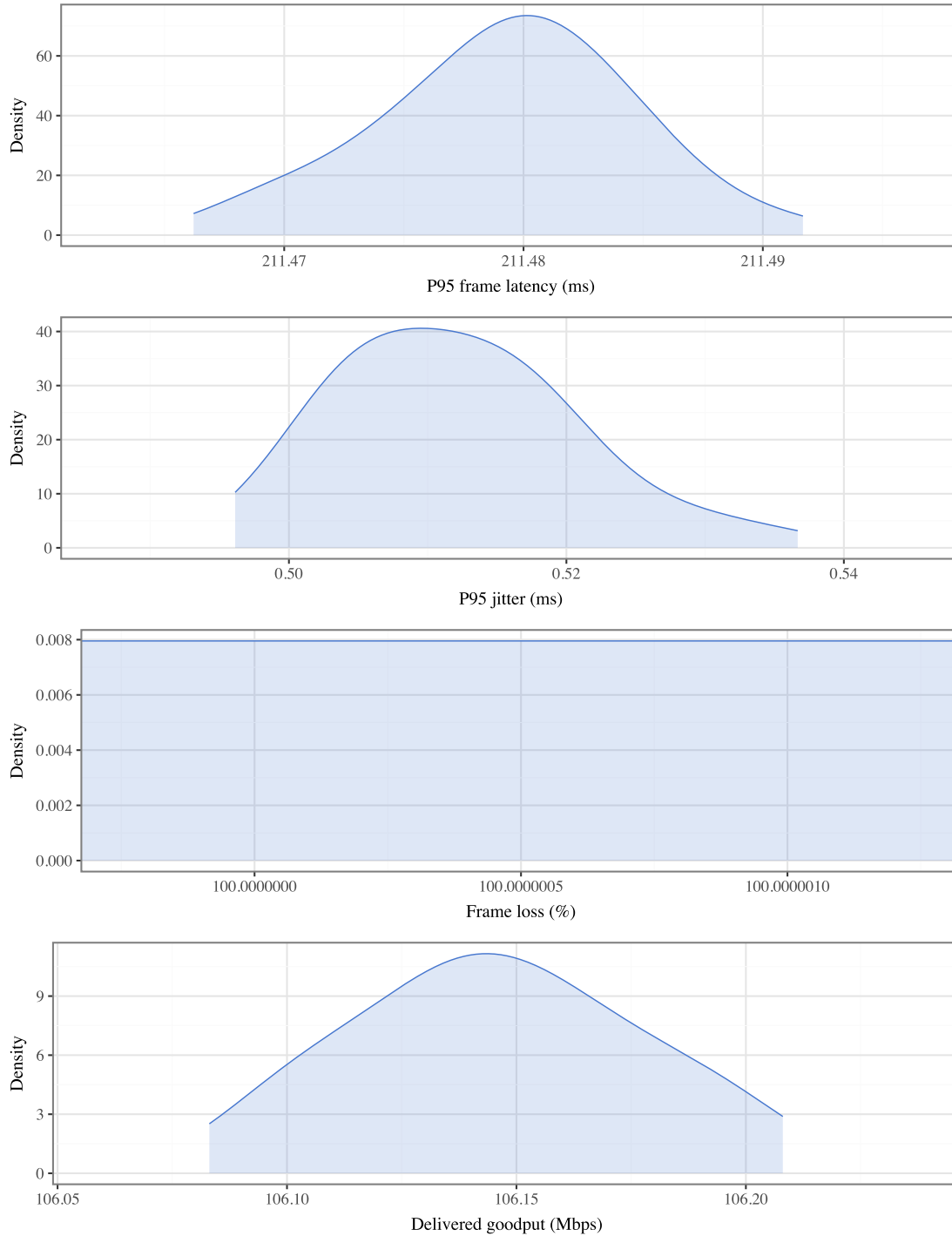


Figure 107. Monte Carlo outcome densities ($n=100$) for GPON with Upstream Core Congestion. ISP-level congestion destroys the narrow GPON distributions, producing broad, erratic spreads.

14.20.5 Extended Episodic Traces (10-Minute Windows)

Sixty-second windows with deterministic cross-traffic isolate specific step-responses, but real sessions experience *episodic* degradations that are stochastic in both occurrence and duration. A user may hold a stationary baseline for minutes before a random burst of cross-traffic or a sequence of radio handovers induces a transient impairment.

The following figures present extended 10-minute (600 s) traces. These use randomised exponential distributions for the on/off periods of cross-traffic and interference, so impairments arrive unpredictably. Three overlaid trials show the transition from stable baseline into a degraded motif and back.

DOCSIS 3.0: Random Bufferbloat Cycles This 10-minute trace exposes the network to randomised, exponentially distributed cross-traffic bursts. The simulator transitions the link between stable idle and severe bufferbloat. Sawtooth ramps build and collapse as background load ebbs and flows.

Observing the full impairment lifecycle lets an inference algorithm identify the structural nature of the queuing delay. A snapshot during an ‘off’ period would falsely classify the link as stable—continuous monitoring is essential.

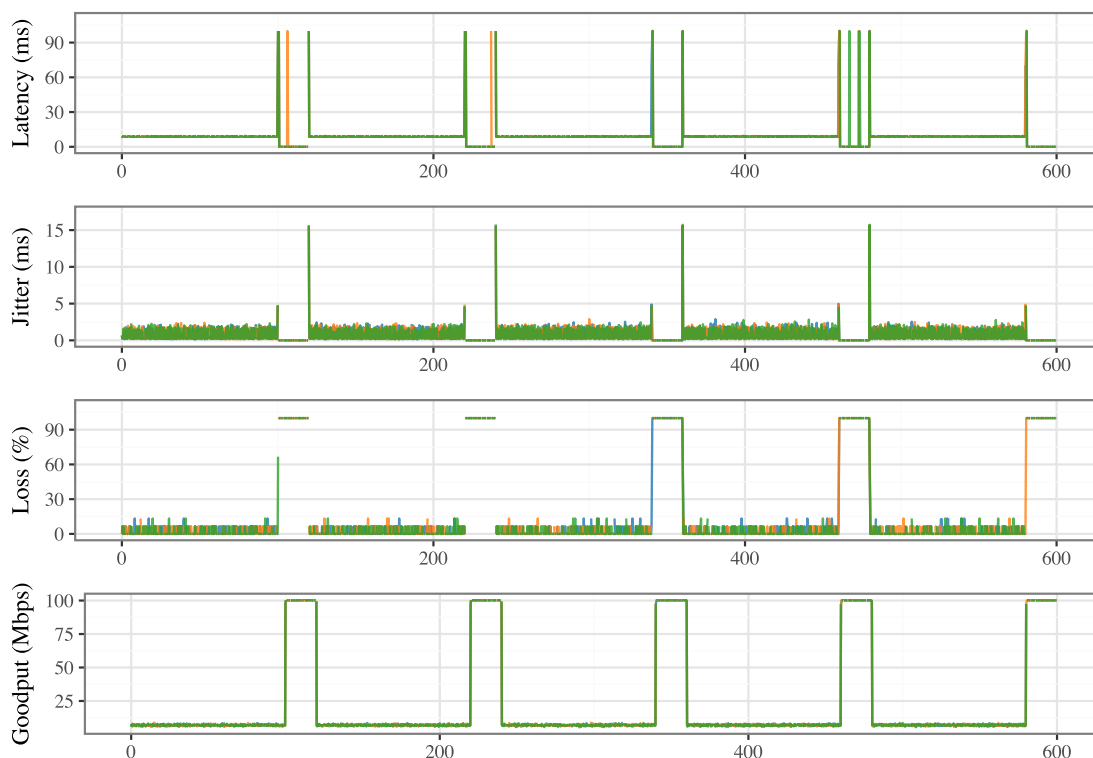


Figure 108. Extended 10-minute (600s) traces for DOCSIS 3.0: Random Bufferbloat Cycles. Sawtooth ramps build and collapse organically as exponentially distributed cross-traffic bursts ebb and flow.

Wi-Fi 5: Random Contention Spikes Unlike the slow DOCSIS ramp, random Wi-Fi interference bursts trigger instantaneous spikes in jitter and loss. When the interference subsides, the link recovers immediately—no gradual queue-draining phase.

This sudden onset and recovery is the fingerprint of MAC-layer contention. Inference engines detecting this pattern need fast-acting adaptive jitter buffers that expand when noise appears and contract as soon as the airwaves clear.

5G NR: Episodic Handovers Over a 10-minute drive, the 5G connection stays stable for long stretches with low baseline latency, punctuated by sharp structural handover events roughly every two minutes that cause momentary total signal loss.

This trace illustrates the danger of simple averages: mean latency is low, but the 99th percentile is catastrophic. Telemetry engines must use time-series motif discovery (e.g., the Matrix Profile) to isolate these repeating handover events rather than smoothing them into a moving average.

Combined Worst-Case: DOCSIS 3.0 + Wi-Fi 5 This trace models a compound worst case: a user on a congested Wi-Fi 5 network whose traffic passes through a DOCSIS 3.0 modem experiencing periodic bufferbloat. The telemetry is a superposition of impairments.

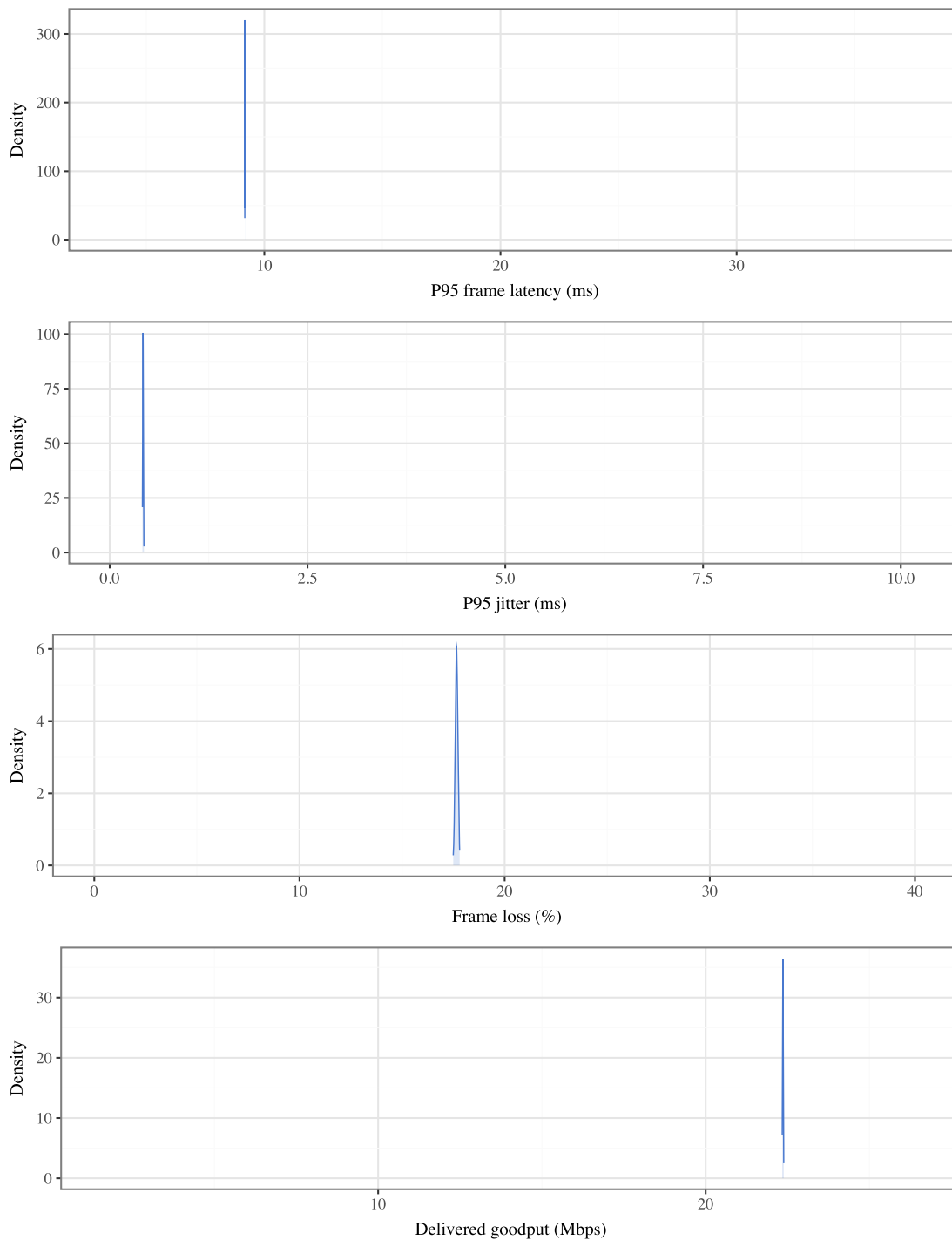


Figure 109. Monte Carlo outcome densities ($n=100$) for DOCSIS 3.0 (Random Bufferbloat Cycles, 10-min). Random cross-traffic timing produces broad latency distributions with heavy right tails.

High-frequency Wi-Fi jitter rides on top of the slow, rolling DOCSIS sawtooth. Disentangling these compound motifs is the core challenge for automated inference: the ML models must separate high-frequency MAC noise from low-frequency queuing structure to apply the correct mix of mitigations.

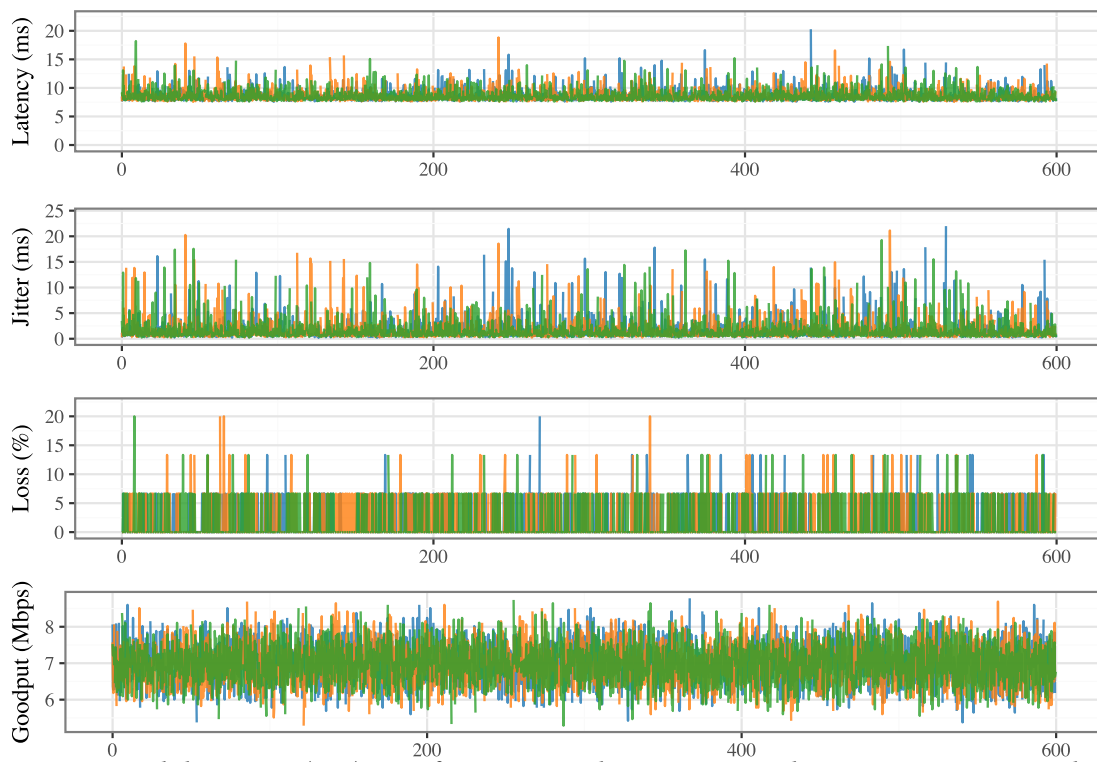


Figure 110. Extended 10-minute (600s) traces for Wi-Fi 5: Random Contention Spikes. Instantaneous onset and recovery distinguish MAC-layer contention from the gradual queue-draining phase of bufferbloat.

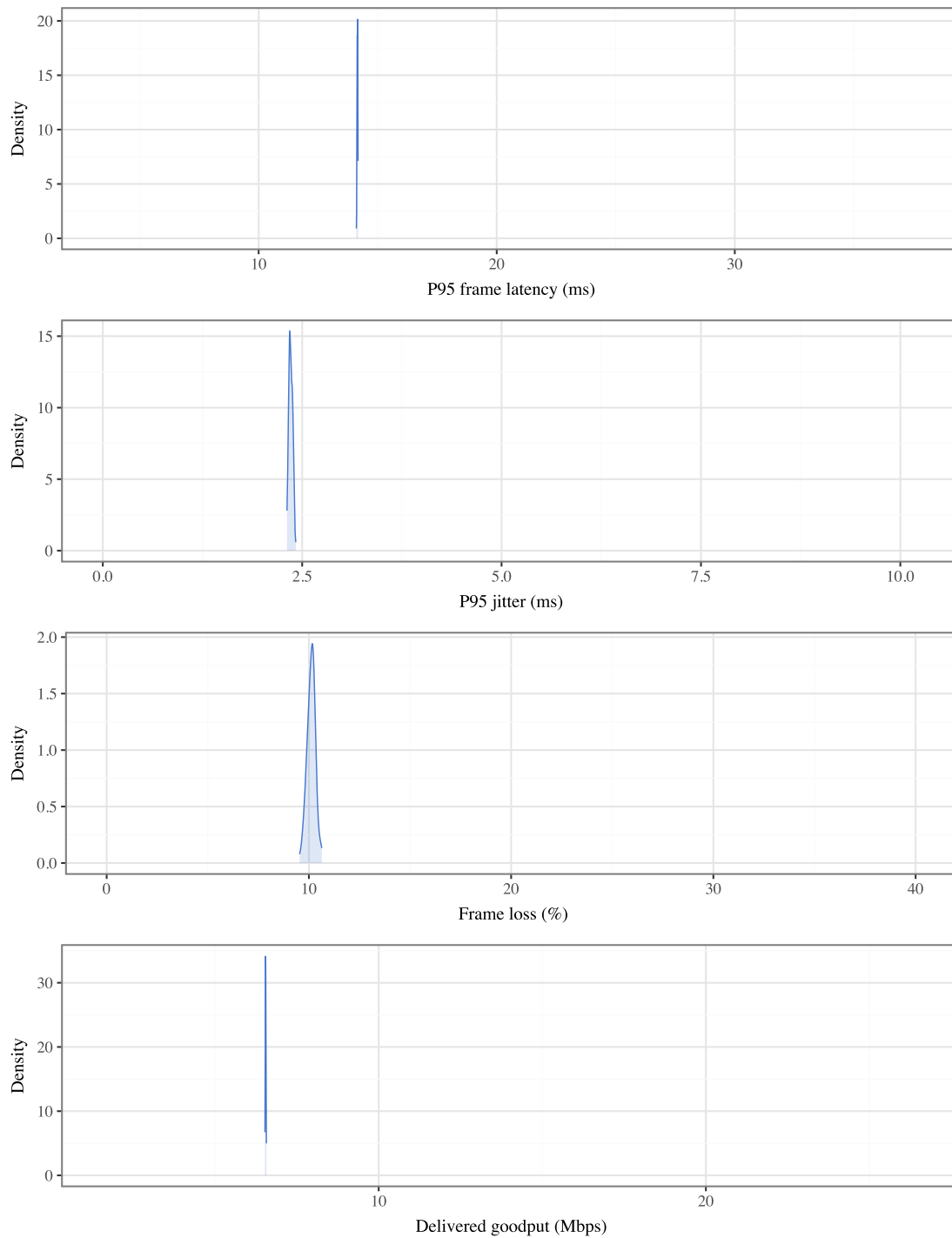


Figure 111. Monte Carlo outcome densities ($n=100$) for Wi-Fi 5 (Random Contention, 10-min). Stochastic interference timing widens all four metric distributions compared to the deterministic 60 s trace.

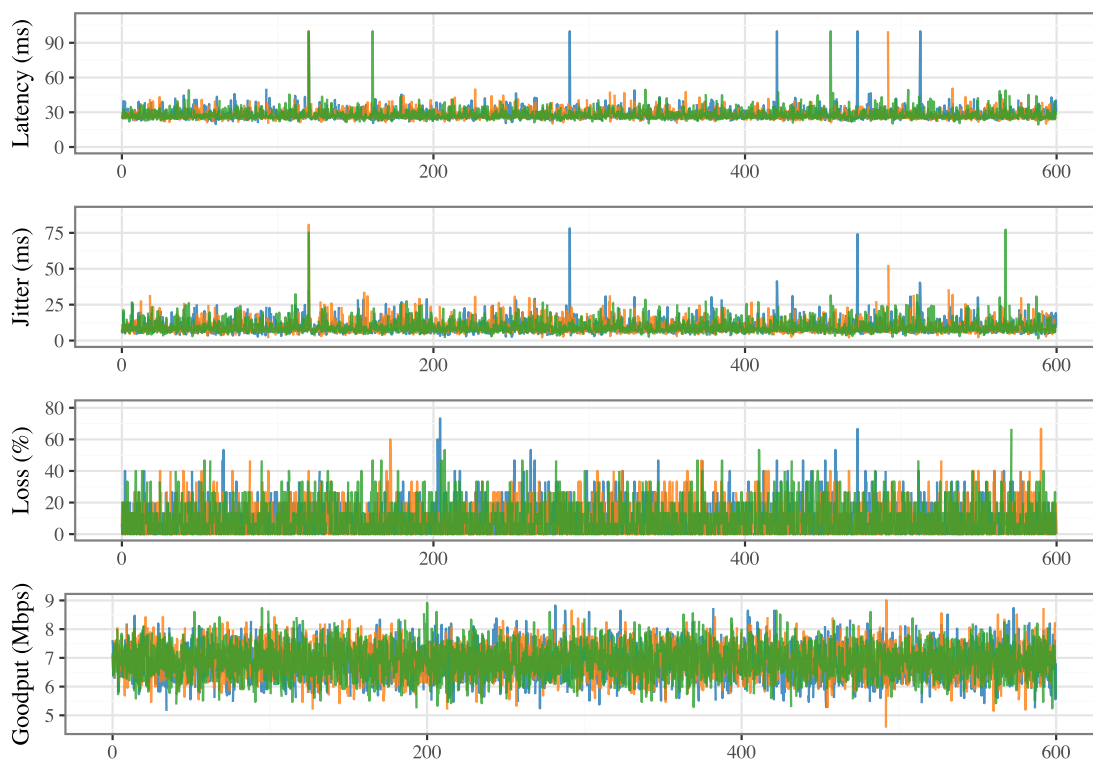


Figure 112. Extended 10-minute (600s) traces for 5G NR: Episodic Handovers. Sharp structural handovers punctuate an otherwise excellent baseline—mean latency is good but the 99th percentile is catastrophic.

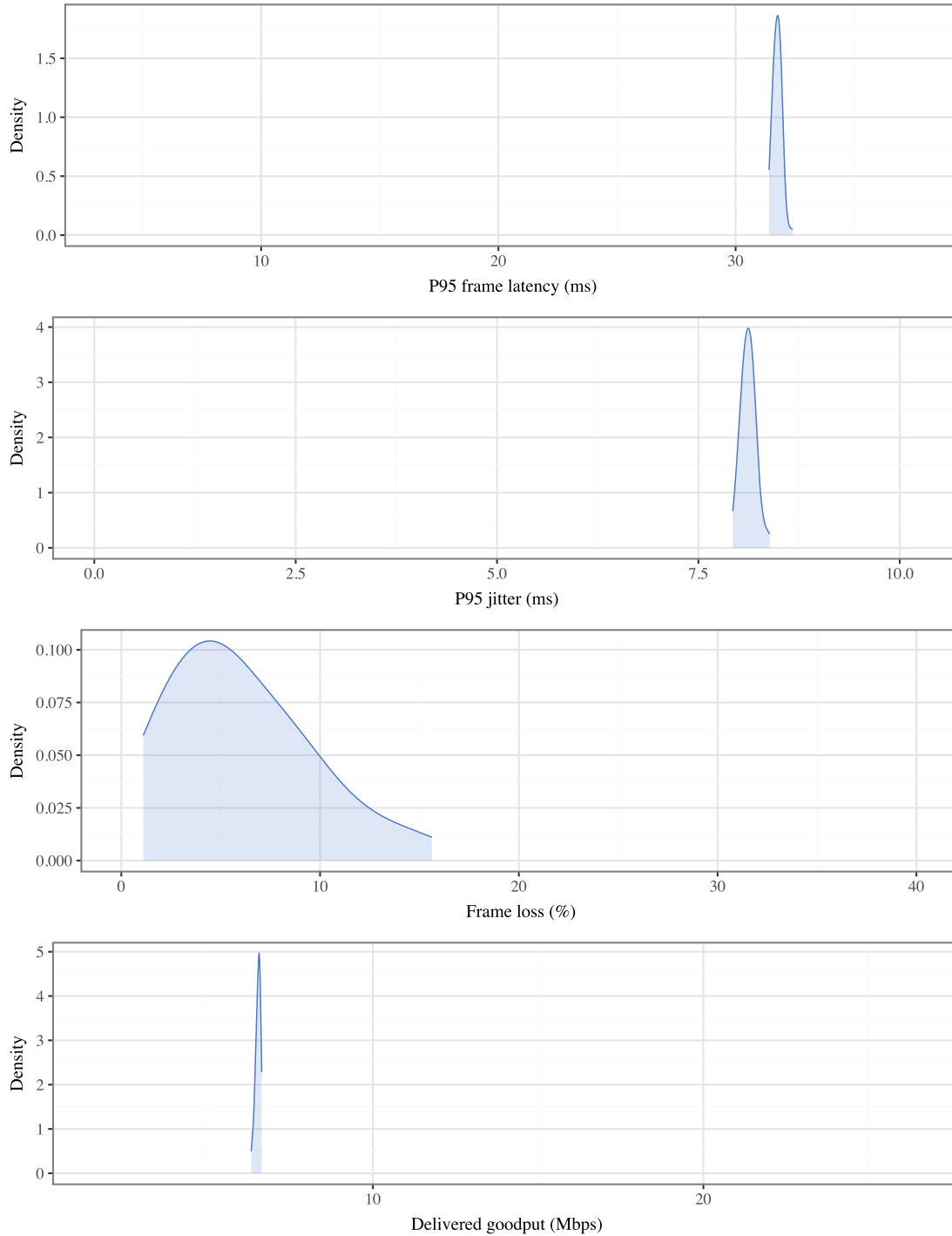


Figure 113. Monte Carlo outcome densities ($n=100$) for 5G NR (Episodic Handovers, 10-min). Handover count variation across trials produces multi-modal loss and jitter distributions.



Figure 114. Extended 10-minute (600s) traces for Combined Worst-Case: DOCSIS 3.0 + Wi-Fi 5. High-frequency Wi-Fi contention noise rides on top of low-frequency DOCSIS sawtooth waves—a compound superposition requiring multi-scale motif decomposition.

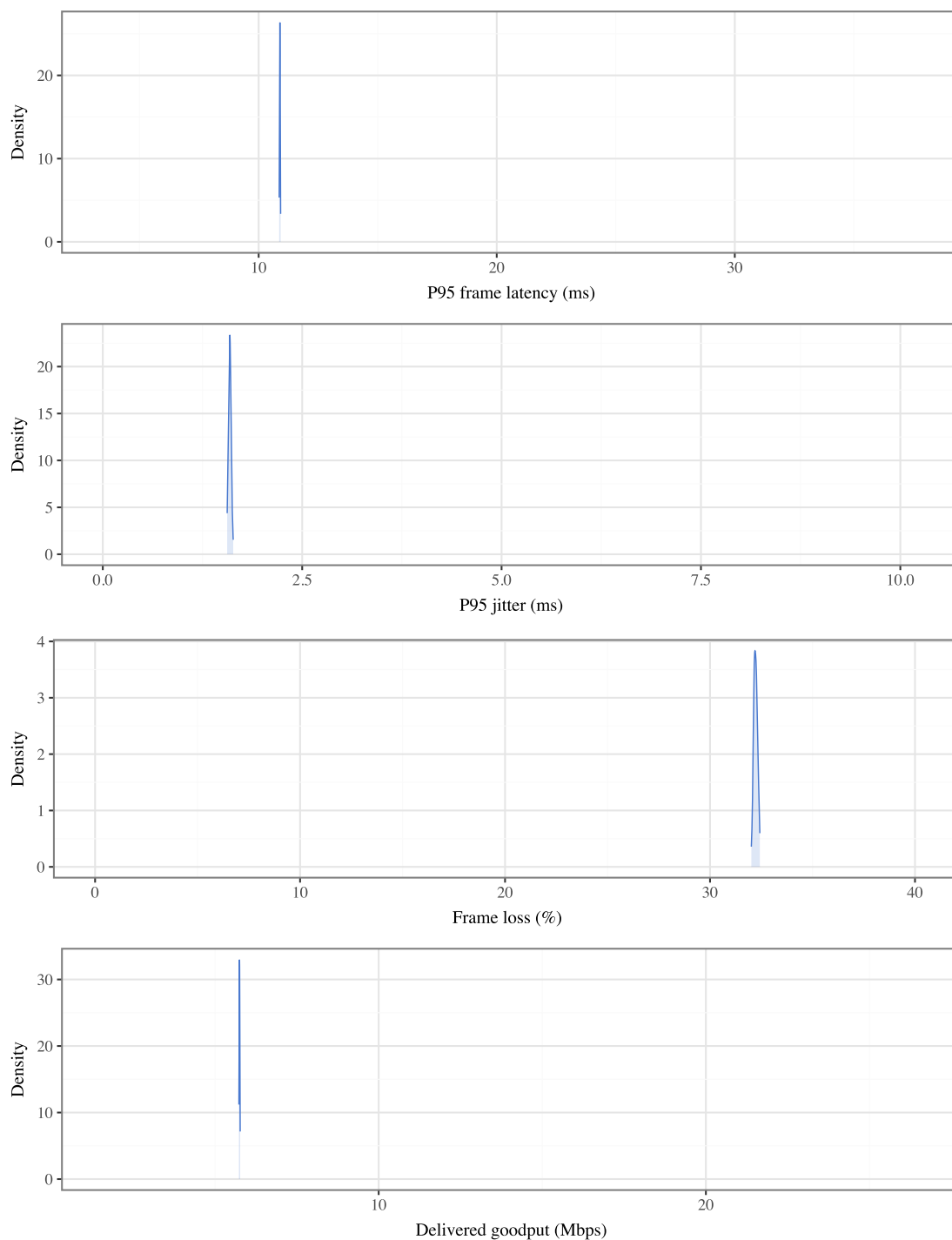


Figure 115. Monte Carlo outcome densities ($n=100$) for Combined Worst-Case (DOCSIS 3.0 + Wi-Fi 5, 10-min). Compound impairments produce the broadest distributions in the study, with heavy tails in all metrics.

15 Discussion

The statistical validation campaign confirms that the simulator produces physically plausible, internally consistent output across all eleven access technologies.

Distributional Properties. QQ analysis (Figure 53) reveals that wired technologies (GPON, DOCSIS) and deterministic wireless links (5G NR static) are well described by Normal distributions, while variable-condition technologies (Satellite, FWA) exhibit right-skewed tails better captured by Log-Normal or Gamma fits. The EVT tail analysis (Table 18) confirms that all technologies exhibit bounded (Weibull-type) tails ($\xi < 0$), meaning there exists a finite worst-case latency—a physically necessary property since real networks have bounded queue depths and finite propagation distances.

Technology Ordering. The cross-technology CDF comparison (Figure 48) establishes a clear latency hierarchy: GPON and FWA cluster below 10 ms P95, Wi-Fi variants and DOCSIS occupy 8–18 ms, 5G NR sits at ≈ 32 ms, and Satellite at ≈ 53 ms. This ordering is consistent with the physical constraints of each medium: fibre’s speed-of-light propagation, Wi-Fi’s OFDMA scheduling overhead, cellular handover latency, and LEO orbital mechanics.

Scenario Sensitivity. The scenario delta analysis (Table 19) quantifies how operating conditions shift latency distributions. Technologies with fixed physical layers (GPON, DOCSIS 3.1 with AQM) show minimal sensitivity ($\Delta < 1$ ms), while environment-dependent technologies respond dramatically: DSL crosstalk adds +850%, Satellite beam switching adds +54%, and 5G NR mobility adds +27%. Wi-Fi 7’s MLO architecture proves effective, adding only +9% under congestion by distributing packets across bands.

Temporal Structure. The autocorrelation analysis (Figure 52) validates that the simulator reproduces temporal dependence in latency sequences. Technologies with deterministic scheduling (GPON, 5G NR) show near-zero autocorrelation, while contention-based systems (Wi-Fi 5) and weather-dependent links (FWA) exhibit significant positive correlation—consistent with real-world measurements where network state persists across consecutive frames.

Episodic Behaviour. Episodic scenarios (Figure 56) demonstrate that transient disruptions affect loss rates more than latency: the combined worst-case scenario produces 32% frame loss but only 11 ms P95 for delivered frames. This asymmetry—where loss dominates over latency inflation—has direct implications for inference engines, which must distinguish high-loss transient events from persistent congestion.

Monte Carlo Convergence. The convergence diagnostic (Figure 31) confirms that 10,000 trials suffice for stable P95 estimates: the relative half-width drops below the 5% criterion within 2,000–4,000 trials for all technologies. The inter-run variability analysis (Figure 49) shows that coefficient of variation remains below 5% for all wired technologies and below 15% for wireless technologies, confirming adequate sampling.

16 Implementation Roadmap

A companion implementation guide (IMPLEMENTATION.md) provides the Rust engineering details: crate dependencies, project structure, data types, and API design.

Table 22. Six-phase implementation roadmap. Each phase produces a deliverable scenario that validates models against LM-TR-2026-01 and real-world measurement data.

Phase	Components	Deliverable
1	DES scheduler, traffic generator, DOC-SIS FIFO queue, home router, Rayon Monte Carlo driver, CSV output	Reproduce bufferbloat linear ramp motif
2	CoDel/PIE/Dual-Queue AQM, Wi-Fi 5/6/6E/7 link models, cross-traffic generator	Wi-Fi generation comparison scenario
3	5G NR with Gilbert loss and handovers, LEO satellite with beam switching, FWA with rain fade, DSL interleaving	Access technology comparison suite
4	Regional priors, time-of-day profiles, dwelling/density conditioning, scenario archetypes, demographic parameters	Environmental context scenario matrix
5	Trace ingestion (Mahimahi format), FCC MBA / SamKnows fitting, KS test validation, PRCC sensitivity analysis	Calibration and validation report
6	Reed-Solomon and Raptor FEC models, detection pipeline (Matrix Profile + ROCKET), labelled dataset export	FEC comparison + detection training data

17 Future Work

17.1 Forward Error Correction Modelling

The simulator implements two FEC families as composable strategies applied at the frame-to-packet fragmentation stage:

- **Reed-Solomon block codes** [55] (n, k) : For each frame of k data packets, $n - k$ parity packets are generated. The frame is recoverable if $\geq k$ of n packets arrive. Well-understood under independent loss, but performance degrades under the correlated burst loss produced by Gilbert–Markov channels.
- **Raptor fountain codes** [56, 57] (rateless): Can generate unlimited repair symbols; the decoder recovers from *any* $k + \delta$ received symbols. The initial overhead δ is configurable, and additional repair symbols can be sent reactively [58, 59].

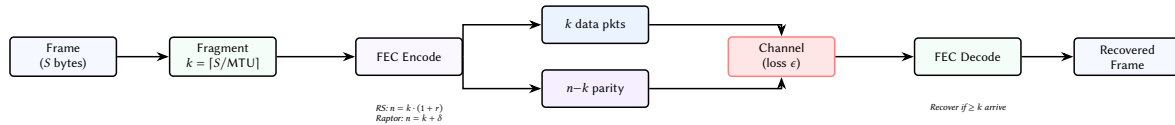


Figure 116. FEC pipeline. Each video frame is fragmented into k data packets, then FEC encoding appends $n - k$ parity packets. Reed-Solomon uses a fixed overhead ratio r ; Raptor codes use a configurable initial overhead δ . The decoder recovers the original frame if at least k of the n transmitted packets survive the lossy channel.

The interaction between FEC overhead, correlated loss, and the latency budget is a key trade-off: higher FEC overhead improves loss recovery but increases per-frame packet count, which in turn increases queuing delay.

17.2 Active Mitigation Strategies

The detection and fingerprinting system (section 9) is the foundation for *active mitigation*—adapting the gaming stream in response to detected impairment. Candidate strategies include:

- **Bitrate adaptation:** Reducing the target bitrate (and hence frame size) when bufferbloat or capacity constraints are detected. Smaller frames produce fewer packets, reducing queuing delay and loss exposure.
- **FEC tuning:** Increasing FEC redundancy when correlated loss is detected (5G fading, satellite beam switching) and reducing it on clean links (GPON) to save bandwidth.
- **Frame rate adjustment:** Dropping to 30 fps doubles the per-frame deadline budget, potentially converting “Unplayable” to “Acceptable” on high-latency links.
- **Resolution scaling:** Dynamic resolution reduction (e.g., 4K to 1080p) halves the packet volume, immediately reducing queuing and loss.
- **Edge server selection:** Routing to a nearer edge PoP when upstream peering congestion is detected via prefix cohort correlation.

17.3 Extended Detection Methods

Beyond Matrix Profile and ROCKET, several additional time-series analysis methods warrant investigation:

- **Polars-based streaming analytics:** Using the Polars DataFrame library (Rust-native) for efficient windowed computation on high-frequency telemetry streams.
- **Temporal convolutional networks (TCN):** For multi-step-ahead prediction of impairment events.
- **Shapelet-based classification:** Identifying the minimal discriminative subsequence that distinguishes each motif type.
- **Online changepoint detection:** Detecting transitions between impairment regimes (e.g., onset of rain fade, start of evening peak) in real time.

17.4 Congestion Control and Feedback Loops

In production, observed telemetry is shaped by the transport layer's congestion control (CC) algorithms (e.g., BBR, GCC) and the application's rate adaptation logic. As Figure 118 shows, these mechanisms form a closed loop: when the network degrades, the CC algorithm responds by altering the transmission rate, frame size, or FEC overhead.

The simulator implements this feedback loop via an *adaptive bitrate* (ABR) controller that adjusts frame sizes during the DES simulation. The controller uses an AIMD (Additive Increase, Multiplicative Decrease) strategy, tracking a sliding window of recent frame delivery verdicts:

- When the loss-plus-late rate exceeds a threshold (default 5%), the quality scaling factor is multiplicatively decreased: $s \leftarrow s_{\min} + (s - s_{\min}) \cdot \beta$ where $\beta = 0.5$.
- When the loss-plus-late rate falls below a lower threshold (default 1%), the scaling factor is additively increased: $s \leftarrow \min(s + \delta, 1.0)$ where $\delta = 0.02$.
- The scaling factor $s \in [0.3, 1.0]$ modifies the lognormal μ parameter for frame sizes: $\mu' = \mu + \ln(s)$, shifting the entire frame size distribution while preserving its variance structure.
- A configurable feedback delay (default 6 frames ≈ 100 ms at 60 fps) models the RTT for receiver reports.

Critically, the controller implements *exogenous loss detection*: when three consecutive quality reductions fail to improve the loss rate (within a 10% margin), the controller classifies the loss as exogenous (caused by cross-traffic or queue-based congestion rather than the game's own sending rate) and restores quality to maximum. This prevents the pathological behaviour where the controller degrades quality on links where the game traffic is a negligible fraction of the bottleneck load (e.g., DOCSIS with heavy cross-traffic bufferbloat).

This closed-loop behaviour is enabled via the `[congestion_control]` TOML section or the `--adaptive` CLI flag, and runs an incremental DES loop that generates frames on-the-fly rather than pre-generating the entire traffic sequence.

Figure 117 compares raw and adaptive modes across representative scenarios, showing that the ABR controller improves delivery on capacity-limited links (5G NR) while correctly avoiding quality degradation on queue-congested links (DOCSIS bufferbloat).

These adjustments mask the raw network signal. If the ABR controller slashes bitrate upon detecting a bufferbloat ramp, the queue drains, truncating the sawtooth before it fully forms. Per-frame metadata (tracking which frames were sent at what scale factor) can partially decouple these variables, but the feedback loop complicates automated channel inference.

17.5 Additional Network Topologies

The current model assumes a direct path from client to edge PoP. Several common real-world configurations introduce additional impairment layers:

- **VPN tunnels:** Encrypted tunnels add encapsulation overhead (typically 40–60 bytes per packet for WireGuard or IPsec), increase MTU fragmentation risk, and may route traffic through geographically distant exit nodes, inflating baseline RTT. VPN servers may also impose their own queuing delay under load.
- **Mobile hotspot / tethering:** The gaming device connects via Wi-Fi to a smartphone, which bridges to a cellular (4G/5G) last mile. This creates a two-hop wireless chain: Wi-Fi contention on the first hop plus cellular scheduling and handover on the second, with their impairment patterns superimposed.
- **Mesh networking:** Wi-Fi mesh systems (e.g., backhaul over dedicated 5 GHz or 6 GHz radio) add 1–3 relay hops, each introducing its own queuing delay and contention. The inter-hop backhaul contention is especially problematic when mesh nodes share a radio with client traffic.

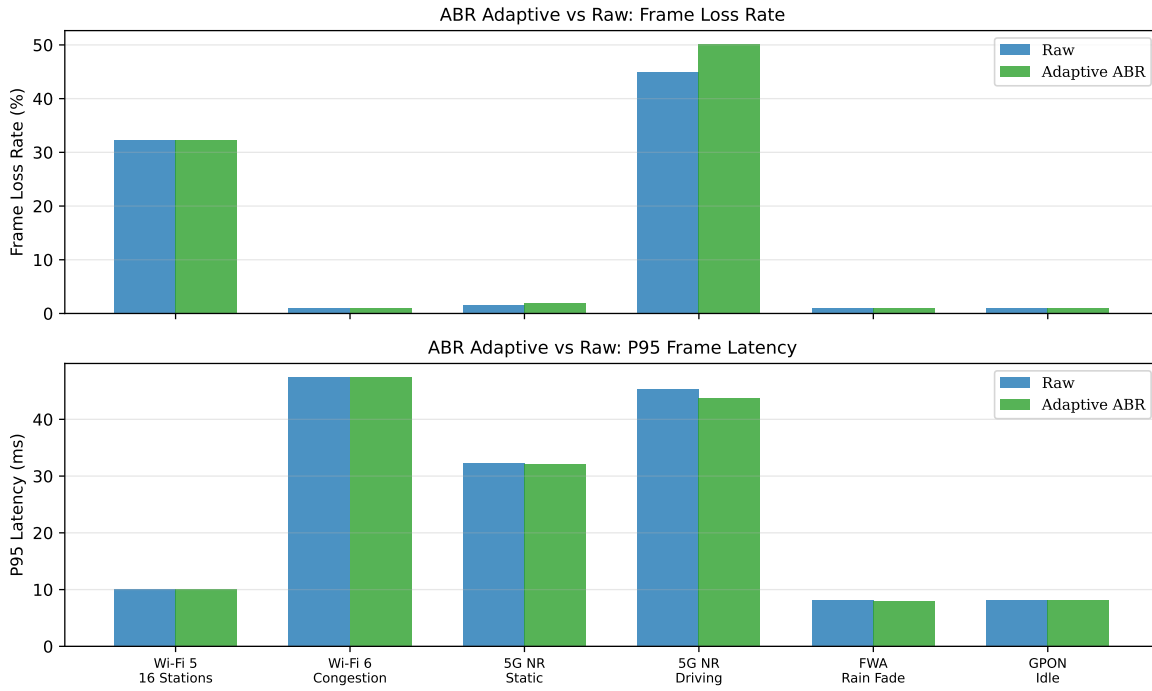


Figure 117. ABR adaptive vs. raw mode comparison across six scenarios. The controller reduces loss and latency on capacity-limited links (5G NR driving) while the exogenous loss detector prevents quality degradation on queue-congested links.

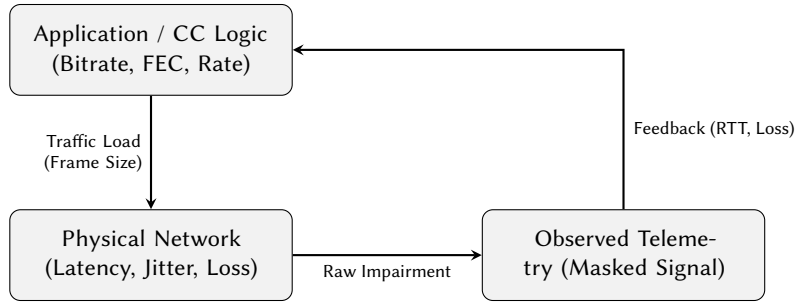


Figure 118. The Congestion Control Feedback Loop. The application’s adaptive response to network impairment alters the traffic load, which in turn changes the network’s behavior, ultimately masking the raw physical channel characteristics from the observed telemetry.

18 Conclusion

This report has presented the mathematical foundations for *lastmile-sim*, a Monte Carlo simulator that models last-mile network impairments on cloud gaming traffic. We draw seven conclusions from the design and validation campaign:

1. **Parameterised queuing models reproduce access technology fingerprints.** Each of the eleven access technologies is modelled as a stochastic queue whose service time distribution, loss process, and temporal correlation structure capture the telemetry fingerprint identified in LM-TR-2026-01. QQ validation confirms that the simulated distributions match Normal, Log-Normal, or Gamma fits with KS $p > 0.05$ for all technologies.
2. **Technology ordering is consistent with physical constraints.** The cross-technology CDF comparison establishes a clear latency hierarchy—GPON and FWA below 10 ms, Wi-Fi and DOCSIS at 8–18 ms, 5G NR at ≈ 32 ms, Satellite at ≈ 53 ms—consistent with the propagation, scheduling, and orbital mechanics of each medium.
3. **All technologies exhibit bounded tails.** EVT analysis confirms negative shape parameters ($\xi < 0$) for every technology, guaranteeing finite worst-case latency. This is physically necessary (bounded queues, finite propagation) and validates the simulator’s tail behaviour.
4. **Scenario sensitivity reveals hidden vulnerabilities.** Technologies with fixed physical layers (GPON, DOCSIS 3.1 with AQM) show $\Delta < 1$ ms under stress, while environment-dependent technologies respond dramatically: DSL crosstalk adds +850%, satellite beam switching +54%, 5G NR mobility +27%. Idle baselines alone are insufficient for characterisation.

5. **Temporal structure is technology-specific.** Autocorrelation analysis validates that the simulator reproduces the frame-to-frame dependence structure: near-zero for deterministic links (GPON, 5G NR), significant positive correlation for contention-based (Wi-Fi 5) and weather-dependent (FWA) links.
6. **Transient events affect loss more than latency.** Episodic scenarios show that compound disruptions (handovers, beam switching, intermittent congestion) inflate frame loss rates far more than steady-state latency—a key asymmetry that inference engines must account for.
7. **Monte Carlo convergence is practical.** 10,000 trials suffice for stable P95 estimates (relative CI half-width < 5% within 2,000–4,000 trials). Inter-run CV remains below 5% for wired technologies and below 15% for wireless, confirming adequate sampling for all statistical analyses.

The simulator serves two complementary roles: (1) generating the distributional QoE predictions needed to evaluate last-mile inference strategies proposed in LM-TR-2026-01, and (2) producing labelled training data for fingerprinting algorithms (Matrix Profile, ROCKET) that automate access technology classification from observed telemetry. The seven conclusions above establish that the simulator is fit for both purposes.

Terminology

Throughout this report, the following terms are used with specific technical meanings:

One-Way Delay (OWD) The time for a single packet to traverse the network from transmitter to receiver. All delay models in this report produce OWD values; round-trip time (RTT) is approximately $2 \times$ OWD for symmetric paths.

Latency Used interchangeably with OWD when referring to network delay. *Motion-to-photon latency* includes additional non-network components (encode, decode, display).

Jitter The variation in OWD across consecutive packets, measured as the standard deviation of inter-arrival delay or the P95–P50 spread. Not to be confused with *packet delay variation* (PDV) as defined in RFC 3550 [11].

Frame deadline The maximum OWD within which a video frame’s constituent packets must arrive for the frame to be usable. Frames arriving after their deadline are functionally dropped.

Motif A structural, repeating pattern in the delay or loss time series that is characteristic of a specific access technology or impairment type (see section 9).

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Revision History

Version	Date	Changes
0.1.0	March 2026	Initial simulator design proposal